MAT3032: Advanced Algebra

Notes written by:

Dr David Fisher
University of Surrey

Minor Modifications Made by the module leader:

Dr James Roberts
29a AA 04
School of Mathematics and Physics
University of Surrey
james.roberts@surrey.ac.uk

MODULE MAT3032 ADVANCED ALGEBRA

Chapter 1

Some History

Familiar objects of study in algebraic structure theory: groups, rings, fields.

The origins of group theory lie in number theory, geometry and equation-solving.

Euler (1761) studied congruences modulo n, developed further by Gauss (1801).

Lagrange (1770) studied permutations of zeros of polynomials. Methods of solution for cubics and quartics were already known. Lagrange wanted to solve the quintic by radicals - later shown by Abel to be impossible.

Cauchy (1844) introduced notation for permutations and investigated their properties.

Möbius (1827) studied geometric properties invariant under particular transformations.

These, and many others, used ideas that would now be seen as group-theoretic - but they did not recognise the unifying abstract concept.

Galois (1832) discovered the relation between solutions of a polynomial equation and a set of permutations now called the Galois group.

Cayley (1849) gave a general definition of a group as a set of distinct symbols, closed under a 'product'.

He recognised that the product must be associative but not necessarily commutative.

His definition implied, but did not explicitly state, the existence of an identity and inverses.

He also showed that every finite group can be represented by permutations (Cayley's Theorem) and illustrated by a table.

Cayley's work had little impact at the time, but he returned to it in 1878

and inspired others including Hölder (1893), who classified groups whose orders were products of up to three primes.

Sylow (1872) proved important structural results about permutation groups.

Lie (1880s) studied continuous groups, with applications to differential equations.

Von Dyck (1882) drew together the various aspects of group theory, recognising that groups could be finite or infinite, and introduced group presentations.

Burnside (1897), Frobenius and others developed group representation theory.

Books on abstract groups started to appear in the early 20th century.

Group theory developed in different directions, e.g. topological groups.

Groups were recognised as the natural mathematical tool for studying all kinds of symmetry and were applied in chemistry, physics and many areas of applied mathematics.

The classification of all finite simple groups was completed in 1981.

From 'A History of Abstract Algebra' by Israel Kleiner

Ring theory has two distinct forms: commutative and noncommutative.

Commutative ring theory was inspired by algebraic number theory, particularly Fermat's Last Theorem: $x^n + y^n = z^n$ has no solutions for $x, y, z \in \mathbb{N}$ if $n \geq 3$.

Polynomials are another motivation for studying commutative rings and fields. These two aspects of commutative ring theory were unified by Noether (1921).

Complex numbers were implicitly used in solving polynomial equations in the 16th and 17th centuries, and were formalised in the 18th century by Wallis, Euler, De Moivre, Argand and others.

Hamilton (1831) defined complex numbers as ordered pairs of reals with $(a, b)(c, d) = (ac - bd, bc + ad)$.

He also (1843) devised the quaternions, a noncommutative structure.

This led to attempts to find more systems of 'hypercomplex numbers', now called 'associative algebras', having the structure of both a ring and a vector space.

Cayley (1850) introduced matrices, which also form a noncommutative associative algebra.

Dedekind (1871) formally introduced the concept of a ring, but called it an 'order'. He also studied ideals and introduced the term 'field' (Körper).

Hilbert used the term 'Zahlring' (number ring), or simply 'ring'.

Wedderburn (1905, 1908) proved theorems on the structure of associative algebras. This work was generalised to rings by Artin (1927) and extended by Jacobson (1945).

1.1 Groups

Recall that a **group** $(G, *)$ is a set G closed under an associative **binary operation** $* : G \times G \to G$, such that there is an identity element $e \in G$ and every element $a \in G$ has an inverse $a^{-1} \in G$. Thus for all $a, b, c \in G$ we have $a * b \in G$, $a * (b * c) = (a * b) * c$, $e * a = a * e = a$ and $a^{-1} * a = a * a^{-1} = e$.

A group is **abelian** if the operation is commutative, i.e. $a * b = b * a$ for all $a, b \in G$.

By default we shall use multiplicative notation, writing $a * b$ as ab, and refer to the group $(G, *)$ simply as G. For any $n \in \mathbb{N}$, a^n means $a * a * \cdots * a$ (n times) and a^{-n} means $(a^n)^{-1}$, or equivalently $(a^{-1})^n$. The usual laws of indices hold, and $a^0 = e$.

Groups can be infinite or finite. In this module we shall mainly consider finite groups. The **order** $|G|$ of a finite group $(G, *)$ is the number of elements in the set G.

A non-empty subset H of G is a **subgroup** of G if it is itself a group under the same operation. This holds if and only if $ab^{-1} \in H$ for all $a, b \in H$ (the subgroup test). If H is finite, it is enough to show that $ab \in H$ for all $a, b \in H$. **Lagrange's theorem** states that when G is finite and H

is a subgroup of G then $|H|$ divides $|G|$. The **trivial subgroup** of a group G is $\{e\}$ and a **proper subgroup** of G is any subgroup which is not the whole of G.

Every element a in a group G **generates** a subgroup $\langle a\rangle = \{a^k \ : \ k \in \mathbb{Z}\}$. The **order** of a, denoted by $o(a)$, is the number of elements in this group, i.e. the smallest $k \in \mathbb{N}$ (if any) such that $a^k = e$. If G is finite then so is $o(a)$, and $a^m = e$ if and only if $o(a) \mid m$. By Lagrange's theorem $o(a)$ divides $|G|$, so $a^{|G|} = e$ for all $a \in G$.

Let G_1, G_2 be groups with identity elements e_1, e_2 respectively. A **group homomorphism** from G_1 to G_2 is a map $\phi : G_1 \to G_2$ such that $\phi(ab) = \phi(a)\phi(b)$ for all $a, b \in G_1$ (note the multiplication on the left side of the equality is that of G_1 and on the right is that of G_2). Its **kernel** $\mathrm{Ker}(\phi)$ is $\{a \in G_1 : \phi(a) = e_2\}$ and its **image** $\phi(G_1)$ is $\{\phi(a) \ : \ a \in G_1\}$. ϕ is **injective** if $\phi(a) = \phi(b) \Rightarrow a = b$, or equivalently $\mathrm{Ker}\,\phi = \{e_1\}$. ϕ is **surjective** if $\phi(G_1) = G_2$.

A **group isomorphism** is a group homomorphism which is **bijective** (both injective and surjective). G_1 and G_2 are then **isomorphic** groups, written as $G_1 \cong G_2$. A **group automorphism** is an isomorphism from a group to itself.

If groups $G_1, \dots, G_n$ are all isomorphic to one another then we say that they are the same group *up to isomorphism*. This means that they are identical as far as their group structure is concerned. Finite groups are isomorphic if and only if their Cayley tables can be put into 1-1 correspondence.

You should be familiar with the following (see Algebra notes (year 1) and Groups and Rings notes (year 2)):

- **Cyclic groups**. A group G is a **cyclic group** with **generator** g if there exists $g \in G$ such that $G = \{g^r : r \in \mathbb{Z}\}$ (or $G = \{r.g : r \in \mathbb{Z}\}$ if is G is additive). In this case we write $G = \langle g\rangle$. Every cyclic group $\langle g\rangle$ is abelian. Any infinite cyclic group $\langle g\rangle$, with identity element e, is isomorphic to $(\mathbb{Z}, +)$ via the map $\psi : g^r \mapsto r$:

$$\psi\downarrow \begin{array}{ccccccccccc} \cdots & g^{-3} & g^{-2} & g^{-1} & e & g & g^2 & g^3 & g^4 & \cdots \\ \cdots & -3 & -2 & -1 & 0 & 1 & 2 & 3 & 4 & \cdots \end{array}$$

  We observe for example that, $\psi(g^2g^{-3}) = \psi(g^{-1}) = -1 = 2 - 3 = \psi(g^2) + \psi(g^{-3})$.

  For each $n \in \mathbb{N}$ there is a cyclic group $C_n = \{g^k \ : \ k \in \mathbb{Z}\}$ of order n, where the *generator* g has order n. All cyclic groups of the same finite order n are isomorphic to one another, and in particular to $(\mathbb{Z}_n, +_n)$ where $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ is the set of integers modulo n and $+_n$ is addition modulo n; if $G = \langle g\rangle$ the isomorphism in this case is the map $\psi : g^r \mapsto r \mod n$. Note that $\mathbb{Z}_n$ is not a subgroup of $\mathbb{Z}$, as the operation is different ($+_n$ rather than $+$).

  If g generates C_n then so does g^m for each integer m that is coprime to n (i.e so that $\gcd(m, n) = 1$)). For example, $(\mathbb{Z}_{10}, +_{10})$ is generated by $1, 3, 7, 9$ and correspondingly if $C_{10} = \langle g\rangle$ then g, g^3, g^7, g^9 are generators of C_{12}. In general the number of generators of C_n is given by the Euler totient function $\varphi(n)$. If $p_1, \dots, p_r$ are the distinct primes that divide n then $\varphi(n) = n\displaystyle\prod_{i=1}^{r}\left(1 - \frac{1}{p_i}\right)$. Recall $\phi(n)$ is the number of positive integers m less than or equal to n with $\gcd(m, n) = 1$. For example, $360 = 2^3 \times 3^2 \times 5$ so

$\varphi(360) = 360\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right) = 96$. Thus C_{360} has 96 distinct generators $g, g^7, g^{11}, \ldots, g^{359}$.

A cyclic group G of order n with generator g has one subgroup of order d, which is itself cyclic, for each divisor d of n and no other subgroups; a generator of the subgroup of order d is $g^{\frac{n}{d}}$. For example, if $C_{30} = \langle g \rangle$ then C_{30} has non-trivial, proper subgroups of orders $2, 3, 5, 6, 10, 15$ with generators $g^{15}, g^{10}, g^6, g^5, g^3, g^2$ respectively.

Every group G of prime order p is cyclic since every non-identity element generates G. Hence, up to isomorphism, C_p is the only group of prime order p.

- **Groups of units**. Suppose $(M, \times)$ is a **monoid**; this means M is a set, with an associative binary operation $\times$ and an identity element $e \in M$; so it has all the properties of a group except that *some elements may not have inverses*. Those elements of M which have inverses are called *units*. They form the group of units $(U(M), \times)$. In particular, the units of $(\mathbb{Z}_n, \times_n)$ form a group U_n of order $\varphi(n)$, e.g. $U_8 = (\{1, 3, 5, 7\}, \times_8)$. The units of any *field* are all the non-zero elements; these always form a cyclic group.

- **Groups of matrices**. The $n \times n$ matrices over any commutative ring R with unity form a multiplicative monoid $M_n(R)$. The units are the invertible matrices and these form the *general linear group* $GL(n, R)$, which is non-abelian in general. Its subgroups include the special linear group $SL(n, R)$ (matrices with determinant 1) and the groups of invertible upper triangular, lower triangular and diagonal matrices.

  When R is finite, e.g. $\mathbb{Z}_n$, the group $GL(n, \mathbb{R})$ has finite order less than $|R|^n$.

  For example, the 2×2 matrices over $\mathbb{Z}_n$ form a multiplicative monoid $M_2(\mathbb{Z}_n)$ of order n^4.

  Let $R_n = \left\{ \begin{pmatrix} u & a \\ 0 & v \end{pmatrix} \ : \ a \in \mathbb{Z}_n,\ u, v \in U_n \right\}$. Any matrix $\begin{pmatrix} u & a \\ 0 & v \end{pmatrix} \in R_n$ has inverse $\begin{pmatrix} u^{-1} & -u^{-1}av^{-1} \\ 0 & v^{-1} \end{pmatrix} \in R_n$. Thus $R_n \subset GL(2, \mathbb{Z}_n)$.

  Also R_n is closed under multiplication, so it is a subgroup of the finite group $GL(2, \mathbb{Z}_n)$. There are n possible values for a and $\varphi(n)$ possible values for each of u and v, so R_n has order $n\,\varphi(n)^2$.

- **Permutation groups**. A **permutation** of a finite set X is a bijective function from X to X. The permutations of a finite set X, under composition, form the **symmetric group** on X, denoted by $\mathrm{Sym}(X)$. For each $n \in \mathbb{N}$, the symmetric group on $X = \{1, \ldots, n\}$ is denoted S_n and consists of all permutations of $\{1, \ldots, n\}$ under composition. The symmetric group S_n has order $n!$ and is non-abelian for all $n \geq 3$. Indeed S_3, of order 6, is the smallest non-abelian group. A permutation σ of $\{1, \ldots, n\}$ can be written as

$$\begin{pmatrix} 1 & 2 & \ldots & n \\ \sigma(1) & \sigma(2) & \ldots & \sigma(n) \end{pmatrix}$$

  The 'product' or multiplication $\sigma\tau$ of two permutations σ and τ is the composite function $\sigma \circ \tau$. In S_n, a **cycle** of length k, or **k-cycle**, is a permutation of the form $i_1 \mapsto i_2 \mapsto \ldots \mapsto i_k \mapsto i_1$, where $i_1, \ldots, i_k$ are distinct elements of $\{1, \ldots, n\}$ and all other elements are fixed. Such a cycle is written as $(i_1\ i_2\ \ldots\ i_k)$. Every permutation in S_n may be expressed as a product of *disjoint* cycles. A **transposition** is a 2-cycle $(i\ j)$, i.e $i \mapsto j \mapsto i$ and all other elements are fixed. Every cycle is a product of transpositions: $(i_1\ i_2\ \ldots\ i_k) = (i_1\ i_k)(i_1\ i_{k-1})\ldots(i_1\ i_2)$. A permutation is **even** or **odd** if it is the product of an even or odd number of transpositions respectively and the **sign** of a permutation σ, denoted $\text{sgn}(\sigma)$ is 1 if σ is even and -1 if it is odd. The *even* permutations, i.e. those which can be written as a product of an even number of transpositions, form a subgroup of S_n called the *alternating group* A_n, which has order $\frac{n!}{2}$. A_n is non-abelian for $n \geq 4$.

- **Dihedral groups**. For each $n \geq 3$, the group D_n consists of all symmetries (rotations and reflections) of a regular n-sided polygon. This is non-abelian. It has order $2n$ (so some books call it D_{2n}) and is isomorphic to S_n if and only if $n = 3$. D_2 is the symmetry group of a rectangle that is not a square; it is isomorphic to the *Klein 4-group* V (or V_4), a non-cyclic group of order 4, in which all elements are self-inverse.

- **Direct products**. Given groups $G_1, \ldots, G_n$, the group $G_1 \times \cdots \times G_n$ consists of all ordered n-tuples $(a_1, \ldots, a_n)$ where $a_i \in G_i$, with operation defined by $(a_1, \ldots, a_n)(b_1, \ldots, b_n) = (a_1b_1, \ldots, a_nb_n)$. We call $G_1 \times \ldots \times G_n$ the **direct product** of $G_1, \ldots, G_n$. In particular, $G_1 \times G_2 = \{(a_1, a_2) \ : \ a_1 \in G_1, a_2 \in G_2\}$. We have $|G_1 \times \cdots \times G_n| = |G_1| \times \cdots \times |G_n|$, provided the order of the direct product is finite.

  Recall that $C_m \times C_n$ is cyclic (isomorphic to C_{mn}) if and only if m and n are coprime. More generally, recall that $C_{n_1} \times \ldots \times C_{n_k}$ is cyclic (isomorphic to $C_{n_1 \ldots n_k}$) if and only if $n_1, \ldots, n_k$ are pairwise coprime. For example, $C_3 \times C_4 \cong C_{12}$. $C_2 \times C_2$ is not cyclic; it is isomorphic to V and D_2.

Exercises 1.1

Multiplicative notation is used in the questions and should be used in your answers.

1. (a) Let H and J be subgroups of a group G. Show that $H \cap J$ is a subgroup of G. Deduce that $H \cap J$ is a subgroup of both H and J.

   (b) Find the possible orders of $H \cap J$ if
   (i) $|H| = 4, |J| = 6$, (ii) $|H| = |J| = 7$, (iii) $|H|$ and $|J|$ are coprime.

2. Let G be a group with identity element e.

   (a) Show that $(ab)^{-1} = b^{-1}a^{-1}$ for all $a, b \in G$.

   (b) Suppose $a^2 = e$ for all $a \in G$. Show that G is abelian.

3. Let G be a multiplicative group. The map $\phi : G \to G$ is given by $\phi(a) = a^2$ for all $a \in G$. Show that ϕ is a group homomorphism if and only if G is abelian.

4. Let a and b be elements of a group G.

   (a) Prove by induction that $(bab^{-1})^n = ba^nb^{-1}$ for all positive integers n.

   (b) If $bab^{-1} = a^r$, prove by induction that $b^nab^{-n} = a^{(r^n)}$ for all $n \in \mathbb{N}$.

5. Let a be a fixed element of a multiplicative group G. The map $\gamma_a : G \to G$ is given by $\gamma_a(x) = axa^{-1}$ for all $x \in G$. Show that γ_a is an automorphism of G.

6. Let $\phi : G \to G$ be a group homomorphism. Show that the set of all elements of G that are fixed by ϕ (i.e. elements a such that $\phi(a) = a$) is a subgroup of G.

7. Make Cayley tables for the cyclic groups C_5 and C_6, letting g be a generator and e the identity.

   For each of these groups, identify all the subgroups and give a generator of each one. Do the same for C_7, C_8 and C_9 without making tables.

8. Prove that a group of even order has an odd number of elements of order 2. Deduce that every 4-element group is isomorphic to C_4 or the Klein 4-group.

9. Find the number of generators of C_{84} and the order of the group of units U_{84}.

10. Express the permutation $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 8 & 6 & 2 & 1 & 7 \end{pmatrix} \in S_8$ as a product of disjoint cycles and as a product of transpositions. Is σ in A_8?

11. The **sign** of a permutation $\sigma \in S_n$, denoted by $\mathrm{sgn}(\sigma)$, is defined to be 1 if σ is even and -1 if σ is odd. Show that the map $\mathrm{sgn} : S_n \to (\mathbb{R}^*, \times)$ is a group homomorphism. Identify its kernel and image.

12. Let R_n be the group of matrices in the example on Page 2. By considering fixed values of u, v and a, find subgroups of R_n of order $n\,\varphi(n)$, $\varphi(n)^2$ and $\varphi(n)$. Which ones are abelian? Find the orders of R_n and each of these subgroups when (a) n is prime, (b) $n = 4$.

13. Let G and H be groups with identity elements e_1 and e_2 respectively.

    (a) Show that if G and H are both abelian then $G \times H$ is abelian.

    (b) Show that the map from $G \times H$ to $H \times G$ given by $(g, h) \mapsto (h, g)$ is a group isomorphism, and so $G \times H \cong H \times G$.

    (c) Show that $G \times \{e_2\}$ is a subgroup of $G \times H$ and is isomorphic to G.

    (d) Identify subgroups of $C_2 \times C_5$ that are isomorphic to C_2 and to C_5.

14. For any non-trivial group G, the **diagonal subgroup** Δ of $G \times G$ is defined as $\{(a, a) : a \in G\}$. Show that Δ is a subgroup of $G \times G$ but is *not* a direct product of subgroups of G.

1.2 Generators and Presentations

Finite groups can be represented by Cayley tables. However, this is inefficient unless the group is very small. It is useful to have a way of specifying all the information that is needed to determine the stucture of a group, without listing all the elements.

Definition 1.1 *Let G be a group and let $S = \{g_1, \ldots, g_k\}$ be a subset of G. The subgroup of G* **generated by** *S, denoted by $\langle S \rangle$ or $\langle g_1, \ldots, g_k \rangle$, consists of all elements of G that can be formed from elements of S, and their inverses, combined by the group operation. S is called a* **generating set** *or a* **set of generators** *for the group $\langle S \rangle$.*

For example, using multiplicative notation, ${g_1}^3 g_2^{-1} {g_1}^4 g_2 g_1^{-2}$ is in the group $\langle g_1, g_2 \rangle$.

$\langle S \rangle$ is the smallest subgroup of G that contains the set S. Equivalently, it is the intersection of all such subgroups. In particular, it may equal the whole of G.

Definition 1.2 *A* **presentation** $\langle S : R \rangle$ *of a group G consists of a set S such that $\langle S \rangle = G$ and a list R of* **relations** *which determine the effect of the group operation on all pairs of elements of G. G is* **free** *on S (or the* **free group** *on S) if the set of relations is empty.*

The relations usually consist of the orders of the generators and as few other equations as are needed to determine how all the elements combine.

- The relation $a^n = e$ means that n is the smallest such positive integer, i.e. a has order n.
- No generator in S is a power of another one. (But some power of one generator may equal some power of another.)
- The order of the group may be given separately if it is not obvious from the presentation.
- The same group may have different presentations.
- A group is abelian if and only if its generators commute with one another.

1.2.1 Examples

(i) A cyclic group of order n with generator g has the presentation $\langle g \ : \ g^n = e \rangle$. An infinite cyclic group with generator g is the free group on $\{g\}$.

(ii) The presentation $\langle a, b \ : \ a^2 = b^2 = e,\ aba = b \rangle$ gives $ba = ab$ and $(ab)^2 = e$, which determines the structure as that of the Klein 4-group V.

(iii) Label the vertices of a regular n-sided polygon from 1 to n, where $n \geq 3$. The dihedral group D_n, of order $2n$, is generated by a rotation through $\dfrac{2\pi}{n}$ represented by the n-cycle $\sigma = (1 \ 2 \ \cdots \ n)$, which has order n, together with a reflection τ in a line of symmetry, which is represented by a product of disjoint transpositions and has order 2. We find that $\tau\sigma = \sigma^{-1}\tau$, or equivalently $\sigma\tau\sigma = \tau$, so D_n has the presentation $\langle \sigma, \tau \ : \ \sigma^n = \tau^2 = \iota,\ \sigma\tau\sigma = \tau \rangle$.

Clearly the names of the elements make no difference to the group structure. The presentation $\langle a, b \ : \ a^n = b^2 = e,\ aba = b \rangle$ also defines D_n.

Here are the Cayley tables for D_3 (isomorphic to S_3) and D_4 (not isomorphic to S_4):

	ι	σ	σ^2	τ	$\sigma\tau$	$\sigma^2\tau$
ι	ι	σ	σ^2	τ	$\sigma\tau$	$\sigma^2\tau$
σ	σ	σ^2	ι	$\sigma\tau$	$\sigma^2\tau$	τ
σ^2	σ^2	ι	σ	$\sigma^2\tau$	τ	$\sigma\tau$
τ	τ	$\sigma^2\tau$	$\sigma\tau$	ι	σ^2	σ
$\sigma\tau$	$\sigma\tau$	τ	$\sigma^2\tau$	σ	ι	σ^2
$\sigma^2\tau$	$\sigma^2\tau$	$\sigma\tau$	τ	σ^2	σ	ι

	ι	σ	σ^2	σ^3	τ	$\sigma\tau$	$\sigma^2\tau$	$\sigma^3\tau$
ι	ι	σ	σ^2	σ^3	τ	$\sigma\tau$	$\sigma^2\tau$	$\sigma^3\tau$
σ	σ	σ^2	σ^3	ι	$\sigma\tau$	$\sigma^2\tau$	$\sigma^3\tau$	τ
σ^2	σ^2	σ^3	ι	σ	$\sigma^2\tau$	$\sigma^3\tau$	τ	$\sigma\tau$
σ^3	σ^3	ι	σ	σ^2	$\sigma^3\tau$	τ	$\sigma\tau$	$\sigma^2\tau$
τ	τ	$\sigma^3\tau$	$\sigma^2\tau$	$\sigma\tau$	ι	σ^3	σ^2	σ
$\sigma\tau$	$\sigma\tau$	τ	$\sigma^3\tau$	$\sigma^2\tau$	σ	ι	σ^3	σ^2
$\sigma^2\tau$	$\sigma^2\tau$	$\sigma\tau$	τ	$\sigma^3\tau$	σ^2	σ	ι	σ^3
$\sigma^3\tau$	$\sigma^3\tau$	$\sigma^2\tau$	$\sigma\tau$	τ	σ^3	σ^2	σ	ι

(iv) Every permutation in S_n can be written as a product of transpositions, so S_n is generated by the set of all transpositions of pairs of elements in $\{1, \ldots, n\}$. A smaller generating set will be found in Exercises 2.3 Question 12.

(v) Let G be a group of order 8. By Lagrange's theorem, every element has order 1, 2, 4 or 8. Only the identity can have order 1. If *all* other elements have order 2 then G is abelian by Exercises 1.1 Question 2 (b).

If any element has order 8, this generates G which is then cyclic, hence abelian.

It follows that if G is non-abelian, all non-identity elements have order 2 or 4 and at least one has order 4. Pick an element a of order 4, so $\langle a \rangle = \{e, a, a^2, a^3\}$.

Let $b \in G, b \notin \langle a \rangle$. Then $a^r b \notin \langle a \rangle$ for any $r \in \mathbb{Z}$, so $G = \{e, a, a^2, a^3, b, ab, a^2b, a^3b\}$. The set $\{a, b\}$ generates G, so if G is non-abelian then $ab \neq ba$.

$b \notin \langle a \rangle$, so $b^2 \notin \{b, ab, a^2b, a^3b\}$. Thus b^2 must be in $\langle a \rangle$.

$b^4 = e$ and $a^2 \neq e$, so $b^2 \notin \{a, a^3\}$. Thus b^2 equals either e or a^2.

$ba \notin \langle a \rangle$ and $ba \notin \{b, ab\}$, so $ba = a^r b$ for some $r \in \{2, 3\}$.

Suppose $b^2 = e$, so $b^{-1} = b$. If $ba = a^2 b$ then $bab = a^2$, so $ab = ba^2$. Thus $a^2 = abba = ba^4b = b^2 = e$. But $a^2 \neq e$. Hence $ba = a^3b$, i.e. $aba = b$.

Then $G = \langle a, b \;:\; a^4 = b^2 = e, aba = b \rangle$, so $G \cong D_4$.

Suppose instead that $b^2 = a^2$. If $ba = a^2b$ then $ba = b^3$ so $a = b^2$, but $a \neq a^2$. Thus again $ba = a^3b$, i.e. $aba = b$. Then $G = \langle a, b \;:\; a^4 = e, b^2 = a^2,\; aba = b \rangle$. In Exercises 1.2 Question 1 you will see a group of order 8 with this presentation. It belongs to the following infinite family of finite groups:

Definition 1.3 *For each* even *integer n, the* **dicyclic group** *Q_n is a group of order $2n$ with presentation $\langle a, b \;:\; a^n = e,\; b^2 = a^{n/2},\; aba = b \rangle$. In particular, Q_4 is the* **quaternion group***, of order 8.*

Here is the Cayley table for Q_4:

	e	a	a^2	a^3	b	ab	a^2b	a^3b
e	e	a	a^2	a^3	b	ab	a^2b	a^3b
a	a	a^2	a^3	e	ab	a^2b	a^3b	b
a^2	a^2	a^3	e	a	a^2b	a^3b	b	ab
a^3	a^3	e	a	a^2	a^3b	b	ab	a^2b
b	b	a^3b	a^2b	ab	a^2	a	e	a^3
ab	ab	b	a^3b	a^2b	a^3	a^2	a	e
a^2b	a^2b	ab	b	a^3b	e	a^3	a^2	a
a^3b	a^3b	a^2b	ab	b	a	e	a^3	a^2

Examples 1.2.1 (v) shows that, up to isomorphism, D_4 and Q_4 are the only non-abelian groups of order 8. We shall show that there are exactly three abelian ones, so altogether there are five distinct (non-isomorphic) groups of order 8.

Exercises 1.2

1. Let $\mathrm{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\mathrm{A} = \begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix}$, $\mathrm{B} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ (where $\mathrm{i}^2 = -1$).

   Show that I, A, A^2, A^3, B, AB, $\mathrm{A}^2\mathrm{B}$, $\mathrm{A}^3\mathrm{B}$ are distinct matrices. Show that this set of eight matrices, under matrix multiplication, satisfies the presentation of Q_4 in Definition 1.3.

2. Identify the group with presentation $\langle a, b \;:\; a^3 = b^2 = e, ab = ba\rangle$ and state its order. Give a presentation in terms of a single generator and express this generator in terms of a and b, showing that it does generate the group.

3. Let $\sigma = (1\ \ 2\ \ 3\ \ 4\ \ 5), \tau = (2\ \ 5)(3\ \ 4) \in S_5$. Describe symmetries of a regular pentagon that can be represented by σ and τ and give a presentation of the group D_5 in terms of σ and τ. List the elements of D_5 in terms of σ and τ. State the order of each element. Identify all the subgroups of D_5.

4. Let G be a non-abelian group of order 6, with identity element e.

   Show that G must have an element a of order 3, and that if $b \notin \langle a\rangle$ then $G = \{e, a, a^2, b, ab, a^2b\}$. Deduce that $aba = b$ and conclude that $G \cong D_3$.

5. By inspection of the Cayley table on the previous page, find the order of each element of Q_4. List all the subgroups of Q_4, stating the order of each one. Are all the subgroups cyclic?

1.3 Cosets, products and quotient groups

Let A, B and C be subsets of a group G. We define AB to mean $\{ab \;:\; a \in A, b \in B\}$. This should not be confused with $A \times B$, which means $\{(a, b) \;:\; a \in A, b \in B\}$.

$(AB)C = \{(ab)c \;:\; a \in A, b \in B, c \in C\} = \{a(bc) \;:\; a \in A, b \in B, c \in C\} = A(BC)$, so this multiplication of subsets is associative.

When one or more subset (but not all of them) has only one element, the set brackets can be omitted: $aB = \{a\}B = \{ab \;:\; b \in B\}$. Similarly, if a and c are fixed, aBc means $\{abc \;:\; b \in B\}$.

If H is a *subgroup* of G then $HH \subset H$ and $H = \{eh \;:\; h \in H\} \subset HH$, so $HH = H$.

Recall that a **left coset** of a subgroup H in G is a set of the form $aH := \{ah \;:\; h \in H\}$, for any $a \in G$. A **right coset** of H in G is a set of the form $Ha := \{ha \;:\; h \in H\}$, for any $a \in G$. Clearly $aH = Ha$ if G is abelian, but left and right cosets may be different in general.

In an *additive* group, the cosets have the form $a + H$ (or equivalently $H + a$), meaning $\{a + h \;:\; h \in H\}$.

The subgroup H itself is a coset of H in G; it is the coset hH, and also Hh, for any $h \in H$. In particular, $H = eH$. The cosets other than H are *not* subgroups of G.

The left cosets of H partition G into disjoint subsets, i.e. every element of G lies in exactly one left coset of H. The same is true for the right cosets of H.

G is the union of all the left (or right) cosets of H, i.e. $G = \bigcup_{a \in G} aH = \bigcup_{a \in G} Ha$.

For $a, b \in G$, the following are equivalent:

a and b are in the same left coset of H; $aH = bH$; $a \in bH$; $b \in aH$; $a^{-1}b \in H$.

For each $a \in G$, the map from H to aH given by $h \mapsto ah$ is easily shown to be a bijection. Thus if H is finite, every left coset of H in G has the same number of elements as H. The same holds for the right cosets.

This shows that when G is finite, the order of H divides the order of G, which is Lagrange's theorem.

The **index** of H in G, denoted by $[G : H]$, is defined to be the number of left (or right) cosets of H in G. In the finite case, it equals $\frac{|G|}{|H|}$.

1.3.1 Example

In D_4, let J be the subgroup $\{\iota, \tau\}$. $[D_4 : J] = 8/2 = 4$. The four left cosets of J in D_4 are J (which equals ιJ and τJ), $\sigma J = \{\sigma, \sigma\tau\}$, $\sigma^2 J = \{\sigma^2, \sigma^2\tau\}$, $\sigma^3 J = \{\sigma^3, \sigma^3\tau\}$.

The right cosets of J in D_4 are J, $J\sigma = \{\sigma, \sigma^3\tau\}$, $J\sigma^2 = \{\sigma^2, \sigma^2\tau\}$, $J\sigma^3 = \{\sigma^3, \sigma\tau\}$.

Both the left and right cosets give a partition of D_4 into disjoint 2-element subsets, but the two partitions are not identical.

Recall that a subgroup H of G is called a **normal subgroup** of G, written $H \lhd G$ ('H is normal in G'), if $aH = Ha$ for all $a \in G$.

It is clear that every subgroup of an *abelian* group is a normal subgroup. However, an abelian subgroup of a non-abelian group is not necessarily normal.

A subgroup of index 2 in a group is always normal, because there are only two left cosets; one is the subgroup itself, and the other consists of all the other group elements. The same is true of the right cosets, so each left coset equals the corresponding right coset. Thus, for instance, $A_n \lhd S_n$ for each $n \in \mathbb{N}$.

If $\phi : G_1 \to G_2$ is a group homomorphism then $\operatorname{Ker}\phi$ is a normal subgroup of G_1.

The condition $aH = Ha$ can be written as $aHa^{-1} = H$, or as $a^{-1}Ha = H$. Hence $H \lhd G$ if and only if H is a subgroup of G such that $aha^{-1} \in H$ for all $a \in G$ and $h \in H$, or equivalently $a^{-1}ha \in H$ for all $a \in G$ and $h \in H$.

Recall that in forming the expression aha^{-1}, we say h is *conjugated* by a. Thus a subgroup H of G is normal in G if and only if it is closed under conjugation by all elements of G. The following simpler condition can be proved:

Proposition 1.1 *Let H be a subgroup of a finite group G. Then $H \lhd G$ if and only if $ghg^{-1} \in H$ for all g in a generating set for G and all h in a generating set for H.*

1.3.2 Example

In D_4, which has generating set $\{\sigma, \tau\}$ let N be the subgroup $\{\iota, \sigma^2\}$ generated by σ^2.

Then $\sigma\sigma^2\sigma^{-1} = \sigma^2 \in N$ and $\tau\sigma^2\tau^{-1} = \sigma^2 \in N$, so $N \lhd D_4$.

This could also be shown by finding the left and right cosets of N and seeing that they are the same.

Let $J = \{\iota, \tau\} \subset D_4$. $\sigma\tau\sigma^{-1} = \sigma\tau\sigma^3 = \sigma^2\tau \notin J$, so J is not normal in D_4, as also shown by Example 1.3.1.

In Exercises 1.1 Question 4 you proved the following equalities, which are often useful when simplifying expressions involving conjugation:

Proposition 1.2 *Let a and b be elements of a group. Then for all* $n \in \mathbb{N}$*,*

(i) $(bab^{-1})^n = ba^nb^{-1}$*,*

(ii) if $bab^{-1} = a^r$ *then* $b^nab^{-n} = a^{(r^n)}$*.*

Next, we consider conditions for the product of two subgroups to be a subgroup.

Proposition 1.3 *Suppose H and J are subgroups of a group G. Then*

(i) HJ *is a subgroup of* G *if and only if* $JH = HJ$*,*

(ii) if either H *or* J *is normal in* G *then* HJ *is a subgroup of* G*,*

(iii) if both H *and* J *are normal in* G *then* HJ *is a normal subgroup of* G*,*

Proof

(i) For all elements $h \in H, j \in J$ we have $h^{-1} \in H, j^{-1} \in J$, so $h^{-1}j^{-1} \in HJ$. If HJ is a subgroup of G then $(h^{-1}j^{-1})^{-1} \in HJ$, so $jh \in HJ$. Thus $JH \subset HJ$. Similarly $HJ \subset JH$, so $JH = HJ$.

Conversely, suppose $JH = HJ$. $e \in HJ$, so HJ is a non-empty subset of G.

Let $a = h_1j_1, b = h_2j_2 \in HJ$. Then $ab^{-1} = h_1j_1{j_2}^{-1}{h_2}^{-1} \in HJH$. Now $HJH = HHJ = HJ$, so $ab^{-1} \in HJ$. Thus by the subgroup test, HJ is a subgroup of G.

(ii) If $H \lhd G$ then $aH = Ha$ for all $a \in G$, so in particular $jH = Hj$ for all $j \in J$. Thus $JH = \bigcup_{j\in J} jH = \bigcup_{j\in J} Hj = HJ$. Hence HJ is a subgroup of G by (i). The same reasoning applies if $J \lhd G$.

(iii) If $H \lhd G$ and $J \lhd G$ then HJ is a subgroup of G by (ii). For all $hj \in HJ$ and $a \in G$ we have $a(hj)a^{-1} = (aha^{-1})(aja^{-1}) \in HJ$, so $HJ \lhd G$. $\square$

Note that $HJ = JH$ holds if, but not only if, $hj = jh$ for all $h \in H$, $j \in J$. In an *abelian* group, we must have $HJ = JH$ so a product of subgroups is always a subgroup. The 'subgroup' HJ may of course be the whole group G.

1.3.3 Example

In D_4 let $J = \{\iota, \tau\}, K = \{\iota, \sigma^3\tau\}$. Then $JK = \{\iota, \sigma, \tau, \sigma^3\tau\}$ and $KJ = \{\iota, \tau, \sigma^3, \sigma^3\tau\}$. These sets are not equal, and neither of them is a group.

If $N = \{\iota, \sigma^2\}$, which is normal in D_4, then $JN = NJ = \{\iota, \sigma^2, \tau, \sigma^2\tau\}$, which *is* a subgroup of D_4.

Proposition 1.4 *If H and J are finite subgroups of a group G,* $|HJ| = \dfrac{|H|.|J|}{|H \cap J|}$.

The proof is deferred until Chapter 2. As a particular case, if $H \cap J = \{e\}$, so $|H \cap J| = 1$, then $|HJ| = |H|.|J|$.

1.3.4 Examples

(i) In C_{24}, with generator g, let $H = \langle g^6 \rangle = \{e, g^6, g^{12}, g^{18}\}$ and $J = \langle g^4 \rangle = \{e, g^4, g^8, g^{12}, g^{16}, g^{20}\}$. Then $H \cap J = \{e, g^{12}\}$ so $|HJ| = \dfrac{4 \times 6}{2} = 12$.

As C_{24} is abelian, $JH = HJ$ so HJ is a subgroup of C_{24}. (In fact $HJ = \langle g^2 \rangle$.)

(ii) Let G be a group of order 21. Each element of G has order dividing 21, so the possible orders of elements are 1, 3, 7, 21. Only the identity e has order 1.

If there is an element of order 21 then this generates G, in which case G is cyclic and has one subgroup of order 3 and one of order 7.

Now suppose $|G| = 21$ and G is not cyclic, so every non-identity element has order 3 or 7 and generates a subgroup of that order. The union of these subgroups is G and, as 3 and 7 are coprime, any two have intersection $\{e\}$.

Each n-element subgroup has $n - 1$ non-identity elements. If there are k subgroups of order 3 and ℓ of order 7 then $2k + 6\ell + 1 = 21$ so $k + 3\ell = 10$. This has integer solutions $(k, \ell) = (1, 3), (4, 2), (7, 1), (10, 0)$.

If $\ell > 0$ then G has at least one subgroup H of order 3 and at least one subgroup J of order 7. As 3 and 7 are coprime, for any such subgroups H and J we have $|H \cap J| = 1$, so in this case $|HJ| = |H|.|J| = 21$ and thus $HJ = G$.

Proposition 1.5

Let H and J be subgroups of a group G with identity element e, such that $G = HJ$ and $H \cap J = \{e\}$. Then

(i) each element of G has a unique *expression as hj where $h \in H, j \in J$,*

(ii) H and J are both normal in G if and only if $hj = jh$ for all $h \in H$ and $j \in J$. When this is the case then $G \cong H \times J$.

Proof

(i) As $G = HJ$, every element of G can be expressed as hj where $h \in H, j \in J$. Suppose an element has two such expressions: $hj = h'j'$. Then $h'^{-1}h = j'j^{-1}$ so this element is in H and in J. Thus $h'^{-1}h = e$, $j'j^{-1} = e$ so $h' = h, j' = j$, i.e. the expression is unique.

(ii) Suppose $H \lhd G$, $J \lhd G$. Let $h \in H, j \in J$ and $x = hjh^{-1}j^{-1}$. As $hjh^{-1} \in J$, we have $x \in J$. Also $jh^{-1}j^{-1} \in H$, so $x \in H$. Hence $x = e$, so $hj = jh$.

Conversely, suppose $hj = jh$ for all $h \in H, j \in J$. Then $G = HJ = JH$. Each $a \in G$ has the form hj and we have $Ha = Hhj = Hj = jH = jhH = hjH = aH$. Thus $H \lhd G$, and similarly $J \lhd G$.

When the condition holds, define a map $\phi : H \times J \to HJ$ by $\phi : (h, j) \mapsto hj$.

Then for all $h_1, h_2 \in H$ and $j_1, j_2 \in J$, $\phi((h_1, j_1)(h_2, j_2)) = \phi(h_1h_2, j_1j_2) = h_1h_2j_1j_2 = h_1j_1h_2j_2 = \phi(h_1, j_1)\phi(h_2, j_2)$, so ϕ is a group homomorphism. By the uniqueness proved in (i), ϕ is bijective so $G \cong H \times J$. □

Definition 1.4 *Suppose H and J are normal subgroups of a group G, such that $G = HJ$ and $H \cap J$ is trivial. Then G is the* **internal direct product** *of H and J.*

Thus when G is the internal direct product of H and J, we have $G = HJ \cong H \times J$ and $G = JH \cong J \times H$.

By induction, Proposition 1.5 can be extended to any number of subgroups. When the following result applies, G is the **internal direct product** of $H_1, H_2, \dots, H_n$:

Proposition 1.6

Let $H_1, H_2, \dots, H_n$ be normal subgroups of a group G with identity element e, such that $G = H_1H_2 \cdots H_n$ and $(H_1 \cdots H_{i-1}) \cap H_i = \{e\}$ for $i = 2, \dots, n$. Then each element of G has a unique *expression as $h_1h_2 \cdots h_n$, where $h_i \in H_i$ for each i, and $G \cong H_1 \times H_2 \times \cdots \times H_n$.*

1.3.5 Examples

(i) In Example 1.3.4 (ii) we saw that any group G of order 21 has subgroups H and J of order 3 and 7, such that $G = HJ$. If H and J are both normal in G then by Proposition 1.5 (ii), G is the internal direct product of H and J, i.e. $G \cong H \times J$.

(ii) In D_4, let $H = \{\iota, \sigma, \sigma^2, \sigma^3\}$ and $J = \{\iota, \tau\}$. Then $D_4 = HJ$ but this is not an internal direct product, since $H \times J \cong C_4 \times C_2$ which is abelian, but D_4 is non-abelian and so $D_4 \not\cong H \times J$. Note that $H \lhd D_4$ but J is not normal in D_4. (We shall see in a later section that D_4 is a **semi-direct product** of H and J.)

Now let H be any *normal* subgroup of a group G. The left and right cosets are equal, so we can simply refer to cosets. We normally use left coset notation.

Then for all $a, b \in G$ we have $(aH)(bH) = a(Hb)H = a(bH)H = (ab)HH = abH$.

Thus there is a well-defined operation on the set of all cosets of H in G given by: (coset containing a)(coset containing b) = (coset containing ab).

This operation is certainly associative. $(eH)(aH) = eaH = aH$ so the identity element for the set of cosets is eH, i.e. H itself. $(a^{-1}H)(aH) = a^{-1}aH = eH = H$ so the coset aH has inverse $a^{-1}H$.

Hence the cosets form a group under this operation.

This group of cosets $\{aH \ : \ a \in G\}$ is the **quotient group** or **factor group** of G by H, denoted by G/H. For finite groups, $|G/H| = |G|/|H|$, the index of H in G.

From the definition of the operation, $(aH)^r = a^r H$ for all $a \in G$ and $r \in \mathbb{Z}$.

In an *additive* group, the cosets have the form $a+H$ and the operation is given by $(a+H)+(b+H) = (a+b)+H$. Then $r.(a+H) = r.a + H$ for all $r \in \mathbb{Z}$.

Recall the **first isomorphism theorem** for groups: if $\phi : G_1 \to G_2$ is a group homomorphism then $\operatorname{Ker}\phi \lhd G_1$ and $G_1/\operatorname{Ker}\phi \cong \phi(G_1)$. This can be understood as follows: the cosets of $\operatorname{Ker}\phi$ in G_1 are in one-to-one correspondence with the elements of the image. All elements in the same coset map to the same element of G_2, i.e. $a \operatorname{Ker}\phi = b \operatorname{Ker}\phi$ if and only if $\phi(a) = \phi(b)$.

1.3.6 Examples

(i) A_3 is the normal subgroup $\{\iota, \sigma, \sigma^2\}$ of S_3. The two distinct cosets A_3 and τA_3 form the quotient group S_3/A_3, which has order 2.

If $H = \{\iota, \tau\}$ then H is not normal in S_3, so there is no quotient group S_3/H.

(ii) Label the vertices of a regular tetrahedron 1, 2, 3, 4. The symmetry group of the tetrahedron (rigid motions combined with reflections in planes of symmetry) is S_4. There are three pairs of non-intersecting opposite edges. Label these pairs p_1, p_2, p_3. Each symmetry σ of the tetrahedron induces a permutation σ' of $\{p_1, p_2, p_3\}$. σ' can be represented as an element of S_3.

Define $\phi : S_4 \to S_3$ by $\phi(\sigma) = \sigma'$. The composite of two symmetries clearly induces the composite of the induced permutations, so ϕ is a group homomorphism. Its kernel is the set of all those elements in S_4 which map every edge onto either itself or the other edge in its pair. These elements are the identity and the products of two disjoint transpositions so $\operatorname{Ker}\phi = \{(1), (1\ \ 2)(3\ \ 4), (1\ \ 3)(2\ \ 4),$
$(1\ \ 4)(2\ \ 3)\}$, which we have seen before as the group D_2 ($\cong V$).

Thus $D_2 \lhd S_4$. All possible permutations of $\{p_1, p_2, p_3\}$ are induced by the symmetries, so $\phi(S_4) = S_3$. Hence by the first isomorphism theorem, $S_4/D_2 \cong S_3$. That is, the cosets of D_2 in S_4 are in one-to-one correspondence with the elements of S_3 and combine in the same way.

(iii) In D_4, let $N = \{\iota, \sigma^2\}$ and $J = \{\iota, \tau\}$. We saw that N is normal in D_4 but J is not. There is no quotient group D_4/J.

The cosets of N in D_4 form the quotient group D_4/N, which is isomorphic to V as shown by the Cayley table:

	N	σN	τN	$\sigma\tau N$
N	N	σN	τN	$\sigma\tau N$
σN	σN	N	$\sigma\tau N$	τN
τN	τN	$\sigma\tau N$	N	σN
$\sigma\tau N$	$\sigma\tau N$	τN	σN	N

The three 4-element subgroups of D_4 all contain N. Let H be any one of these. Then $N \triangleleft H$, and H/N is a subgroup of D_4/N.

If $H = \{\iota, \sigma, \sigma^2, \sigma^3\}$ then $H/N = \{N, \sigma N\}$, which we can see from the table is a subgroup of D_4/N. The following result generalises this observation.

Proposition 1.7 (Subgroup correspondence theorem) *Let N be a normal subgroup of a group G. The subgroups of G/N are the groups H/N for all subgroups H of G which contain N. Furthermore, $H/N \triangleleft G/N$ if and only if $N \triangleleft H \triangleleft G$, and then $(G/N)/(H/N) \cong G/H$.*

Proof

If $N \triangleleft G$ and N is contained in a subgroup H of G then $N \triangleleft H$ by Exercises 1.3 Question 4 (a), so H/N is a group. It is clearly contained in G/N.

Conversely, let S be a subgroup of G/N and let $H = \{h \in G \ : \ hN \in S\}$.
N is the identity of G/N so $N \in S$. Hence $nN \in S$ for all $n \in N$, so $N \subset H$.

Let $h_1, h_2 \in H$, so $h_1N, h_2N \in S$. Then $(h_1h_2^{-1})N = (h_1N)(h_2^{-1}N) = (h_1N)(h_2N)^{-1} \in S$, so $h_1h_2^{-1} \in H$, so H is a subgroup of G and we have $S = \{hN \ : \ h \in H\} = H/N$.

If $N \triangleleft H \triangleleft G$ then for all $g \in G$, $h \in H$ we have $ghg^{-1} = h'$ for some $h' \in H$, so $(gN)(hN)(gN)^{-1} = h'N \in H/N$ for all $gN \in G/N, hN \in H/N$. Thus $H/N \triangleleft G/N$.

Conversely, if $H/N \triangleleft G/N$ then certainly $N \triangleleft H$. For all $gN \in G/N$ and $hN \in H/N$ we have $(gN)(hN)(gN)^{-1} \in H/N$, so $ghg^{-1}N \in H/N$. Thus $ghg^{-1} \in H$ for all $g \in G, h \in H$ and hence $H \triangleleft G$.

Define $\psi : G/N \to G/H$ by $\psi(gN) = gH$. It is left as an exercise to show that ψ is well-defined and is a group homomorphism, with kernel H/N and image G/H. By the first isomorphism theorem, $(G/N)/(H/N) \cong G/H$. $\square$

The result $(G/N)/(H/N) \cong G/H$ is often called the *third isomorphism theorem*. (See the exercises for the second one.)

Proposition 1.8 *Let N be a normal subgroup of a group G. If G/N is cyclic then G has a subgroup isomorphic to G/N.*

Proof

Let gN have order m in G/N and let r be the order of g in G. Then $g^r = e$, so $(gN)^r = g^rN = N$. Thus $m \mid r$, i.e. $r = km$ for some $k \in \mathbb{N}$.

$(g^k)^m = g^r = e$, and no smaller power of g^k can equal e, so g^k has order m in G. Thus $\langle g^k \rangle$ is a cyclic subgroup of G of order m.

If G/N is cyclic then it has a generator gN of order $|G/N|$, and then $\langle g^k \rangle \cong G/N$. $\square$

1.3.7 Example

Suppose a group G of order 20 has a normal subgroup N of order 4. Then G/N has order 5, which is prime, so G/N is cyclic (generated by some coset gN). By Proposition 1.8, G has a cyclic subgroup of order 5.

Exercises 1.3

1. Let H and J be subgroups of a group G, such that $|H| = 12$ and $|J| = 8$. Find all the possible values of $|HJ|$. Is HJ necessarily a group?

2. Let H and J be subgroups of a group G such that $|H|^2 > |G|$ and $|J|^2 > |G|$. Show that $H \cap J$ is non-trivial. Deduce that a group of order 15 cannot have more than one subgroup of order 5.

3. Suppose a non-cyclic group G of order 55 has r subgroups of order 5 and s subgroups of order 11. Show that $2r + 5s = 27$. Find the possible values of r and s. In each case, state the number of *elements* in G of order 5 and order 11.

4. (a) Suppose $N \lhd G$ and H is a subgroup of G which contains N. Show that $N \lhd H$.

   (b) Suppose $H \lhd N$ and $N \lhd G$. By considering subgroups of D_4, or otherwise, show that it is not necessarily the case that $H \lhd G$.

5. Let g be a generator of C_n. Make Cayley tables for the quotient group C_n/H in each of the cases (a) $n = 6,\ H = \langle g^3 \rangle$, (b) $n = 8,\ H = \langle g^2 \rangle$

6. In S_3 let A and B be the subgroups $\{\iota, \sigma, \sigma^2\}$ and $\{\iota, \tau\}$ respectively.

   (a) Make a Cayley table for the quotient group S_3/A.

   (b) Show that $S_3 = AB$ but $S_3 \ncong A \times B$. Which condition in Proposition 1.5 fails to hold?

7. Suppose $N \lhd G$. Show that

   (a) if G is abelian then G/N is abelian,

   (b) if G is cyclic with generator g then G/N is cyclic with generator gN,

   (c) if the coset aN has order m in G/N then $a^m \in N$.

8. Let $\mathrm{X} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\mathrm{Y} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ be matrices in $M_2(\mathbb{F}_5)$, so all calculations are carried out modulo 5. Find the orders of X and Y. Find the order of the group generated by $\{\mathrm{X}, \mathrm{Y}\}$ and give a presentation of it. Generalise this example.

9. The group D_6, of order 12, has presentation $\langle a, b \ :\ a^6 = b^2 = e,\ aba = b \rangle$.

   Let $H = \{e, a^2, a^4, b, a^2b, a^4b\}$ and $J = \{e, a^3\}$. Show that H and J are normal subgroups of D_6, $H \cap J = \{e\}$ and $HJ = D_6$. Deduce that $D_6 \cong D_3 \times C_2$.

10. By considering two-element normal subgroups of D_4 and of C_6, show that if $N \lhd G$ then it may or may not be the case that $G \cong N \times G/N$.

11. Q_4 is the quaternion group, for which the table is on Page 5. Let $H = \{e, a^2\}$.

    (a) Show that H is a normal subgroup of Q_4, but Q_4 does not have a subgroup isomorphic to Q_4/H.

    (b) Identify all the non-trivial proper subgroups of Q_4/H.

12. Let N and H be normal subgroups of a group G, such that $N \subset H$.

    Define a map $\psi : G/N \to G/H$ by $\psi(gN) = gH$. Show that ψ is well-defined (i.e. that if $g_1N = g_2N$ then $g_1H = g_2H$) and is a surjective group homomorphism with kernel H/N.

    Deduce that $(G/N)/(H/N) \cong G/H$, completing the proof of Proposition 1.7.

13. By considering Examples 1.3.6 (iii), show that the converse of Proposition 1.8 is not true.

14. Let $\phi : G_1 \to G_2$ be a group homomorphism.

    (a) Show that the order of $\phi(G_1)$ divides both $|G_1|$ and $|G_2|$.

    (b) Show that there is no non-trivial group homomorphism from S_3 to C_7.

    (c) Let $a \in G_1$. Show that the order of $\phi(a)$ in G_2 divides the order of a in G_1, and is equal to it if ϕ is injective.

15. Let H and N be subgroups of a group G, such that $N \lhd G$, so HN is a subgroup of G by Proposition 1.3 (ii). Show that $N \lhd HN$.

    Define a map $\phi : H \to HN/N$ by $\phi(h) = hN$. Show that ϕ is a group homomorphism and find its kernel and image. Deduce the *second isomorphism theorem*: $H/(H \cap N) \cong HN/N$.

1.4 Finite Abelian Groups

We know that $C_m \times C_n \cong C_{mn}$ if and only if m and n are coprime. Thus a cyclic group of composite order is isomorphic to a direct product of cyclic groups of prime power order. For example, $360 = 2^3 \times 3^2 \times 5 = 8 \times 9 \times 5$. As $8, 9$ and 5 are pairwise coprime, $C_{360} \cong C_8 \times C_9 \times C_5$. This is *not* isomorphic to $C_2 \times C_{180}$, since 2 and 180 are not coprime.

We now show that every finite abelian group can be decomposed as a direct product of cyclic groups.

Suppose $\{g_1, \dots, g_k\}$ is a generating set for an *abelian* group G. The generators commute with one another, so $G = \{{g_1}^{n_1}{g_2}^{n_2} \cdots {g_k}^{n_k} \ : \ n_1, \dots, n_k \in \mathbb{Z}\}$.

Proposition 1.9 *Suppose an abelian group G has a finite generating set $\{g_1, \dots, g_k\}$ and let $\{m_1, \dots, m_k\}$ be a set of integers whose greatest common divisor is 1. Then G has a generating set of order k which contains ${g_1}^{m_1} \cdots {g_k}^{m_k}$.*

Proof

There is at least one j for which $gcd(m_1, m_j) = 1$. By relabelling if necessary, we may assume $j = 2$ so there are integers a, b such that $am_1 + bm_2 = 1$. Then any element ${g_1}^p{g_2}^q$ can be written as $({g_1}^{m_1}{g_2}^{m_2})^{ap+bq}({g_1}^b{g_2}^{-a})^{m_2p-m_1q}$. Also $({g_1}^{m_1}{g_2}^{m_2})^r({g_1}^b{g_2}^{-a})^s$ is equal to ${g_1}^{m_1r+bs}{g_2}^{m_2r-as}$. Thus $\langle g_1, g_2\rangle = \langle {g_1}^{m_1}{g_2}^{m_2},\ {g_1}^b{g_2}^{-a}\rangle$.

$gcd(1, m_3) = 1$ and $(1 - m_3)1 + 1m_3 = 1$ so the above reasoning applied to $\langle {g_1}^{m_1}{g_2}^{m_2}, g_3\rangle$ with $a = 1 - m_3, b = 1$ gives $\{{g_1}^{m_1}{g_2}^{m_2}{g_3}^{m_3},\ {g_1}^{m_1}{g_2}^{m_2}{g_3}^{m_3-1}\}$ as a generating set. Then $\langle g_1, g_2, g_3\rangle = \langle {g_1}^{m_1}{g_2}^{m_2}{g_3}^{m_3},\ {g_1}^{m_1}{g_2}^{m_2}{g_3}^{m_3-1},\ {g_1}^b{g_2}^{-a}\rangle$.

Continuing this process gives a generating set $\{{g_1}^{m_1}\cdots{g_k}^{m_k}, \ldots\}$ with k elements. □

Proposition 1.10 (Fundamental theorem of finite abelian groups) *Every finite abelian group G is isomorphic to a direct product of cyclic groups in the following two ways, each of which is unique up to the order of the factors:*

(1) $G \cong C_{p_1^{n_1}} \times C_{p_2^{n_2}} \times \cdots \times C_{p_r^{n_r}}$ where $p_1, \ldots, p_r$ are primes (not necessarily distinct) such that $|G| = {p_1}^{n_1}{p_2}^{n_2}\cdots{p_r}^{n_r}$;

(2) $G \cong C_{q_1} \times C_{q_2} \times \cdots \times C_{q_s}$ where $s = 1$ or $q_i \mid q_{i+1}$ for $i = 1, \ldots, s-1$.

Proof

Let $n(G)$ be the order of the smallest generating set for an abelian group G. The result is clearly true when $n(G) = 1$, as then G is cyclic. Assume it is true for $n(G) < k$. Now suppose $n(G) = k$.

Let t be the smallest order of any element in a generating set of order k and let $\{g_1, \ldots, g_k\}$ be such a set, where $o(g_1) = t$. Let $H = \langle g_2, \ldots, g_k\rangle$, so $G = \langle g_1\rangle H$. As G is abelian, $\langle g_1\rangle$ and H are normal subgroups of G. By the hypothesis H is isomorphic to a direct product of cyclic groups.

Suppose a is a non-identity element of $\langle g_1\rangle \cap H$, so $a = {g_1}^{r_1} = {g_2}^{r_2}\cdots{g_k}^{r_k}$ for some $r_1, \ldots, r_k \in \mathbb{Z}$ where $0 < r_1 < t$. Let $d = gcd(r_1, \ldots, r_k)$, so $\dfrac{r_1}{d}, \ldots, \dfrac{r_k}{d}$ are coprime. Let $x = {g_1}^{-r_1/d}{g_2}^{r_2/d}\cdots{g_k}^{r_k/d}$. Then $x^d = a^{-1}a = e$ and $d \le r_1 < t$, so $o(x) < t$. By Proposition 1.9 G has a k-element generating set containing x. This contradicts the minimality of $o(g_1)$. We conclude that $\langle g_1\rangle \cap H$ is trivial, so $G \cong \langle g_1\rangle \times H$ and hence G is isomorphic to a direct product of cyclic groups. The result follows by induction.

We know that a cyclic group of composite order is isomorphic to a direct product of cyclic groups of prime power order. This can be used, as shown in the examples, to obtain the decompositions (1) and (2). □

1.4.1 Examples

(i) Every abelian group of order 8 is isomorphic to one of $C_2 \times C_2 \times C_2,\ C_2 \times C_4,\ C_8$, so up to isomorphism there are five groups of order 8 (these three, D_4 and Q_4).

(ii) $54 = 2 \times 3^3$ so the distinct abelian groups of order 54 (up to isomorphism) are $C_2 \times C_{27}$ ($\cong C_{54}$), $C_2 \times C_3 \times C_9$ ($\cong C_3 \times C_{18}$) and $C_2 \times C_3 \times C_3 \times C_3$ ($\cong C_3 \times C_3 \times C_6$).

(iii) $252 = 2^2 \times 3^2 \times 7$ so the distinct abelian groups of order 252 (up to isomorphism) are

$C_2 \times C_2 \times C_3 \times C_3 \times C_7, \quad C_2 \times C_2 \times C_9 \times C_7, \quad C_4 \times C_3 \times C_3 \times C_7, \quad C_4 \times C_9 \times C_7.$

The decomposition (2) can be obtained from (1) as follows. Arrange powers of the same prime in the same column, then multiply along the rows, as follows:

$$\begin{array}{lllll} 2 \times 3 & = 6 & 2 & = 2 & 3 \quad\quad\quad = 3 \\ 2 \times 3 \times 7 & = 42 & 2 \times 9 \times 7 & = 126 & 3 \times 4 \times 7 = 84 \end{array}$$

Thus $C_2 \times C_2 \times C_3 \times C_3 \times C_7 \cong C_6 \times C_{42}$, $\quad C_2 \times C_2 \times C_9 \times C_7 \cong C_2 \times C_{126}$,

$C_4 \times C_3 \times C_3 \times C_7 \cong C_3 \times C_{84}$. As 4, 7, 9 are coprime, $C_4 \times C_9 \times C_7 \cong C_{252}$.

Exercises 1.4

1. For each of the following, find an isomorphic group which is either a single cyclic group or a direct product of the form $C_{q_1} \times C_{q_2} \times \cdots \times C_{q_s}$ where $q_i \mid q_{i+1}$.

   (a) $C_5 \times C_{11}$, (b) $C_2 \times C_3 \times C_8$,

   (c) $C_3 \times C_8 \times C_4 \times C_9$, (d) $C_5 \times C_6 \times C_7 \times C_8$.

2. Find, up to isomorphism, all abelian groups of order

   (a) 10, (b) 18, (c) 45, (d) 120, (e) 540.

   Express each as a direct product in the two ways given by Proposition 1.10.

3. Find, up to isomorphism, the number of groups of orders 11, 12 and 13. (You are given that every non-abelian group of order 12 is of a type that has already been mentioned in these notes.)

4. List as many groups of order 24 as you can think of. Which ones are abelian?

5. Find, up to isomorphism, the number of abelian groups

   (a) of order $\prod_{i=1}^{r} p_i$ where $p_1, \ldots, p_r$ are distinct primes,

   (b) of order p^k where p is prime, for $k = 1, 2, 3, 4, 5, 6$.

6. (a) Suppose g_i generates a cyclic group of order q_i, for $i = 1, \ldots, s$. Find the order of $(g_1, g_2, \ldots, g_s)$ in the direct product group $\langle g_1 \rangle \times \langle g_2 \rangle \times \cdots \times \langle g_s \rangle$.

   (b) Suppose G is a non-trivial group in which every non-identity element has order 2 (i.e. all elements are self-inverse). Exercises 1.1 Question 2 shows that G is abelian.

   Using the result from part (a) above, show that $G \cong C_2 \times \cdots \times C_2$ and hence that $|G|$ is a power of 2.

1.5 Automorphisms and semi-direct products

Recall that an **automorphism** of a group G is a bijective homomorphism from G to itself. Clearly every group has at least one automorphism, namely the identity map. If ϕ and ψ are automorphisms of G then so are their composite and their inverses, so the automorphisms of G form a group under composition.

By Exercises 1.1 Question 5, the map $\gamma_a : x \mapsto axa^{-1}$ (conjugation by a) is an automorphism of G. We have $\gamma_a {\gamma_b}^{-1} = \gamma_{ab^{-1}}$ (see exercises), so the automorphisms of this type also form a group.

Definition 1.5 *The group of all automorphisms of a group G is denoted by* $\mathrm{Aut}(G)$*.*

An automorphism of the form $x \mapsto axa^{-1}$ is called an **inner automorphism***. The subgroup of* $\mathrm{Aut}(G)$ *consisting of all inner automorphisms is denoted by* $\mathrm{Inn}(G)$*.*

If G is abelian, $axa^{-1} = x$ so the only inner automorphism is the identity map, i.e. $\mathrm{Inn}(G)$ is a trivial group.

In general it is not easy to find $\mathrm{Aut}(G)$, but there are some straightforward cases.

Proposition 1.11 *Let g be a generator of C_n. There are $\varphi(n)$ automorphisms of C_n, given by $g \mapsto g^m$ for each $m \in \{1, \dots, n-1\}$ such that $gcd(n, m) = 1$.* $\mathrm{Aut}(C_n) \cong U_n$*.*

Proof

Let $C_n = \langle g \rangle = \{e, g, \dots, g^{n-1}\}$. For any $m \in \{1, \dots, n-1\}$, define $\psi_m : C_n \to C_n$ by $\psi_m(g^r) = g^{mr}$. Then $\psi_m(g^r g^s) = \psi_m(g^{r+s}) = g^{mr+ms} = g^{mr} g^{ms} = \psi_m(g^r)\psi_m(g^s)$, so ψ_m is a homomorphism. In particular $\psi_m(g) = g^m$, and the effect of ψ on g determines its effect on all elements of C_n.

ψ_m is bijective if and only if there is a map $\psi_m^{-1} : g \mapsto g^k$ such that $\psi_m^{-1}\psi_m(g) = g$, so $g^{mk} = g$, so $mk \equiv 1 \pmod{n}$. This holds if and only if m mod n is a unit of $\mathbb{Z}_n$. The number of such m is $\varphi(n)$.

Define $\theta : U_n \to \mathrm{Aut}(C_n)$ by $\theta(m) = \psi_m$. Then $\psi_{m_1 m_2}(g) = g^{m_1 m_2} = (g^{m_1})^{m_2} = \psi_{m_1}\psi_{m_2}(g)$, so $\theta(m_1 m_2) = \theta(m_1)\theta(m_2)$, so θ is a group homomorphism. ψ_m is the identity map if and only if $m = 1$, so $\mathrm{Ker}\,\theta = \{1\}$ and hence θ is injective. As $|U_n| = \varphi(n) = |\theta(U_n)|$, θ is bijective and is thus an isomorphism. $\square$

1.5.1 Examples

(i) Let g be a generator of C_{18}. There are six automorphisms of C_{18}, given by $g \mapsto g$ (the identity map), $g \mapsto g^5$, $g \mapsto g^7$, $g \mapsto g^{11}$, $g \mapsto g^{13}$, $g \mapsto g^{17}$. These form a group isomorphic to $U_{18} = (\{1, 5, 7, 11, 13, 17\}, \times_{18})$.

(ii) $\varphi(360) = 96$, so there are 96 automorphisms of C_{360}.

(iii) Consider the Klein 4-group $V = \{e, a, b, c\}$ where $c = ab$ and $a^2 = b^2 = c^2 = e$. If $\theta : V \to V$ is an automorphism then $\theta(e) = e$ and $\theta(a), \theta(b), \theta(c)$ must all have order 2. Thus θ permutes $\{a, b, c\}$. We can check that each permutation does give an automorphism, so $\mathrm{Aut}(V) \cong S_3$.

Definition 1.6 *Suppose a group G has subgroups H and J such that $HJ = G$, $H \cap J = \{e\}$ and $H \lhd G$. Then G is the (internal)* **semi-direct product** *$H \rtimes J$.*

Note that if also $J \lhd G$ then G is the internal direct product of H and J, so $G \cong H \times J$.

If $G = H \rtimes J$ then each $j \in J$ determines an automorphism $\gamma_j : H \to H$ by $\gamma_j(h) = jhj^{-1}$. We say *J acts on H* by conjugation. The map from J to $\mathrm{Aut}(H)$ by $j \mapsto \gamma_j$ is readily shown to be a group homomorphism.

1.5.2 Example

Suppose a group G of order 12 has cyclic subgroups H and J of orders 3 and 4 with generators h and j respectively. As these orders are coprime, $|H \cap J| = 1$ so $|HJ| = 12$, hence $G = HJ = \{h^r j^s \ : \ h \in \{0, 1, 2\}, j \in \{0, 1, 2, 3\}\}$.

Suppose $H \lhd G$. Then $jhj^{-1} \in H$ and we have $G = H \rtimes J$. As conjugation by j is an automorphism of H, it must map h to a generator of H. Thus $jhj^{-1} = h$ or h^2.

If $jhj^{-1} = h$ then $jh = hj$, in which case $J \lhd G$ and $G \cong H \times J$ by Proposition 1.5 (ii), so $G \cong C_{12}$.

If $jhj^{-1} = h^2$ then $jh = h^2 j$, so $hjh = h^3 j = j$.

G then has presentation $\langle h, j \ : \ h^3 = j^4 = e,\ hjh = j\rangle$. It is not immediately obvious that such a group exists, but it can in fact be constructed as the subgroup of $GL(2, \mathbb{C})$ generated by $\begin{pmatrix} 0 & \mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}$ and $\begin{pmatrix} \omega & 0 \\ 0 & \omega^2 \end{pmatrix}$ (where $\omega = \mathrm{e}^{2\pi \mathrm{i}/3}$), these matrices having orders 4 and 3 respectively.

$H \cong C_3$, so $\mathrm{Aut}(H) = \{\iota, \psi\}$ where $\psi : h \mapsto h^2$. In the two groups that we have found, conjugation by j corresponds to ι and ψ respectively.

Finally suppose $J \lhd G$ and $H \not\lhd G$. Then conjugation by h is an automorphism of J, so it must map j to a generator of J. Now if $hjh^{-1} = j$ we have $G \cong C_{12}$ as before. Thus $hjh^{-1} = j^3$. As $h^3 = j^4 = e$, by Proposition 1.2 (ii) we have $j = h^3 j h^{-3} = j^{(3^3)} = j^{27} = j^3$ and so $j^2 = e$, contradicting the assumption that j has order 4. Hence no such group of order 12 exists.

It can be shown that given any two groups H and J and a homomorphism θ from J to $\mathrm{Aut}(H)$, a group called the *external* semi-direct product $H \rtimes_\theta J$ can be constructed. Different homomorphisms can give different semi-direct products of the same two groups. See Exercises 1.5 Question 12 for details.

Exercises 1.5

1. Using answers from Exercises 1.1,
   (a) find all the automorphisms of C_{16}, defining each one by its effect on a generator g,
   (b) find the order of $\mathrm{Aut}(C_{84})$.
2. Find the automorphism group of $(\mathbb{Z}, +)$.

3. Determine whether the map $A \mapsto A^{-1}$ is an automorphism of $GL(n, \mathbb{R})$.

4. For a fixed element a in a group G, let γ_a denote the inner automorphism $x \mapsto axa^{-1}$.

   (a) Show that $\gamma_a {\gamma_b}^{-1} = \gamma_{ab^{-1}}$. Deduce that the inner automorphisms of G form a subgroup of $\mathrm{Aut}(G)$. Show further that $\mathrm{Inn}(G) \lhd \mathrm{Aut}(G)$.

   (b) Show that the map $\psi : G \to \mathrm{Aut}(G)$ given by $\psi : a \mapsto \gamma_a$ is a group homomorphism and show that its kernel is the **centre** $Z(G) := \{a \in G \ : \ ax = xa \text{ for all } x \in G\}$. Deduce that $G/Z(G) \cong \mathrm{Inn}(G)$. Show that if $Z(G)$ is trivial then $\mathrm{Aut}(G)$ has a subgroup isomorphic to G.

5. Consider the group S_3, with presentation $\langle \sigma, \tau \ : \ \sigma^3 = \tau^2 = \iota, \tau\sigma = \sigma^2\tau \rangle$. Let ϕ be an automorphism of S_3. Clearly ϕ is determined by its effect on the generators σ and τ.

   (a) Using the fact that ϕ preserves the orders of elements, as shown in Exercises 1.3 Question 14, show that there are at most six automorphisms of S_3.

   (b) Use Question 4 (b) above to show that $\mathrm{Aut}(S_3) \cong S_3$ and all automorphisms of S_3 are inner.

6. Consider the dihedral group D_n with presentation $\langle \sigma, \tau \ : \ \sigma^n = \tau^2 = \iota, \sigma\tau\sigma = \tau \rangle$. Express D_n as the semi-direct product of two subgroups.

7. Verify that the matrices in Example 1.5.2 satisfy the given presentation.

8. Suppose a group G of order 20 has cyclic subgroups H and J of orders 5 and 4 with generators h and j respectively, where $H \lhd G$. Show that $G = H \rtimes J$ and that $jhj^{-1} = h^r$ where $r \in \{1, 2, 3, 4\}$.

   Determine which value of r corresponds to the subgroup of $GL(2, \mathbb{F}_5)$ generated by the matrices $X = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $Y = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

9. Suppose a group G of order 24 has cyclic subgroups H and J of orders 8 and 3. Show that if $H \lhd G$ then G is abelian. Investigate the case where $J \lhd G$.

10. A **characteristic subgroup** of a group G is a subgroup H such that $\phi(H) = H$ for every automorphism $\phi : G \to G$.

    (a) Show that every subgroup of a cyclic group is a characteristic subgroup.

    (b) Show that every characteristic subgroup of a group G is a normal subgroup.

    (c) Let H and J be subgroups of G such that J is characteristic in H and also H is characteristic in G. Show that J is characteristic in G. Explain where the reasoning fails if 'characteristic' is replaced by 'normal'.

11. Suppose $N \lhd G$, G/N is cyclic and $|N|, |G/N|$ are coprime.

    Show that $G = N \rtimes Q$ where Q is a cyclic subgroup of G.

12. Let H and J be groups (not necessarily subgroups of the same group) and let θ be a homomorphism from J to $\mathrm{Aut}(H)$. Define $H \rtimes_\theta J$ to be the Cartesian product set $H \times J$ with the operation given by $(h_1, j_1)(h_2, j_2) = (h_1\theta_1(h_2), j_1 j_2)$, where θ_1 is the map $\theta(j_1)$.

    Show that this operation is associative, (e_H, e_J) is an identity element and $({\theta_1}^{-1}(h_1^{-1}), j_1^{-1})$ is the inverse of (h_1, j_1). Hence $H \rtimes_\theta J$ is a group (called the **external semi-direct product** of H and J). Show that the external direct product $H \times J$ is a special case of this construction.

    Show further that if $\overline{H} = \{(h, e_J) \ : \ h \in H\}$ and $\overline{J} = \{(e_H, j) \ : \ j \in J\}$ then $H \rtimes_\theta J$ is the internal semi-direct product of $\overline{H}$ and $\overline{J}$.

13. Investigate the automorphism group of the dihedral group D_4.

Chapter 2

2.1 Group actions

Definition 2.1 *Let $(G,*)$ be a group with identity element e. Let X be a set.*

A (left) **action** *of G on X is a rule for combining any element of G with any element of X, given by $(g,x) \mapsto g(x)$, such that for all $x \in X$ and $g, g_1, g_2 \in G$,*

(i) $g(x) \in X$, *(ii)* $e(x) = x$, *(iii)* $g_1(g_2(x)) = (g_1 * g_2)(x)$.

If $X = G$ then we say that G **acts on itself**.

A *right action* can be defined similarly: $(x,g) \mapsto (x)g \in X$ where $(x)e = x$, $((x)g_1)g_2 = (x)(g_1*g_2)$. In this module, all group actions are left actions unless otherwise stated.

For any $g \in G$, we refer to $g(x)$ as the (left) action of g on the element x. We shall normally use multiplicative notation, so part (iii) of the definition is written as $g_1(g_2(x)) = g_1g_2(x)$. If G is an additive group, it becomes $g_1(g_2(x)) = (g_1+g_2)(x)$.

Note that $g(x)$ is *not* an element of G unless X happens to be a subset of G. Even then, $g(x)$ does not necessarily mean g combined with x by the operation in G. The meaning of $g(x)$ will be defined in each example.

2.1.1 Examples

(i) Any subgroup of S_n acts on the set $\{1,\ldots,n\}$ by permuting the elements.

(ii) The group $GL(n,\mathbb{R})$ acts on $\mathbb{R}^n$ in the 'obvious' way: $\mathrm{A}(\mathbf{x}) = \mathrm{A}\mathbf{x}$.

(iii) Let $(H,*)$ be a subgroup of a group $(G,*)$.

For each $h \in H$ and $g \in G$, let $h(g) = h*g$. Show that this defines an action of H on G.

Since $h, g \in G$, we have $h(g) \in G$. Also $e(g) = e*g = g$ for all $g \in G$.

If $h_1, h_2 \in H$ then $h_1(h_2(g)) = h_1*(h_2*g) = (h_1*h_2)*g = (h_1*h_2)(g)$.

Hence $h(g) = h*g$ defines a group action, called the action of H on G by left multiplication. In particular, taking $H = G$, we see that the group G acts on itself in this natural way.

(iv) A special case of the previous example: any subgroup of the additive group $\mathbb{R}$, such as $\mathbb{Z}$, acts on $\mathbb{R}$ by translation, i.e. if $n \in \mathbb{Z}$, $a \in \mathbb{R}$ then $n(a) = n+a$. In this case we have $0(a) = a$ and $m(n(a)) = m+(n+a) = (m+n)+a = (m+n)(a)$, where the last expression means the action of $m+n$ on a, *not* a multiple of a.

(v) Let G be a multiplicative group. The rule $g(S) = gS$ for all $g \in G$ and $S \in X$ defines an action of G on X when X is any of the following:

the set of all subsets of G; the set of all subsets of G having some fixed order;
the set of all left cosets of a subgroup in G.

Unless G is trivial, this rule does *not* define an action of G on the set of all *subgroups* of G, since if S is a subgroup of G then gS is not a subgroup of G for any $g \notin S$.

Definition 2.2 *Let G be a group acting on a set X.*

The **orbit** *of an element $x \in X$ is the set $\mathcal{O}(x) := \{y \in X \ : \ y = g(x) \text{ for some } g \in G\}$.*

The **stabiliser** *of an element $x \in X$ is the set $G_x := \{g \in G \ : \ g(x) = x\}$.*

The **stabiliser** *of a subset $Y \subset X$ is the set $\widetilde{G}_Y := \{g \in G \ : \ g(x) = x \text{ for all } x \in Y\}$.*

The **fixed set** *of an element $g \in G$ is the set $\mathrm{Fix}(g) := \{x \in X \ : \ g(x) = x\}$.*

The **fixed set** *of a subgroup $H \subset G$ is the set $\mathrm{Fix}(H) := \{x \in X \ : \ g(x) = x \text{ for all } g \in H\}$.*

Thus the orbit $\mathcal{O}(x)$ is the set of all elements of X to which x can be mapped by the action of elements in G. The stabiliser G_x consists of all elements of G which keep x fixed. The notation $\mathrm{Stab}(x)$ is sometimes used instead of G_x.

The stabiliser of a set $Y \subset X$ consists of all elements in G which fix *everything* in Y, i.e. $\widetilde{G}_Y = \bigcap\limits_{x \in Y} G_x$.

In particular, $\widetilde{G}_X$ is the set of all elements in G which fix every element in X.

The notation $\widetilde{G}_Y$ avoids confusion with the use of G_Y for the stabiliser of an individual set Y, in cases where the group G acts on a set of sets.

The fixed set of a group element g consists of all elements of X that are fixed by g.

The fixed set of a subgroup $H \subset G$ consists of all elements of X fixed by *every* element of the subgroup, i.e. $\mathrm{Fix}(H) = \bigcap\limits_{g \in H} \mathrm{Fix}(g)$. In particular, $\mathrm{Fix}(G)$ is the set of all elements of X that are fixed by every element of G.

Remember that orbits and fixed sets are contained in the set X that is being acted on, while stabilisers are contained in the group G which acts on X.

All these sets depend on the group, the set that it acts on and the way that it acts. If there is any doubt, the notation can be suitably modified; e.g. if groups G and H both act on X we can write $\mathcal{O}_G(x)$ and $\mathcal{O}_H(x)$ for the orbits of x under G and H.

2.1.2 Example

Let $ABCD$ be a square. Label the lines AB, BC, CD, DA, AC, BD as 1, 2, 3, 4, 5, 6 respectively. Let O be the centre of the square.

The symmetries of the square act as permutations of $X = \{1, 2, 3, 4, 5, 6\}$, as follows:

The identity map ι leaves each line fixed.

A quarter-turn about O, such that B moves to where A was: $\sigma = (1 \ \ 2 \ \ 3 \ \ 4)(5 \ \ 6)$.

A half-turn about O: $\sigma^2 = (1 \ \ 3)(2 \ \ 4)$.

A quarter-turn in the opposite sense to σ: $\sigma^3 = (1 \ \ 4 \ \ 3 \ \ 2)(5 \ \ 6)$.

A reflection in the line joining the mid-points of AB and CD: $\tau = (2 \ \ 4)(5 \ \ 6)$.

A reflection in BD: $\sigma\tau = (1 \ \ 2)(3 \ \ 4)$.

A reflection in the line joining the mid-points of AD and BC: $\sigma^2\tau = (1 \quad 3)(5 \quad 6)$.

A reflection in AC: $\sigma^3\tau = (1 \quad 4)(2 \quad 3)$.

These symmetries form a subgroup G of S_6. We see that G is isomorphic to D_4.

For this action, the orbits are $\mathcal{O}(1) = \mathcal{O}(2) = \mathcal{O}(3) = \mathcal{O}(4) = \{1, 2, 3, 4\}$, $\mathcal{O}(5) = \mathcal{O}(6) = \{5, 6\}$.

The stablilisers of individual elements of X are $G_1 = G_3 = \{\iota, \tau\}$, $G_2 = G_4 = \{\iota, \sigma^2\tau\}$, $G_5 = G_6 = \{\iota, \sigma^2, \sigma\tau, \sigma^3\tau\}$.

We can also find stabilisers of subsets of X, e.g. $\widetilde{G}_{\{1,3\}} = \{\iota, \tau\}$, $\widetilde{G}_{\{1,2,3\}} = \{\iota\}$.

The fixed sets are $\mathrm{Fix}(\iota) = X$, $\mathrm{Fix}(\sigma^2) = \mathrm{Fix}(\sigma\tau) = \mathrm{Fix}(\sigma^3\tau) = \{5, 6\}$, $\mathrm{Fix}(\sigma^2\tau) = \{2, 4\}$, $\mathrm{Fix}(\tau) = \{1, 3\}$, $\mathrm{Fix}(\sigma) = \mathrm{Fix}(\sigma^3) = \emptyset$ (empty, i.e. *no* points are fixed).

We can also find fixed sets of subgroups of G, e.g. $\mathrm{Fix}(\{\iota, \sigma\tau\}) = \{5, 6\}$

Proposition 2.1 *Let G be a group acting on a set X.*

(i) For each element $x \in X$, the stabiliser G_x is a subgroup of G.

(ii) For each subset Y of X, the stabiliser $\widetilde{G}_Y$ is a subgroup of G.

Proof

$x = e(x)$, so $e \in G_x$, so G_x is non-empty. It is a subset of G by definition.

Let $g_1, g_2 \in G_x$, so $g_1(x) = g_2(x) = x$. Then ${g_2}^{-1}(x) = {g_2}^{-1}(g_2(x)) = e(x) = x$.

Thus $g_1{g_2}^{-1}(x) = g_1(x) = x$, so $g_1{g_2}^{-1} \in G_x$. Hence G_x is a subgroup of G.

The intersection of subgroups is a subgroup, so $\widetilde{G}_Y = \bigcap\limits_{x \in Y} G_x$ is a subgroup of G. □

Given an action of a group G on a set X, define a relation $\sim$ on X as follows: $x_1 \sim x_2$ if and only if there is an element $g \in G$ such that $g(x_1) = x_2$. It is straightforward to show that $\sim$ is an equivalence relation on X, and that the equivalence class of $x \in X$ is precisely the orbit $\mathcal{O}(x)$.

Any equivalence relation partitions X into disjoint subsets, so each element of X lies in exactly one orbit.

Proposition 2.2 (The orbit-stabiliser theorem)

Let G be a group acting on a set X. For each $x \in X$, there is a bijection between the orbit $\mathcal{O}(x)$ and the set of all left cosets of the stabiliser G_x in G, given by $g(x) \mapsto gG_x$, such that g and h are in the same left coset if and only if $g(x) = h(x)$.

If G is finite then we have the **orbit-stabiliser equation** $|\mathcal{O}(x)| \,.\, |G_x| = |G|$.

Proof

Let $x \in X$ and consider the left cosets of its stabiliser in G, i.e. the sets gG_x for $g \in G$. These are disjoint subsets which form a partition of G.

We have $h(x) = g(x) \Leftrightarrow g^{-1}h(x) = x \Leftrightarrow g^{-1}h \in G_x \Leftrightarrow h \in gG_x$, so two elements of G are in the same left coset of G_x if and only if their actions on x are the same. Thus for each $x \in X$ there is a bijection (i.e. a one-to-one correspondence) between the orbit $\mathcal{O}(x)$ and the set of all left cosets of G_x in G.

It follows that if G is finite, the number of elements in $\mathcal{O}(x)$ equals the number of left cosets of G_x in G. This in turn equals the index $[G : G_x] = |G|/|G_x|$, so $|\mathcal{O}(x)| \,.\, |G_x| = |G|$. $\square$

We considered *left* cosets because the elements of G act on elements of X on the left.

The orbit-stabiliser equation shows that the number of elements in each orbit divides the order of G. The orbits need not all have the same number of elements.

Proposition 2.3 (The orbit decomposition theorem) *Let G be a group acting on a finite set X. Let $\mathcal{O}(x_1), \ldots, \mathcal{O}(x_n)$ be the* distinct *orbits which have more than one element. Then* $|X| = |\mathrm{Fix}(G)| + \sum\limits_{i=1}^{n} [G : G_{x_i}]$.

Proof X is the union of the orbits, which are disjoint, so if X is finite then its order is the sum of the orders of the distinct orbits. An orbit has a single element if and only if that element is fixed by all $g \in G$, so the union of all the one-element orbits is $\mathrm{Fix}(G)$. Each other orbit has order $[G : G_{x_i}]$. $\square$

2.1.3 Example

In Example 2.1.2, $|\mathcal{O}(1)| = 4$, $|G_1| = 2$ and $4 \times 2 = 8 = |G|$.

Similarly $|\mathcal{O}(5)| = 2$, $|G_5| = 4$ and $2 \times 4 = 8 = |G|$.

This verifies the orbit-stabiliser equation in this case.

The left cosets of G_1 are: $G_1 = \{\iota, \tau\}$, $\sigma G_1 = \{\sigma, \sigma\tau\}$, $\sigma^2 G_1 = \{\sigma^2, \sigma^2\tau\}$, $\sigma^3 G_1 = \{\sigma^3, \sigma^3\tau\}$. These cosets are in one-to-one correspondence with the elements 1, 2, 3, 4 of $\mathcal{O}(1)$. Both elements of G_1 map 1 to 1; both elements of σG_1 map 1 to 2; both elements of $\sigma^2 G_1$ map 1 to 3; both elements of $\sigma^3 G_1$ map 1 to 4.

The left cosets of G_5 in G are: $G_5 = \{\iota, \sigma^2, \sigma\tau, \sigma^3\tau\}$, $\sigma G_5 = \{\sigma, \sigma^3, \tau, \sigma^2\tau\}$.

These cosets are in one-to-one correspondence with the elements 5 and 6 of $\mathcal{O}(5)$. All elements of G_5 map 5 to 5; all elements of σG_5 map 5 to 6.

This demonstrates the bijection described in the orbit-stabiliser theorem.

The distinct orbits are $\mathcal{O}(1) = \{1, 2, 3, 4\}$ and $\mathcal{O}(5) = \{5, 6\}$. (We could also call the first one $\mathcal{O}(2), \mathcal{O}(3)$ or $\mathcal{O}(4)$ and the second one $\mathcal{O}(6)$.)

$\mathrm{Fix}(G) = \emptyset$ so $|\mathrm{Fix}(G)| = 0$. $|G_1| = 2$ so $[G : G_1] = 4$. $|G_5| = 4$ so $[G : G_5] = 2$.

$|X| = 6 = 0 + 4 + 2$, verifying the orbit decomposition theorem for this example.

2.1.4 Example

As a further illustration of the use of group actions, we can now prove Proposition 1.4. Consider the set (not necessarily a group) G/J of all left cosets of J in G. H acts on G/J by $h(aJ) = haJ$

for all $h \in H$ and $aJ \in G/J$. The stabiliser of J under this action is $H_J = \{h \in H \ : \ hJ = J\} = \{h \in H : \ : \ h \in J\} = H \cap J$.

The orbit of J is $\mathcal{O}(J) = \{hJ \ : \ h \in H\}$. The cosets in $\mathcal{O}(J)$ have union HJ. They are disjoint and each has order $|J|$, so there are $\dfrac{|HJ|}{|J|}$ of them, i.e. this is $|\mathcal{O}(J)|$.

Now by the orbit-stabiliser theorem $\dfrac{|HJ|}{|J|}|H \cap J| = |H|$, so $|HJ| = \dfrac{|H|.|J|}{|H \cap J|}$. $\square$

Proposition 2.4 *Let G be a group acting on a finite set X of order m. Then*

(i) The action of G permutes the elements of X.

(ii) There is a group homomorphism from G to S_m with kernel $\widetilde{G}_X$.

(iii) $G/\widetilde{G}_X$ is isomorphic to a subgroup of S_m.

Proof

(i) Let $X = \{x_1, \ldots, x_m\}$. If $g(x_j) = g(x_k)$ then $g^{-1}(g(x_j)) = g^{-1}(g(x_k))$, so $g^{-1}g(x_j) = g^{-1}g(x_k)$, so $e(x_j) = e(x_k)$ and thus $x_j = x_k$.

Hence $g(x_1), \ldots, g(x_m)$ are distinct, so they are $x_1, \ldots, x_m$ in some order.

(ii) The action of G on X determines an action of G on $\{1, \ldots, m\}$, where $g(i)$ is such that $g(x_i) = x_{g(i)}$.

By (i), $\sigma_g : i \mapsto g(i)$ is a permutation of $\{1, \ldots, m\}$. Define $\phi : G \to S_m$ by $\phi(g) = \sigma_g$.

Let $g, h \in G$. For each $i \in \{1, \ldots, m\}$, $\sigma_{gh}(i) = gh(i) = g(h(i)) = \sigma_g\sigma_h(i)$

so $\phi(gh) = \phi(g)\phi(h)$. Hence ϕ is a group homomorphism.

$\operatorname{Ker}\phi = \{g \in G \ : \ \sigma_g = \iota\} = \{g \in G \ : \ g(x_i) = x_i \text{ for } i = 1, \ldots, m\} = \widetilde{G}_X$.

(iii) By the first isomorphism theorem, $G/\operatorname{Ker}\phi \cong \phi(G)$, a subgroup of S_m. $\square$

Note that this result shows, in particular, that $\widetilde{G}_X$ is a normal subgroup of G.

We now consider an action which can be very useful in analysing the structure of a group. Let H be a subgroup of a finite group G. Then G is the union of the disjoint left cosets of H. The number of these cosets equals the index $[G : H]$.

Write G/H for the *set* of all left cosets of H in G. This is not a group unless $H \lhd G$.

Suppose the distinct cosets are $a_1H, \ldots, a_mH$. (One of these is H itself.)

G acts on G/H by $g(aH) = gaH$. (It is straightforward to show that this *is* a group action.) For any $g \in G$. we have $ga_iH = ga_jH$ if and only if $a_iH = a_jH$, so the cosets are permuted by this action.

Proposition 2.5 *Let H be a proper subgroup of index m in a finite group G.*

Then there is a group homomorphism from G to S_m with its kernel contained in H.

Proof

G acts on G/H by $g(aH) = gaH$. The stabiliser of G/H under this action is $\widetilde{G}_{G/H} = \{g \in G : gaH = aH \text{ for all } a \in G\}$. $|G/H| = m$ so by Proposition 2.4 there is a homomorphism $\phi : G \to S_m$ with kernel $\widetilde{G}_{G/H}$.

If $g \in \widetilde{G}_{G/H}$ then g keeps *every* coset fixed, so in particular $gH = H$, hence $g \in H$. Thus $\widetilde{G}_{G/H} \subset H$, i.e. $\operatorname{Ker}\phi \subset H$. $\square$

The map ϕ in the above proof is often described as a **representation on cosets**.

2.1.5 Example

Let G be a group of order 1092. Suppose G has a subgroup H of order 182, so H has index 6 in G. Then there is a homomorphism from G to S_6 whose kernel is contained in H. As $1092 > 6!$ this homomorphism cannot be injective so its kernel is a non-trivial normal subgroup of G.

As a special case, taking $H = \{e\}$, we see that there is an injective homomorphism from G to S_n where $n = |G|$. This gives:

Proposition 2.6 (Cayley's theorem) *Every group of finite order n is isomorphic to a subgroup of S_n.*

Cayley's theorem simply describes the natural action of G on itself. If $G = \{a_1, \dots, a_n\}$ then the row headed a_i in a Cayley table contains $a_i a_1, \dots, a_i a_n$, which are $a_1, \dots, a_n$ in some order. Let σ_i be the permutation of $\{1, \dots, n\}$ such that $a_i a_j = a_{\sigma_i(j)}$ for $j = 1, \dots, n$. Then $\{\sigma_1, \dots, \sigma_n\}$ is a subgroup of S_n which is isomorphic to G. If a_i is not the identity then for each j we have $a_{\sigma_i(j)} \neq a_j$, i.e. $\operatorname{Fix}(\sigma_i) = \emptyset$.

Proposition 2.7 *Let p be the smallest prime number that divides the order of a finite group G. Then any subgroup of index p in G is normal in G.*

Proof

Let H have index p in G. Then by Proposition 2.5 there is a homomorphism from G to S_p with kernel $K \subset H$. The image is a subgroup of S_p which is isomorphic to G/K. Hence $|G/K|$ divides $|S_p| = p\,!$

Also $|G/K|$ divides $|G|$, whose factors are powers of primes greater than or equal to p, so $|G/K| = p = |G/H|$. Hence $|K| = |H|$ and $K \subset H$, so $K = H$. As the kernel of a homomorphism is a normal subgroup, $H \lhd G$. $\square$

2.1.6 Examples

(i) If G is even then 2 is the smallest prime dividing $|G|$, so (as we already knew) any subgroup of index 2 is normal in G.

(ii) In a group of order 45, if there is a subgroup of index 3 (i.e. order 15) then it is a normal subgroup.

(iii) In a group of order p^k where p is prime, any subgroup of order p^{k-1} is normal.

Proposition 2.8 *Let G be a group of order p^2, where p is prime. Then G is isomorphic to either C_{p^2} or $C_p \times C_p$, and so G is abelian.*

Proof

Every non-identity element of G has order p or p^2. If G has an element of order p^2, this generates G and so $G \cong C_{p^2}$.

Otherwise, the p^2-1 elements of order p generate $p+1$ subgroups of order p, hence index p, in G. By Proposition 2.7 these are normal in G. Let H and J be any two of them. We have $H \cap J = \{e\}$ so $|HJ| = |H|.|J| = p^2 = |G|$ and thus $HJ = G$. By Proposition 1.5, $G \cong H \times J \cong C_p \times C_p$. $\square$

2.1.7 Example

Up to isomorphism, the only two groups of order 9 are $C_3 \times C_3$ and C_9.

Exercises 2.1

1. Show that the rule $x(z) = \mathrm{e}^{\mathrm{i}x}z$ defines an action of the group $(\mathbb{R}, +)$ on $\mathbb{C}$.
   For this action, find $\mathcal{O}(1)$, G_1 and $\mathrm{Fix}(0)$.

2. $\mathbb{R}^*$ is the multiplicative group of all non-zero real numbers and V is a real vector space. Show that $a(v) = av$ defines a group action of $\mathbb{R}^*$ on V.

3. Show that $n(x) = nx$ does not define a group action of $(\mathbb{Z}, +)$ on $\mathbb{R}$.

4. Let H be a subset of a non-abelian group G. Show that $h(g) = gh$ does not define a left action of H on G, but $h(g) = gh^{-1}$ does.

5. S_3 acts on $X = \{1, 2, 3\}$ by permutation. Find the orbit of each element of X, the stabiliser of each element of X and the fixed set of each element of S_3.

6. $O(2, \mathbb{R})$, the group of all orthogonal 2×2 real matrices, acts on $\mathbb{R}^2$ by $\mathrm{A}(\mathbf{v}) = \mathrm{A}\mathbf{v}$ for any column vector $\mathbf{v}$. Describe the orbit of $\mathbf{v}$.

7. H and J are subgroups of a group G. Show that the rule $k(h, j) = (hk^{-1}, kj)$, where $h, k \in H$, $j, k \in J$, defines an action of $H \cap J$ on $H \times J$.

8. ABC is an equilateral triangle. L, M and N are the mid-points of AB, BC, CA.

   The lines AB, BC, CA, CL, AM, BN are labelled 1, 2, 3, 4, 5, 6 respectively.

   Let $\sigma = (1 \quad 2 \quad 3)(4 \quad 5 \quad 6)$ and $\tau = (2 \quad 3)(5 \quad 6)$.

   (a) Find all the symmetries of the figure, expressing each as a permutation. Identify the symmetry group G.

   (b) G acts on $X = \{1, 2, 3, 4, 5, 6\}$ by permutation. For this action, find the orbit of each element of X, the stabiliser of each element of X and the fixed set of each element of G.

(c) Verify the orbit-stabiliser equation for $\mathcal{O}(1)$. Demonstrate the bijection between the orbit of 1 and the set of left cosets of G_1 in G.

(d) Verify the orbit decomposition theorem for this action.

9. Let G be a group acting on a set X. We defined a relation $\sim$ by $x_1 \sim x_2$ iff $g(x_1) = x_2$ for some $g \in G$. Show that $\sim$ is an equivalence relation on X and that the equivalence class of an element $x \in X$ is $\mathcal{O}(x)$.

10. Let G be a finite group acting on a set X. Suppose x and y are in the same orbit. Use results proved in Section 2.1 to show that the stabilisers G_x and G_y have the same order. Are they necessarily the same subgroup of G?

11. An action of a group G on a set X is said to be **transitive** on X if for every pair of elements $x, y \in X$ there is some $g \in G$ such that $g(x) = y$. Show that G is transitive on X if and only if $\mathcal{O}(x) = X$ for all $x \in X$.

    If G is finite and acts transitively on itself, show that $G_x = \{e\}$ for all $x \in G$.

12. Show that $g(x) = gxg^{-1}$ defines an action of a group G on itself. For this action, show that $\widetilde{G}_G = \mathrm{Fix}(G)$.

13. Use Proposition 2.7 to find orders of subgroups which are guaranteed to be normal, if they exist, in groups of order (a) 21, (b) 143, (c) 175.

14. List (up to isomorphism) all the groups of orders 25 and 49.

15. Let G be a finite group and let X be the set of all subsets of G that have order n. Show that the rule $g(S) = gS$ for each $S \in X$ defines an action of G on X.

    Suppose $y \in G_S x$. Show that $y \in S$ if and only if $x \in S$. Deduce that for each $x \in S$, either $G_S x \subset S$ or $G_S x \cap S = \emptyset$. Hence show that S is a union of right cosets of G_S in G, so $|G_S|$ divides $|S|$.

16. A group G of prime order p acts on a set X. Show that every orbit has order 1 or p. Deduce that if p divides $|X|$ then p divides $|\mathrm{Fix}(G)|$.

17. Let G be a group whose order is divisible by a prime p. **Cauchy's theorem** states that G has at least one element of order p. Prove this as follows:

    Let $X = \{(a_0, \dots, a_{p-1}) \ : \ a_0, \dots, a_{p-1} \in G, \ \prod_{r=0}^{p-1} a_r = e\}$ where e is the identity element of G.

    (a) Show that $|X| = |G|^{p-1}$.

    (b) Show that the following rule defines an action ('by cyclic permutation') of the cyclic group C_p, with generator g, on the set X:
    For $k = 0, \dots, p-1$, let $g^k(a_0, a_1, \dots, a_{p-1}) = (a_k, a_{k+1} \dots, a_{k+p-1})$ where all subscripts are evaluated modulo p.

    (c) Identify $\mathrm{Fix}(C_p)$ and use Question 16 to show that the number of solutions of $x^p = e$ in G is a multiple of p. Deduce that G has elements of order p and that the number of these is congruent to -1 (modulo p).

(d) Show that every group of order 99 has a *normal* subgroup of order 11.

2.2 Conjugacy

Definition 2.3 *Let G be a group. $g(x) = gxg^{-1}$ is the* **conjugation action** *of G on G. The* **conjugacy class** *of an element $x \in G$ is the set $x^G := \{gxg^{-1} \ : \ g \in G\}$.*

By Exercises 2.1 Question 12, conjugation *is* a group action. The conjugacy class of x is the orbit of x under this action, so G is partitioned into disjoint conjugacy classes. By the orbit-stabiliser theorem, the order of any conjugacy class divides $|G|$.

The conjugacy class of the identity element e is $\{geg^{-1} \ : \ g \in G\}$, which equals $\{e\}$, so no other class contains e. Thus $\{e\}$ is the only conjugacy class which is actually a subgroup of G.

Unlike cosets, conjugacy classes do not all have the same order in general.

The stabiliser of $x \in G$ under the conjugation action is $\{g \in G \ : \ gxg^{-1} = x\}$, i.e. $\{g \in G \ : \ gx = xg\}$, the set of all elements which commute with the given element x. By Proposition 2.1, this is a subgroup of G. It has the following name:

Definition 2.4 *Let x be an element of a group G. The* **centraliser** *of x in G is the subgroup $C_G(x) := \{g \in G \ : \ gx = xg\}$.*

The stabiliser $\widetilde{G}_G$ of the whole group G is the intersection of the centralisers of all the elements of G, so it is $\{g \in G \ : \ gxg^{-1} = x$ for all $x \in G\}$ or equivalently $\{g \in G \ : \ gx = xg$ for all $x \in G\}$, the set of all elements of G which commute with *every* element of G. This also has a name:

Definition 2.5 *The* **centre** *of a group G is $Z(G) := \{g \in G \ : \ gx = xg \ \forall \ x \in G\}$.*

It follows from Proposition 2.4 (ii) that $Z(G)$ is a normal subgroup of G.

Clearly $Z(G)$ is abelian, and G is itself abelian if and only if $Z(G) = G$.

$Z(G)$ is contained in $C_G(x)$, and hence is a subgroup of it, for each $x \in G$.

By the orbit-stabiliser theorem, if G is finite then the number of elements in the conjugacy class of x is equal to $\dfrac{|G|}{|C_G(x)|}$ and hence this number divides $|G|$.

We have $x \in Z(G) \Leftrightarrow C_G(x) = G \Leftrightarrow$ the conjugacy class of x has order 1, so:

Proposition 2.9 *The centre of a group is the union of all the one-element conjugacy classes.*

Using this result and the orbit decomposition equation, we get the **class equation** for a finite group G: $\quad |G| = |Z(G)| + \displaystyle\sum_{x \in T} \frac{|G|}{|C_G(x)|}$ where the set T (a *transversal* of G) consists of one representative from each conjugacy class of order greater than 1.

Exercises 2.1 Question 17 proves the following result, for which we now give another proof using the class equation.

Proposition 2.10 (Cauchy's theorem) *Let p be a prime factor of the order of a group G. Then G has an element of order p.*

Proof

First suppose G is abelian, so by Proposition 1.10 G is isomorphic to a direct product of cyclic groups: say $G \cong C_1 \times C_2 \times \cdots C_r$. We may assume p divides $|C_1|$. As C_1 is cyclic, it has a subgroup of order p with generator g, say. Then $(g, e, \ldots, e)$ has order p in G.

Now suppose G is non-abelian. The result is certainly true for the smallest non-abelian group S_3, which has elements of orders 2 and 3. Make the inductive hypothesis that it holds for all groups of order $< |G|$.

We have $|G| = pm$ for some $m > 1$. The class equation gives

$|G| = |Z(G)| + \sum\limits_{x \in T} [G : C_G(x)]$. If $x \in T$ then clearly $C_G(x) \neq G$.

If p divides $[G : C_G(x)]$ for all $x \in T$ then p divides $|Z(G)|$. As $Z(G)$ is abelian, it has an element of order p as shown above.

Otherwise p divides $|C_G(x)|$ for some $x \in T$, so by the hypothesis there is an element of order p in $C_G(x)$ and hence in G. The result follows by induction. □

Proposition 2.11 *Let G be a non-abelian finite group. Then $G/Z(G)$ is not cyclic.*

Proof

Let $Z = Z(G)$. Suppose G/Z is cyclic, so it is generated by some coset aZ.

$(aZ)^k = a^k Z$ for all $k \in \mathbb{N}$, so $G/Z = \{Z, aZ, a^2Z, \ldots, a^{m-1}Z\}$ where $|G/Z| = m$.

G is the union of the disjoint cosets in G/Z, so every element of G is in one of $Z, aZ, \ldots, a^{m-1}Z$. Thus any two elements of G can be written as $a^k y, a^\ell z$ where $k, \ell \in \mathbb{N}$ and $y, z \in Z$.

Powers of a commute with each other, and y, z commute with all elements, so $(a^k y)(a^\ell z) = (a^\ell z)(a^k y)$. Thus G is abelian (and hence G/Z is in fact trivial).

It follows by contraposition that if G is non-abelian, G/Z is not cyclic. □

2.2.1 Examples

(i) Suppose G is a non-abelian group of order pq where p, q are prime. $Z(G)$ is a proper subgroup of G, so it has order $1, p$ or q. Thus $|G/Z(G)| = pq, q$ or p. Any group of prime order is cyclic so by Proposition 2.11 the only possibility is $|G/Z(G)| = pq$ and $|Z(G)| = 1$, i.e. $Z(G)$ is trivial.

(ii) Suppose $|G| = 21$. The order of a conjugacy class divides $|G|$, so it can be 1, 3, 7 or 21. But at least one class $\{e\}$ has order 1, so there is no class of order 21. If G is abelian, $Z(G) = G$ and all the classes have order 1. If G is non-abelian then $|Z(G)| = 1$ by (i) above, so all classes except $\{e\}$ have order 3 or 7.

Suppose there are m conjugacy classes of order 3 and n of order 7. They partition G so $3m+7n+1 = 21$, i.e. $3m + 7n = 20$. The only integer solution is $m = n = 2$.

A normal subgroup $H \lhd G$ can be defined as a subgroup which is closed under conjugation by all elements of G, i.e. $aha^{-1} \in H$ for all $a \in G$ and $h \in H$. Thus for any $h \in H$, all elements in the conjugacy class of h are in H, so:

Proposition 2.12 *Let H be a subgroup of a group G. Then $H \lhd G$ if and only if H is a union of conjugacy classes in G.*

2.2.2 Example

We saw above that if G is a non-abelian group of order 21, there is one conjugacy class $\{e\}$ of order 1, two of order 3 (say A_1, A_2) and two of order 7 (say B_1, B_2).

Any normal subgroup of G is a union of some of these disjoint classes, it must contain $\{e\}$ and its order must divide 21. Thus the only possible non-trivial normal subgroup of G is $\{e\} \cup A_1 \cup A_2$. *If* this is a subgroup then it is normal in G.

Warning: Not every union of conjugacy classes is necessarily a subgroup. For example, in S_3, $\{\iota\} \cup \{\tau, \sigma\tau, \sigma^2\tau\}$ is a union of conjugacy classes but is not a subgroup, as it is not closed: $(\sigma\tau)\tau = \sigma$, which is not in it.

As conjugation is an automorphism, if H is a subgroup of G and $g \in G$ then gHg^{-1} is also a subgroup of G and is isomorphic to H. We have $H \lhd G$ if and only if no other subgroup of G is conjugate to H. Furthermore, $H \lhd G$ if (but not only if) no other subgroup of G has the same order as H.

$g(H) = gHg^{-1}$ defines an action of G on the set of all subgroups of G (see Exercises 2.2 Question 9). The subgroups of G fall into disjoint orbits with respect to this action. If H and J are in the same orbit then they are **conjugate subgroups** of G, i.e. $J = gHg^{-1}$ for some $g \in G$.

For this action, G_H (the stabiliser of the element H in the set of all subgroups of G) equals $\{g \in G \ : \ gHg^{-1} = H\}$. By Proposition 2.1 this is a subgroup of G. Now $gHg^{-1} = H$ if and only if $gH = Hg$, so the stabiliser consists of all elements of G for which the left and right cosets of H coincide. It is given the following name:

Definition 2.6 *The* **normaliser** *of a subgroup H in a group G is the subgroup $N_G(H) := \{g \in G \ : \ gH = Hg\}$.*

Exercises 2.2

1. Using the tables on Pages 5 and 6 of these notes, find the centraliser and conjugacy class of (a) σ in S_3, (b) a in Q_4.

2. In D_4 (table on Page 5), conjugate σ by each element and hence find the conjugacy class of σ. Do the same for each element that is not in a conjugacy class that you have already found. Hence find all the conjugacy classes in D_4.

   Identify $Z(D_4)$ and all the normal subgroups of D_4.

3. Let G be a group of order p^n, where p is prime. Prove that the centre of G cannot have order p^{n-1}.

4. Prove that every subgroup of $Z(G)$ is a normal subgroup of G.

5. Let G be a finite non-abelian group. Show that $|Z(G)| \leq \frac{1}{4}|G|$.

   If $|Z(G)| = \frac{1}{n}|G|$, show that no conjugacy class in G has more than n elements.

6. Let G be a group acting on a finite set X. Suppose a and b are conjugate elements of G. Show that if $x \in \mathrm{Fix}(a)$ then $g(x) \in \mathrm{Fix}(b)$ for some $g \in G$. Hence show that $|\mathrm{Fix}(a)| = |\mathrm{Fix}(b)|$.

7. H and J are subgroups of a group G, such that $H \lhd G$ and $H \cap J = \{e\}$.

   For $g \in G, h \in H, j \in J$, let $g(hj) = ghg^{-1}j$.

   (a) Show that this rule defines a group action of G on HJ.

   (b) Show that $G_{hj} = C_G(h)$ and describe $\widetilde{G}_{HJ}$.

8. Find the possible numbers of conjugacy classes, and their orders, in a non-abelian group G of order 253. Deduce that if $N \lhd G$ then $|N| = 23$.

9. Show that $g(H) = gHg^{-1}$ (conjugation of subgroups by elements) is an action of a group G on the set of all its subgroups. Identify $\mathrm{Fix}(G)$ for this action.

10. Let H be a subgroup of a group G. Show that

    (a) $H \lhd N_G(H)$, (b) $H \lhd G$ if and only if $N_G(H) = G$,

    (c) the number of subgroups of G that are conjugate to H is $\dfrac{|G|}{|N_G(H)|}$,

    (d) If $H \neq G$ then G is not a union of subgroups conjugate to H.

2.3 Conjugacy in permutation groups

Proposition 2.13

Let H be a subgroup of S_n. Then either $H \subset A_n$ or $|H \cap A_n| = \frac{1}{2}|H|$.

Proof If all the permutations in H are even then clearly $H \subset A_n$.

Otherwise, let $\pi_1, \ldots, \pi_r$ be all the even permutations in H and let $\rho_1, \ldots, \rho_s$ be all the odd permutations in H. Then $\rho_1\pi_1, \ldots, \rho_1\pi_r$ are r distinct odd permutations in H, so $r \leq s$. Also $\rho_1\rho_1, \ldots, \rho_1\rho_s$ are s distinct even permutations in H, so $s \leq r$. Hence $r = s$ so exactly half the permutations in

H lie in A_n, i.e. $|H \cap A_n| = \dfrac{1}{2}|H|$. $\square$

In particular, taking $H = S_n$, Proposition 2.13 confirms that $|A_n| = \dfrac{1}{2}n!$

Recall the following from Groups & Rings. Let $\rho_1, \rho_2, \ldots, \rho_s$ be disjoint cycles in S_n, such that ρ_i has length k_i. Then the order of $\rho_1\rho_2\cdots\rho_s$ is $lcm(k_1, \ldots, k_s)$.

Definition 2.7 *Let $\sigma \in S_n$ be a permutation written as a product of disjoint cycles, including* 1-*cycles for fixed elements. Let m_i be the number of cycles of length i, for $1 \le i \le n$. The* **cycle type** *of σ is the expression $1^{m_1}2^{m_2}\cdots n^{m_n}$.*

The **cycle index** *of σ is the number of disjoint cycles: $cyc(\sigma) := \sum_{i=1}^{n} m_i$.*

When writing the cycle type, we normally include only the factors in which $m_i > 0$.

2.3.1 Examples

(i) The permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 3 & 2 & 5 & 1 & 8 & 7 & 6 \end{pmatrix} \in S_8$ is the product of disjoint cycles $(1\ 4\ 5)(2\ 3)(6\ 8)(7)$ (including a 1-cycle for the fixed element 7). σ has order $lcm(3, 2, 2, 1) = 6$, cycle type $1^1 2^2 3^1$ and cycle index 4.

(ii) A 5-cycle in S_9 can be written as $(a\ \ b\ \ c\ \ d\ \ e)(f)(g)(h)(i)$ so it has cycle type $1^4 5^1$ and cycle index 5. Its order is also 5.

(iii) The maximum order of any permutation in S_n is given by the largest lcm of natural numbers $k_1, \ldots, k_s$ such that $k_1 + \cdots + k_s = n$, i.e. $k_1, \ldots, k_s$ form a *partition* of n. Thus in S_{10}, the largest possible order of any element is $lcm(2, 3, 5) = 30$. Such an element has cycle type $2^1 3^1 5^1$ and cycle index 3.

No element of S_{10} can have order 28, since any natural numbers whose lcm is 28 will add up to more than 10.

Proposition 2.14 *Let ρ be a k-cycle $(i_1\ \ i_2\ \ \cdots\ \ i_k)$ in S_n and let π be any element of S_n. Then $\pi\rho\pi^{-1}$ is the k-cycle $(\pi(i_1)\ \ \pi(i_2)\ \ \cdots\ \ \pi(i_k))$.*

The proof is left as an exercise. To illustrate it, let $\rho = (1\ 3\ 4\ 6\ 7)$ and $\pi = (1\ 2)(4\ 5\ 6\ 8)$ in S_8. Then $\pi\rho\pi^{-1} = (\pi(1)\ \ \pi(3)\ \ \pi(4)\ \ \pi(6)\ \ \pi(7)) = (2\ 3\ 5\ 8\ 7)$.

Proposition 2.15 *Permutations are conjugate in S_n if and only if they have the same cycle type.*

Proof

Let $\sigma \in S_n$ be a product of disjoint cycles $\rho_1 \cdots \rho_s$ (including those of length 1).

Let π be any element of S_n. Then $\pi\sigma\pi^{-1} = \pi(\rho_1\cdots\rho_s)\pi^{-1} = (\pi\rho_1\pi^{-1})\cdots(\pi\rho_s\pi^{-1})$. By Proposition 2.14, $\pi\rho_i\pi^{-1}$ is a cycle of the same length as ρ_i. Thus $\pi\sigma\pi^{-1}$ has the same cycle type as σ.

Conversely, suppose $\tau = \rho'_1 \cdots \rho'_s$, where ρ'_i is a cycle of the same length as ρ_i for $i = 1, \ldots, s$. Construct a permutation π as follows: for each $j \in \{1, \ldots, n\}$, let ρ_r be the cycle in which j occurs. Let $\pi(j)$ be the number in the corresponding position in ρ'_r. Then $\pi\rho_r(j) = \rho'_r\pi(j)$, so $\pi\rho_r\pi^{-1} = \rho'_r$ for each r.

Thus $\pi\sigma\pi^{-1} = \pi(\rho_1 \cdots \rho_s)\pi^{-1} = (\pi\rho_1\pi^{-1}) \cdots (\pi\rho_s\pi^{-1}) = \rho'_1 \cdots \rho'_s = \tau$,

so σ and τ are conjugate in S_n. □

2.3.2 Example

By Proposition 2.15, the conjugacy classes in S_3 are $\{\iota\}$, $\{\sigma, \sigma^2\}$ and $\{\tau, \sigma\tau, \sigma^2\tau\}$, consisting of the elements of cycle types 1^3, 3^1, 1^12^1 respectively.

The only one-element conjugacy class is $\{\iota\}$, which is the centre of S_3.

Inspecting the unions of conjugacy classes, we see that those which are subgroups of S_3 are $\{\iota\}$, $\{\iota, \sigma, \sigma^2\}$ and S_3 itself. Hence (as we already knew) these are all the normal subgroups of S_3.

We saw that the conjugacy class of an element x has $\dfrac{|G|}{|C_G(x)|}$ elements so, for example, the centraliser of τ in S_3 has order $6/3 = 2$. This can be confirmed from the Cayley table: $C_{S_3}(\tau) = \{\iota, \tau\}$.

The proof of Proposition 2.15 gives a method for finding, in two-row notation, a permutation π which conjugates one to give another of the same cycle type. Simply write one permutation above the other in cycle notation, aligning the cycles of equal length. Rearrange the columns so that the top row is in order.

2.3.3 Example

$\sigma = (1\ \ 2\ \ 4)(3\ \ 5)$ and $\tau = (2\ \ 4\ \ 5)(1\ \ 3)$ both have cycle type 2^13^1, so they are conjugate in S_5.

Let $\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 1 & 5 & 3 \end{pmatrix}$. Then $\pi\sigma\pi^{-1} = \tau$. (Note: π is not unique.)

Proposition 2.16

Let σ be a permutation in S_n which has cycle type $1^{m_1}2^{m_2} \cdots n^{m_n}$. The number of elements in the conjugacy class of σ in S_n is $\dfrac{n!}{1^{m_1}2^{m_2} \cdots n^{m_n} m_1! m_2! \cdots m_n!}$.

The number of permutations in S_n that commute with σ is $1^{m_1}2^{m_2} \cdots n^{m_n} m_1! \cdots m_n!$

Proof

Let $\sigma = \rho_1 \cdots \rho_s$ as a product of disjoint cycles (including those of length 1). Then the conjugates of σ are all the products of disjoint cycles of the same lengths as $\rho_1, \ldots, \rho_s$.

The numbers $1, \ldots, n$ can be formed into such a product of disjoint cycles in $n!$ ways. However, these do not all represent different permutations.

Suppose σ has cycle type $1^{m_1}2^{m_2}\cdots n^{m_n}$. Each k-cycle can be written in k ways, giving k^{m_k} equivalent expressions. Also the order in which the k-cycles appear can be permuted in $m_k!$ ways. Hence the number of distinct elements in the conjugacy class of σ is $\dfrac{n!}{1^{m_1}2^{m_2}\cdots n^{m_n}m_1!m_2!\cdots m_n!}$.

This is $\dfrac{|S_n|}{|C_{S_n}(\sigma)|}$, so the number of permutations which commute with σ is thus
$|C_{S_n}(\sigma)| = 1^{m_1}2^{m_2}\cdots n^{m_n}m_1!m_2!\cdots m_n!$ □

2.3.4 Examples

(i) A 5-cycle in S_9 has cycle type $1^4 5^1$, so the the conjugacy class containing the 5-cycles has $\dfrac{9!}{1^4 5^1 4!} = 3024$ elements. This is the number of 5-cycles in S_9. There are $5.4! = 120$ elements in S_9 which commute with any given 5-cycle.

(ii) Suppose we want to find the number of elements of S_6 with cycle index 3. These can have the form $(a)(b)(c\ \ d\ \ e\ \ f)$, $(a)(b\ \ c)(d\ \ e\ \ f)$, $(a\ \ b)(c\ \ d)(e\ \ f)$, so the cycle types are $1^2 4^1, 1^1 2^1 3^1, 2^3$. Thus the required number is $\dfrac{720}{4.2!} + \dfrac{720}{2.3} + \dfrac{720}{2^3.3!} = 225$.

(iii) Find the conjugacy classes in S_5.

Cycle Type	Example	Odd/Even	|centraliser|	|conjugacy class|
5^1	$(1\ 2\ 3\ 4\ 5)$	Even	5	24
$2^1 3^1$	$(1\ 2)(3\ 4\ 5)$	Odd	6	20
$1^1 4^1$	$(1)(2\ 3\ 4\ 5)$	Odd	4	30
$1^1 2^2$	$(1)(2\ 3)(4\ 5)$	Even	$4.2! = 8$	15
$1^2 3^1$	$(1)(2)(3\ 4\ 5)$	Even	$3.2! = 6$	20
$1^3 2^1$	$(1)(2)(3)(4\ 5)$	Odd	$2.3! = 12$	10
1^5	$(1)(2)(3)(4)(5)$	Even	$5! = 120$	1
				120

S_5 is the union of the seven conjugacy classes. There is only one 1-element class, so $|Z(S_6)| = 1$.

The possible orders of normal subgroups of S_5 (unions of conjugacy classes whose order divides 120) are 1, $1 + 15 + 24 = 40$, $1 + 15 + 20 + 24 = 60$, and 120.

The 60-element subgroup is A_5. It can be shown that S_5 has no 40-element normal subgroup.

Suppose H is a subgroup of a group G. If elements x and y are conjugate in H, i.e. $axa^{-1} = y$ for some $a \in H$, then they are certainly conjugate in G since $a \in G$. The converse is not true in general; elements of H may be conjugate in G but not in H.

For any $\sigma \in S_n$ we have $C_{A_n}(\sigma) = C_{S_n}(\sigma) \cap A_n$. By Proposition 2.13 either $C_{S_n}(\sigma) \subset A_n$, in which case $|C_{A_n}(\sigma)| = |C_{S_n}(\sigma)|$, or $|C_{A_n}(\sigma)| = \dfrac{1}{2}|C_{S_n}(\sigma)|$.

It follows from the orbit-stabiliser equation that the conjugacy class of σ in A_n is either equal to that in S_n or is half the size. That is, each conjugacy class in S_n is either a single conjugacy class in A_n or the union of two equal-sized conjugacy classes in A_n.

Hence the possible orders of the conjugacy classes in A_5 are $24, 20, 15, 1,\quad 24, 10, 10, 15, 1,$ $12, 12, 10, 15, 1,\quad 12, 12, 10, 10, 15, 1$. In no case can we find a union of conjugacy classes, including the trivial class, whose order divides 60. Thus A_5 has no non-trivial proper normal subgroups.

Exercises 2.3

1. (a) Show that a k-cycle is even if and only if k is odd.

   (b) Show that a permutation $\sigma \in S_n$ is even if and only if $n - cyc(\sigma)$ is even.

   (c) In how many ways can a given k-cycle be expressed in cycle notation? (e.g. $(1\ \ 2\ \ 3) = (2\ \ 3\ \ 1) \neq (1\ \ 3\ \ 2)$.)

   (d) Let $1 \leq k \leq n$. How many distinct k-cycles are there in S_n? In A_n?

2. In S_n, let ρ be the k-cycle $(i_1\quad i_2\quad \cdots\quad i_k)$ and π any permutation. Let $i'_j = \pi(i_j)$. Find $\pi\rho\pi^{-1}(i'_j)$ in the cases $1 \leq j < k$ and $j = k$. Hence prove Proposition 2.14.

   Use this result to find $\pi\rho\pi^{-1}$ when $\rho = (1\quad 3\quad 6\quad 4\quad 2\quad 5)$ and $\pi = (1\quad 4\quad 6\quad 2)(3\quad 5)$, in S_6.

3. Determine whether $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 1 & 5 & 4 & 7 & 6 & 8 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 8 & 2 & 1 & 7 & 6 & 5 & 3 \end{pmatrix}$ are conjugate elements in S_8.

4. Write down a permutation π such that $\pi\sigma\pi^{-1} = \tau$ in each of the following cases:

   (a) $\sigma = (1\ \ 3\ \ 5\ \ 4)$, $\tau = (1\ \ 2\ \ 4\ \ 6)$ in S_6.

   (b) $\sigma = (1\ \ 3)(4\ \ 5\ \ 6)$, $\tau = (1\ \ 3\ \ 6)(2\ \ 4)$ in S_6.

   (c) $\sigma = (1\ \ 4)(2\ \ 5)(3\ \ 7)$, $\tau = (2\ \ 4)(3\ \ 5)(6\ \ 7)$ in S_7.

   Find the order, cycle type and cycle index of π in each case.

5. Find the largest possible order of an element in S_n for each of $n = 1, \ldots, 12$. Find out about Landau's function.

6. Show that S_{10} has a subgroup of order 24, but no cyclic subgroup of this order.

7. List all the elements of A_4, stating the order of each one. Show that A_4 has no subgroup isomorphic to C_6 or S_3, hence no subgroup of order 6.

8. Show that $(1\ \ 2\ \ 3)$ and $(1\ \ 3\ \ 2)$ are not conjugate in A_4, as follows: suppose $\pi(1\ \ 2\ \ 3)\pi^{-1} = (1\ \ 3\ \ 2)$. Then $(\pi(1)\quad \pi(2)\quad \pi(3)) = (1\ \ 3\ \ 2)$, which can be written in two other ways. Show that π must be a transposition, so $\pi \notin A_4$.

9. List all the possible cycle types in S_4 and find the number of elements of each cycle type. State the order of each conjugacy class in S_4. Hence find the possible orders of normal subgroups of S_4. Also find the orders of the centralisers.

10. (a) Find the number of elements of S_7 with cycle index 4.

    (b) Let p be prime. Show that the only elements of S_p that commute with a p-cycle are the powers of this p-cycle.

11. Let $n \geq 3$. Show that every 3-cycle in S_n is in A_n. By considering products of pairs of transpositions of the form $(i \quad j)(i \quad k)$ or $(i \quad j)(k \quad \ell)$, show that every element of A_n is a product of 3-cycles.

12. (a) Express $(1 \;\; i)(1 \;\; j)(1 \;\; i)$ as a single transposition. Deduce that S_n is generated by the set of all transpositions of the form $(1 \quad i)$ and state the order of this set.

    (b) Let ρ be the $(n-1)$-cycle $(2 \;\; \cdots \;\; n)$ and τ the transposition $(1 \;\; 2)$.
    Show that $\rho^r \tau \rho^{-r} = (1 \quad r+2)$ for $r = 0, \ldots, n-2$. Deduce that S_n has a two-element generating set.

2.4 Burnside's formula

Combinatorics is the branch of mathematics that deals with counting the number of arrangements and selections of elements of a set. You should recognise some basic results: given a set with n distinct elements, the number of combinations (unordered selections) of r elements is given by the binomial coefficient $\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$. The number of permutations (ordered selections) of r elements out of n is equal to $\dfrac{n!}{(n-r)!}$.

Proposition 2.17 (Burnside's formula) *For a finite group G acting on a finite set X, the number of distinct orbits is* $\dfrac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)|$.

Proof

Let $S = \{(g, x) \in G \times X \;:\; g(x) = x\}$. Then $|S| = \sum_{g \in G} |\operatorname{Fix}(g)| = \sum_{x \in X} |G_x|$,

so $\dfrac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)| = \sum_{x \in X} \dfrac{|G_x|}{|G|} = \sum_{x \in X} \dfrac{1}{|\mathcal{O}(x)|}$ by the Orbit-Stabiliser theorem.

For each orbit, the term $\dfrac{1}{|\mathcal{O}(x)|}$ occurs $|\mathcal{O}(x)|$ times in this sum, which is thus equal to $1+1+\cdots+1$ where 1 occurs as many times as there are distinct orbits in X. Hence the sum equals the number of orbits. $\square$

This formula says that the number of orbits is the mean number of elements of X fixed by an element of G. It is generally agreed that this result should not be attributed to William Burnside (1852 - 1927), but his name has become attached to it.

Burnside's formula can be used to solve combinatorial problems in ways which are often shorter than direct enumeration of all cases. We consider colouring problems of the following kind: a figure in two or three dimensions is to have all its vertices, edges or faces marked using a set of colours. A particular way of doing this is called an **allocation** of colours. It is normally assumed that each colour may be used any number of times.

If q different colours are available, the distinguishable ways of colouring the figure are called **q-colourings**.

If one allocation can be obtained from the other by a symmetry of the figure then the allocations are regarded as indistinguishable, i.e. they are the same q-colouring,

2.4.1 Example

Label the edges of a square 1, 2, 3, 4. Suppose each edge is to be coloured blue, green or red. The allocations (1 blue, 2 green, 3 red, 4 green) and (1 green, 2 red, 3 green, 4 blue) are regarded as the same 3-colouring, as rotating the square maps one onto the other. However, (1 blue, 2 red, 3 green, 4 green) is a different 3-colouring.

The symmetry group of the figure acts on the set of all possible allocations of colours. Two allocations correspond to the same q-colouring if and only if they are in the same orbit, so a symmetry maps one to the other. Thus the number of q-colourings equals the number of orbits.

To find this number by Burnside's formula, we need to know the size of the fixed set of each symmetry. Labelling the features to be coloured as $1, \dots, n$, the symmetries can be expressed as permutations of $\{1, \dots, n\}$ so they form a subgroup of S_n.

A product of disjoint cycles fixes precisely those allocations in which all points involved in the same cycle have the same colour. Thus $|\mathrm{Fix}(\sigma)| = q^{cyc(\sigma)}$, so:

Proposition 2.18

Suppose n features of a figure are to be coloured using q colours. Let $G \subset S_n$ be the symmetry group that acts. Then the number of distinct q-colourings is $\dfrac{1}{|G|} \sum\limits_{\sigma \in G} q^{cyc(\sigma)}$.

2.4.2 Example

A regular hexagon is drawn in a plane and each vertex is coloured red, blue or green. Each colour may be used any number of times.

In how many distinguishable ways can this be done if

(a) the hexagon can only be rotated in its plane, (b) it can also be turned over?

(c) Also find the number of colourings if eight colours are available and each colour may be used only once.

(a) The symmetries in the plane are rotations ρ^k through $\dfrac{2k\pi}{6}$ for $k = 0, \dots, 5$, forming the group C_6. This group acts on the set of vertices and on the set of all 3^6 allocations of colours to the vertices.

ρ^0 is the identity map ι, which fixes all 3^6 allocations. As a product of cycles it is $(1)(2)(3)(4)(5)(6)$, which has cycle index 6.

$\rho = (1\ \ 2\ \ 3\ \ 4\ \ 5\ \ 6)$ and $\rho^5 = (1\ \ 6\ \ 5\ \ 4\ \ 3\ \ 2)$ have cycle index 1. They each fix 3 allocations: all vertices red, all blue or all green.

$\rho^2 = (1\ \ 3\ \ 5)(2\ \ 4\ \ 6)$ and $\rho^4 = (1\ \ 5\ \ 3)(2\ \ 6\ \ 4)$ have cycle index 2. They each fix $3^2 = 9$ allocations: all those where alternate vertices are the same colour.

$\rho^3 = (1\ \ 4)(2\ \ 5)(3\ \ 6)$ has cycle index 3 and fixes the $3^3 = 27$ allocations in which opposite vertices have the same colour.

Thus the number of 3-colourings is $\dfrac{1}{6}(3^6 + 2(3^1) + 2(3^2) + 3^3) = 130$.

(b) Turning the hexagon over corresponds to reflecting it in a line of symmetry. The symmetry group is now D_6, which has order 12. The three reflections in diagonals, such as $(1)(4)(2\ \ 6)(3\ \ 5)$, each have cycle index 4. The three reflections in lines through mid-points of opposite sides, such as $(1\ \ 2)(3\ \ 6)(4\ \ 5)$, have cycle index 3.

Adding these to the results obtained before, the number of orbits (distinguishable 3-colourings) is now $\dfrac{1}{12}(3^6 + 2(3^1) + 2(3^2) + 3^3 + 3(3^4) + 3(3^3)) = 92$.

(c) If eight colours are available and each can be used only once, the number of ways of assigning colours to vertices is $n = 8 \times 7 \times 6 \times 5 \times 4 \times 3$

The identity fixes all of these and each other symmetry fixes none, so the numbers of distinct colourings are (a) $\dfrac{1}{6}n = 3360$, (b) $\dfrac{1}{12}n = 1680$.

2.4.3 Example

A cuboid has square ends of side 5 cm. Its length is 10 cm. Find the number of distinct q-colourings of (a) its faces, (b) its vertices.

First note that there are eight symmetries of the cuboid, since any of the four long faces can be taken as the base and then rotated through 180°, or not, in its plane. Thus the symmetry group that acts on the cuboid has order 8. We consider this group as a subgroup of S_n, where n is the number of features to be coloured so $n = 6$ for the faces and $n = 8$ for the vertices.

(a) Label the long faces 1 to 4 (with 1 at the top, 3 at the bottom) and the square ends 5 and 6. Then the eight symmetries, as permutations in S_6, are:

$(1)(2)(3)(4)(5)(6)$, $(1\ \ 2\ \ 3\ \ 4)(5)(6)$, $(1\ \ 3)(2\ \ 4)(5)(6)$, $(1\ \ 4\ \ 3\ \ 2)(5)(6)$,

$(1)(3)(2\ \ 4)(5\ \ 6)$, $(1\ \ 3)(2)(4)(5\ \ 6)$, $(1\ \ 2)(3\ \ 4)(5\ \ 6)$, $(1\ \ 4)(2\ \ 3)(5\ \ 6)$.

These have cycle indices $6, 3, 4, 3, 4, 4, 3, 3$, so the number of q-colourings of the faces is $\dfrac{1}{8}(q^6 + 3q^4 + 4q^3)$.

(b) Label the vertices 1 to 4 around one square end and 5 to 8 around the other, so the long edges join 1 to 5, etc. Then the eight symmetries, as permutations in S_8, are:

$(1)(2)(3)(4)(5)(6)(7)(8)$, $(1\ \ 2\ \ 3\ \ 4)(5\ \ 6\ \ 7\ \ 8)$, $(1\ \ 3)(2\ \ 4)(5\ \ 7)(6\ \ 8)$, $(1\ \ 4\ \ 3\ \ 2)(5\ \ 8\ \ 7\ \ 6)$,

$(1\ \ 5)(2\ \ 8)(3\ \ 7)(4\ \ 6)$, $(1\ \ 6)(2\ \ 5)(3\ \ 8)(4\ \ 7)$, $(1\ \ 7)(2\ \ 6)(3\ \ 5)(4\ \ 8)$, $(1\ \ 8)(2\ \ 7)(3\ \ 6)(4\ \ 5)$. These have cycle indices $8, 2, 4, 2, 4, 4, 4, 4$, so the number of q-colourings of the edges is $\frac{1}{8}(q^8 + 5q^4 + 2q^2)$.

Exercises 2.4

1. Deduce from Burnside's formula that the number of conjugacy classes in a finite group G is $\frac{1}{|G|}\sum_{g\in G}|C_G(g)|$. Verify this for S_3.

2. A hexagon $ABCDEF$ has all its sides equal in length. The interior angles at A, B, D and E are each $150°$ and and those at C and F are each $60°$. The hexagon is drawn on transparent plastic and cut out.

   Find the number of q-colourings of (a) the vertices of the hexagon,
   (b) the edges of the hexagon,
   (c) the six triangles formed by joining AD, AE, BD and BE.

3. A regular pentagon has each vertex coloured in one of q colours, repeated colours being allowed. What is the number of q-colourings if the pentagon can be (a) rotated (in its plane) but not turned over,
   (b) rotated and/or turned over.

4. Each face of a regular tetrahedron (triangular pyramid) is to be painted blue, green, red or yellow. Any colour can be used on any number of faces. You are given that the group of rotations of the tetrahedron is the alternating group A_4. By considering the action of this group, find the number of 4-colourings.

   What is the number of colourings if each colour can be used only once?

5. If each of q colours may be used any number of times, find the number of q-colourings of
   (a) the faces, (b) the vertices, (c) the edges
   of a triangular prism.

6. Find the number of q-colourings of the edges of the cuboid in Example 2.4.3.

7. A square piece of transparent plastic can be rotated in its plane or turned over. It is divided into regions in such a way that the number of q-colourings is $\frac{1}{4}(q^6 + q^5 + q^4 + q^3)$. Find the number of regions. Sketch the figure and identify its symmetry group.

2.5 p-groups

Definition 2.8 *Let p be prime. A* **p-group** *is a group of order p^k for some $k \in \mathbb{N}$. A* **p-subgroup** *is a subgroup which is a p-group.*

Proposition 2.19 *Let G be a p-group.*

(i) If G acts on a finite set X then $|\mathrm{Fix}(G)| \equiv |X|$ (mod p).

(ii) $Z(G)$ is non-trivial.

Proof

(i) Fix(G) is the union of the one-element orbits. Suppose $x \in X$ is such that $|\mathcal{O}(x)| > 1$. Then G_x is a proper subgroup of G so $[G : G_x] = p^\ell$ for some $\ell \in \mathbb{N}$. By the orbit decomposition theorem, $|X| = |\mathrm{Fix}(G)| + \sum[G : G_x]$ so p divides $|X| - |\mathrm{Fix}(G)|$, i.e. $|\mathrm{Fix}(G)| \equiv |X|$ (mod p).

(ii) For the action of G on itself by conjugation, Fix(G) $= Z(G)$ so by (i), $|Z(G)| \equiv p^k$ (mod p). Hence p divides $|Z(G)| - p^k$, so p divides $|Z(G)|$, so $|Z(G)| > 1$. □

Proposition 2.20 *Let H be a p-subgroup of a finite group G. Then $[N_G(H) : H] \equiv [G : H]$* (mod p).

Proof

H acts on the set G/H of left cosets by $h(aH) = haH$. For this action we have

$$\begin{aligned} aH \in \mathrm{Fix}(H) &\Leftrightarrow haH = aH \text{ for all } h \in H \\ &\Leftrightarrow a^{-1}Ha = H \Leftrightarrow a \in N_G(H). \end{aligned}$$

Thus Fix(H) $= \{aH \ : \ a \in N_G(H)\} = N_G(H)/H$, since $H \lhd N_G(H)$ by Exercises 2.2 Question 10 (a).

Hence, using Proposition 2.19 (i), $[N_G(H) : H] = |\mathrm{Fix}(H)| \equiv_p |G/H| = [G : H]$. □

Definition 2.9 *Let G be a finite group of order $p^k m$, where p is prime and $p \nmid m$.*

A **p-Sylow** *(or* **Sylow p-subgroup***) of G is a subgroup of G which has order p^k.*

We shall denote the number of p-Sylows in a group by w_p.

Proposition 2.21 (The Sylow theorems)

Suppose $|G| = p^k m$, where p is prime and $p \nmid m$. Then

(i) G has at least one p-Sylow,

(ii) every p-subgroup of G is contained in a p-Sylow of G,

(iii) all p-Sylows of G are conjugate,

(iv) w_p, the number of p-Sylows of G, is congruent to 1 (modulo p) and divides m,

(v) G has a subgroup of index w_p, namely the normaliser of any p-Sylow of G.

Proof

(i) and (ii): By Cauchy's theorem, G has at least one element of order p. This element generates a subgroup of order p.

Let H be any subgroup of order p^ℓ where $1 \leq \ell < k$. Then p divides $[G : H]$, so by Proposition 2.20 p divides $[N_G(H) : H]$. Thus $N_G(H)/H$ has a subgroup of order p. By Proposition 1.7 this must have the form J/H where $H \lhd J \subset N_G(H)$, so $|J| = p|H| = p^{\ell+1}$. It follows that G has subgroups of order $p, p^2, \dots, p^k$ and that each of these is contained in a subgroup of order p^k, i.e. a p-Sylow of G.

(iii) Suppose P and P' are p-Sylows of G. P' acts on the set G/P of all m left cosets of P by $g(aP) = gaP$. By Proposition 2.19 (i), $|\mathrm{Fix}(P')| \equiv m \pmod p$ so $|\mathrm{Fix}(P')| \neq 0$. Hence there is an element $bP \in G/P$ such that for all $g \in P'$ we have $gbP = bP$, so $b^{-1}gb \in P$, so $g \in bPb^{-1}$. Thus P' is contained in bPb^{-1}. But also $|P| = |P'|$ so $P' = bPb^{-1}$, i.e. P and P' are conjugate.

(iv) Let w_p be the number of p-Sylows. This is the number of subgroups of G that are conjugate to any p-Sylow P, so by Exercises 2.2 Question 10 (c) $w_p = [G : N_G(P)]$. Let $q = [N_G(P) : P]$, so $qw_p = m$, so w_p divides m. By Proposition 2.20, $m = q + kp$ for some $k \in \mathbb{N}$, so $w_p = 1 + \dfrac{k}{q}p$. Now $p \nmid m$ so $p \nmid q$, so $\dfrac{k}{q} \in \mathbb{Z}$. Thus $w_p \equiv_p 1$.

(v) We have seen that $w_p = [G : N_G(P)]$, i.e. $N_G(P)$ has index w_p in G. □

2.5.1 Example

Suppose $|G| = 600 = 2^3 \times 3 \times 5^2$. Then G has 2-Sylows, 3-Sylows and 5-Sylows.

The 2-Sylows have order 8. By Proposition 2.21 (iii), $w_2 \equiv 1 \pmod 2$ and $w_2 \mid 75$. Hence w_2 is one of 1, 3, 5, 15, 25 or 75.

The 3-Sylows have order 3. $w_3 \equiv 1 \pmod 3$ and $w_3 \mid 200$. Hence $w_3 = 1$, 4, 10, 25, 40 or 100.

The 5-Sylows have order 25. $w_5 \equiv 1 \pmod 5$ and $w_5 \mid 24$. Hence $w_5 = 1$ or 6.

If $w_5 = 6$ then G has at least one subgroup of index 6, i.e. order 100, namely the normaliser of a 5-Sylow of G.

Proposition 2.22 *Let G be a finite group and let $p_1, \dots, p_r$ be the distinct primes that divide $|G|$.*

(i) G has exactly one p_i-Sylow if and only if that subgroup is normal in G.

(ii) If G is abelian, it has only one p_i-Sylow for each i.

(iii) If G has just one p_i-Sylow for each i then G is the internal direct product of its p_i-Sylows.

Proof

(i) By Proposition 2.21 (ii), G has only one p_i-Sylow if and only if the p_i-Sylow has no conjugates except itself, which holds if and only if the p_i-Sylow is normal in G.

(ii) If G is abelian, every subgroup of G is normal so the p_i-Sylow is unique.

(iii) Suppose there is only one p_i-Sylow P_i for each i. Then each P_i is normal in G by (i). As the orders of the P_i are coprime, $(P_1 \cdots P_{i-1}) \cap P_i$ is trivial for $i = 2, \dots, r$.

Also $|G| = |P_1| \cdots |P_r|$, which equals $|P_1 \cdots P_r|$ by repeated use of Proposition 1.4.

Hence by Proposition 1.6, G is the internal direct product of $P_1, \dots, P_r$, so $G \cong P_1 \times \cdots \times P_r$. □

2.5.2 Example

Suppose $|G| = 99 = 3^2 \times 11$. Then G has 3-Sylows and 11-Sylows.

The 3-Sylows have order 9. $w_3 \equiv 1 \pmod 3$ and $w_3 \mid 11$. Hence $w_3 = 1$ so the 3-Sylow is normal in G. Call it H. By Proposition 2.8, H is abelian

The 11-Sylows have order 11. $w_{11} \equiv 1 \pmod{11}$ and $w_{11} \mid 9$. Hence $w_{11} = 1$ so the 11-Sylow is normal in G. Call it J. As 11 is prime, J is abelian.

As 9 and 11 are coprime, $H \cap J = \{e\}$ so by Proposition 1.5 $G = HJ \cong H \times J$.

Thus all groups of order 99 are abelian, and isomorphic to either $C_9 \times C_{11}$ or $C_3 \times C_3 \times C_{11}$.

Proposition 2.23 *Let p and q be distinct primes, where $p > q$. Then every abelian group of order pq is cyclic.*

A non-abelian group of order pq exists if and only if $q \mid (p-1)$, in which case the group is unique up to isomorphism and has one p-Sylow and p q-Sylows.

Proof

By Proposition 1.10 the only abelian groups of order pq are $C_p \times C_q$ and C_{pq}, but these are isomorphic since p and q are coprime.

The number w_p of p-Sylows in G is congruent to 1 (mod p) and divides q. As $q < p$, we must have $w_p = 1$. If also $w_q = 1$ then $G \cong C_p \times C_q \cong C_{pq}$.

Thus if G is non-abelian, $w_q > 1$ and $w_q \mid p$, so $w_q = p$. Also $w_q \equiv 1 \pmod q$ so $p = cq + 1$ for some $c \in \mathbb{N}$. Thus $q \mid (p-1)$.

Let a and b be elements of order p, q respectively in G. $\langle a \rangle$ has index q, so is normal in G by Proposition 2.7; thus $bab^{-1} = a^m$ for some m. Then $a = b^q a b^{-q} = a^{m^q}$ by Proposition 1.2 (ii), so $a^{m^q - 1} = e$, so $p \mid m^q - 1$, i.e. $m^q \equiv 1 \pmod p$.

As $q \mid (p-1)$, there is an element of order q in $(\mathbb{Z}_p{}^*, \times_p)$ so a value of $m \neq 1$ exists. Then G has presentation $\langle a, b \ : \ a^p = b^q = e,\ ba = a^m b \rangle$. Although m is not unique, it can be shown that all possible values of m give isomorphic groups. $\square$

When $q \mid (p-1)$, a non-abelian group of order pq can be constructed as follows. Let H be the subgroup of order q in $\mathbb{Z}_p{}^*$. Let $G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \ : \ a \in H, b \in \mathbb{Z}_p \right\}$ under matrix multiplication modulo p. Then $|G| = pq$ and we can easily check that G is a non-abelian subgroup of the group of units of $M_2(\mathbb{Z}_p)$.

2.5.3 Examples

(i) $15 = 3 \times 5$ and $3 \nmid (5-1)$ so, up to isomorphism, there is only one group of order 15, namely the cyclic group C_{15}.

(ii) $21 = 3 \times 7$. Every abelian group of order 21 is cyclic.

$3 \mid (7-1)$ so there is a non-abelian group of order 21. This has one normal subgroup of order 7 and seven subgroups of order 3.

This group can be represented by the matrices $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$ where $a \in \{1, 2, 4\}$ and $b \in \{0, 1, \ldots, 6\}$ under matrix multiplication modulo 7. The subgroup of order 7 is generated by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. The subgroups of order 3 are generated by the matrices $\begin{pmatrix} 2 & b \\ 0 & 1 \end{pmatrix}$ for $b = 0, \ldots, 6$.

(iii) Suppose $|G| = 363 = 3 \times 11^2$. $w_3 \equiv 1 \pmod 3$ and $w_3 \mid 121$, so $w_3 = 1$ or 121.

$w_{11} \equiv 1 \pmod{11}$ and $w_{11} \mid 3$, so $w_{11} = 1$. Hence the 11-Sylow H (of order 121) is normal in G.

If $w_3 = 1$ then the 3-Sylow J is also normal in G. Then $G \cong H \times J$, giving the two abelian possibilities $C_3 \times C_{121}$ and $C_3 \times C_{11} \times C_{11}$.

It follows that *if* a non-abelian group of order 363 exists, it has 121 3-Sylows. These contain a total of 243 elements, leaving room for the one 11-Sylow.

Such a group can be constructed as follows. Recall from Groups & Rings that for any prime p and any $n \in \mathbb{N}$, there exists a finite field $\mathbb{F}_{p^n}$ having p^n elements. (If $n > 1$, this field is *not* the integers modulo p^n.) All the non-zero elements are units of $\mathbb{F}_{p^n}$ and form a cyclic multiplicative group $\mathbb{F}_{p^n}{}^*$, of order $p^n - 1$.

$|\mathbb{F}_{121}{}^*| = 120$ so $\mathbb{F}_{121}{}^*$ has a subgroup H of order 3.

Let $G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \ : \ a \in H, b \in \mathbb{F}_{121} \right\}$. Then G is a non-abelian subgroup of $GL(2, \mathbb{F}_{121})$ and $|G| = 363$.

The following result is often useful in conjunction with the Sylow theorems:

Proposition 2.24 *Suppose a group G has a cyclic normal subgroup H and a subgroup J such that $\varphi(|H|)$ and $|J|$ are coprime. Then $hj = jh$ for all $h \in H, j \in J$.*

In particular, if H is a cyclic normal subgroup of G and $\varphi(|H|)$ is coprime to $|G|$ then $H \subset Z(G)$.

Proof

For each $a \in G$, the map $\gamma_a : x \mapsto axa^{-1}$ is an automorphism of H. Define $\theta : J \to \mathrm{Aut}(H)$ by $\theta : j \mapsto \gamma_j$ for all $j \in J$. Then for all $j, k \in J$, $\theta(jk) = \gamma_{jk} = \gamma_j \gamma_k = \theta(j)\theta(k)$, so θ is a group homomorphism and thus $\theta(J)$ is a subgroup of $\mathrm{Aut}(H)$.

As H is cyclic, $|\mathrm{Aut}(H)| = \varphi(|H|)$ so $|\theta(J)|$ divides $\varphi(|H|)$.

Also $J/\mathrm{Ker}\,\theta \cong \theta(J)$ so $|\theta(J)|$ also divides $|J|$.

Hence if $\varphi(|H|)$ and $|J|$ are coprime then $|\theta(J)| = 1$, so γ_j is the identity map on H for all $j \in J$. Thus $jhj^{-1} = h$, i.e. $jh = hj$, for all $h \in H$ and $j \in J$.

If $J = G$, we have $gh = hg$ for all $h \in H, g \in G$ so $H \subset Z(G)$. □

2.5.4 Example

Suppose $|G| = 297 = 3^3 \times 11$. Then $w_{11} \equiv 1 \pmod{11}$ and $w_{11} \mid 27$, so $w_{11} = 1$.

Hence G has a normal subgroup H of order 11, which is cyclic since 11 is prime.

$\varphi(11) = 10$, which is coprime to 297, so by Proposition 2.24 $H \subset Z(G)$, i.e. every element of H commutes with every element of G.

2.5.5 Example

Let G be a group of order $255 = 3 \times 5 \times 17$. $w_{17} \equiv 1 \pmod{17}$ and $w_{17} \mid 15$. Thus $w_{17} = 1$, so G has a normal subgroup H of order 17. As 17 is prime, H is cyclic.

$|G/H| = 15$ so, as in Examples 2.5.3 (i), G/H is cyclic. Thus by Proposition 1.8, G has a subgroup $J \cong C_{15}$.

$\varphi(17) = 16$, which is coprime to 15 so $hj = jh$ for all $h \in H, j \in J$. As H and J have coprime orders, $|H \cap J| = 1$ so $|G| = |HJ|$, so $G = HJ$. For any $g = hj \in G$ and $j' \in J$ we have $gj'g^{-1} = hjj'j^{-1}h^{-1} = jj'j^{-1} \in J$, so $J \lhd G$. Hence by Proposition 1.5, $G \cong H \times J \cong C_{17} \times C_{15} \cong C_{255}$.

Thus, up to isomorphism, the only group of order 255 is the cyclic group C_{255}.

Exercises 2.5

1. For each prime p that divides the order of a group G, state the orders of its p-Sylows and the possible numbers of them when $|G| =$

   (a) 75, (b) 126, (c) 175, (d) 400.

2. Show that every group of order 133 is abelian. How many 7-Sylows has it?

3. G is a group of order 45. Show that G has normal subgroups of orders 5 and 9. Hence show that, up to isomorphism, there are just two groups of order 45 and both are abelian.

4. In S_7, let $\sigma = (1\ 2\ 3\ 4\ 5\ 6\ 7)$. Find τ such that $\tau\sigma\tau^{-1} = \sigma^2$. Show that the group generated by $\{\sigma, \tau\}$ is non-abelian, of order 21. Give a presentation of this group.

5. Find a non-abelian group of matrices which has order 711. Hence show that there are at least three non-isomorphic groups of this order. Find a normal subgroup of the matrix group.

6. G is a group of order $1617 = 3 \times 7^2 \times 11$. Show that G has a normal subgroup J of order 49 and that G/J is cyclic. Use Proposition 1.8 to deduce that G has a subgroup H isomorphic to C_{33}. Show that if this subgroup is normal in G then it is contained in $Z(G)$, and that $G \cong H \times J$ in this case.

7. G is a group of order $315 = 3^2 \times 5 \times 7$. Suppose G has exactly one subgroup, N, of order 7. Show that G/N has a normal subgroup of order 9 and hence that G has a normal subgroup of order 63.

8. Show that up to isomorphism there is only one group of order 455, and this group is cyclic.

9. G is a group of order $p^3 - p$, where p is prime. Show that the number of p-Sylows of G is 1 or $p+1$. If it is $p+1$, show that G has a subgroup of order $p^2 - p$.

10. G is a non-abelian group of order $18 = 2 \times 3^2$.

    (a) Give a reason why G has a normal subgroup N of order 9 and an element a of order 2.

    (b) Suppose N is cyclic with generator b. Show that $aba^{-1} = b^r$ for some $r \in \{0, \dots, 8\}$. Deduce that $b = b^{(r^2)}$ and hence find the value of r. Give a presentation of G and state the usual name for this group.

    (c) By considering orders of elements of G, show that $|\mathrm{Aut}(G)| = 54$. Given that $Z(G) = \{e\}$, use Exercises 1.5 Question 4 to find $|\mathrm{Inn}(G)|$.

2.6 Simple Groups

Definition 2.10 *A* **simple group** *is a non-trivial group which has no non-trivial proper normal subgroups.*

Proposition 2.25 *A finite abelian group is simple if and only if its order is prime.*

Proof

Suppose G is abelian. Let a be a non-identity element of G. If a does not generate G then $\langle a \rangle$ is a proper normal subgroup of G, so G is not simple.

If $G = \langle a \rangle$ then G is cyclic so it has a normal subgroup of every order dividing $|G|$.

Thus G is not simple unless its order is a prime p, in which case $G \cong C_p$ which certainly *is* simple. $\square$

For each $n \in \mathbb{N}$, the symmetric group S_n has the normal subgroup A_n so S_n is not a simple group.

We saw in Examples 1.3.6 (ii) that $D_2 = \{(1), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ is a normal subgroup of S_4. Clearly $D_2 \subset A_4$, so $D_2 \lhd A_4$. Thus A_4 has a non-trivial normal subgroup, so A_4 is not a simple group.

Proposition 2.26 *Let H be a subgroup of S_n, such that $|H| > 2$. If H is a simple group then $H \subset A_n$.*

Proof

If $H \not\subset A_n$ then by Proposition 2.13, $H \cap A_n$ is a subgroup of index 2 in H. It is therefore a proper normal subgroup of H and, when $|H| > 2$, non-trivial. Thus for H to be simple we must have $H \subset A_n$. $\square$

Proposition 2.27 *Let G be a group of order $2m$, where m is odd and $m > 1$. Then G is not a simple group.*

Proof

G has even order so by Exercises 1.1 Question 8, there is an element $a \in G$ such that $a \neq e$ and $a^2 = e$.

By Cayley's theorem, G is isomorphic to a subgroup $\overline{G}$, say, of S_{2m}. Let $\sigma \in \overline{G}$ be the permutation corresponding to a. By Exercises 1.1 Question 14, an isomorphism preserves the order of an element so σ has order 2. This is the lcm of the orders of the disjoint cycles whose product is σ, so σ is a product of disjoint transpositions.

As shown just before Proposition 2.7, $\mathrm{Fix}(\sigma) = \emptyset$ so σ is a product of m transpositions, hence σ is an odd permutation. Thus $\overline{G} \not\subset A_{2m}$, so by Proposition 2.26 $\overline{G}$ is not simple and hence nor is G. $\square$

2.6.1 Example

By Proposition 2.27 there are no simple groups of order 6, 10, 14, 18, 22, 26, 30, ...

It can be shown that, apart from the cyclic groups of prime order, all other finite simple groups have even order. By Proposition 2.27, their orders are multiples of 4.

The classification of all finite simple groups has been a major achievement in group theory.

In Examples 2.3.4 (iii) we saw that A_5 is a simple group. In fact, we have:

Proposition 2.28 *A_n is a simple group for every integer $n \geq 5$.*

Proof

We first consider A_6. The elements of A_6 which permute $\{1, \ldots, 5\}$ and fix 6 form a subgroup H which is isomorphic to A_5 and hence simple.

Suppose A_6 has a non-trivial normal subgroup N and let σ be a non-identity element of N which keeps some element fixed; assume without loss of generality that $\sigma(6) = 6$. Then $H \cap N$ is a non-trivial normal subgroup of H so, as H is simple, $H \cap N = H$, i.e. $H \subset N$.

Thus N contains a 3-cycle. As $N \lhd A_6$ and all 3-cycles are conjugate in A_6, it follows from Exercises 2.3 Question 11 that $N = A_6$.

Suppose $\sigma \in N$ keeps no element of $\{1, \ldots, 6\}$ fixed. The possible cycle types for σ are $(a \ \ b)(c \ \ d \ \ e \ \ f)$, in which case $\sigma^2 = (c \ \ f \ \ e \ \ d)$, or $(a \ \ b \ \ c)(d \ \ e \ \ f)$, in which case taking $\tau = (b \ \ c \ \ d)$ gives $\sigma\tau\sigma\tau^{-1} = (b \ \ f \ \ c \ \ e \ \ d)$. In both cases we have an element of N which keeps some element fixed, so as shown above $N = A_6$. We have thus shown that A_6 is simple.

Now suppose $n \geq 7$ and make the inductive hypothesis that A_n is simple for $5 \leq k < n$. Let N be a non-trivial normal subgroup of A_n. If $\sigma \in N$ and $\sigma(n) = n$ then $\sigma \in A_{n-1}$ so by the same reasoning as for A_6 above, $N = A_n$. Suppose σ fixes none of $1, \ldots, n$; say without loss of generality $\sigma(1) = 2$. Let $\rho = (2 \ \ 3 \ \ 4)$ and $\tau = \rho\sigma\rho^{-1}\sigma^{-1}$. Then $\tau(2) = 3$ so τ is a non-identity element of N. Also τ is a product of two 3-cycles ρ and $\sigma\rho^{-1}\sigma^{-1}$, so it must fix at least one element. We again conclude that $N = A_n$, so A_n is simple. The result follows by induction on n. $\square$

Thus $A_5, A_6, A_7, \ldots$ are non-abelian simple groups of order $60, 360, 2520, \ldots$

Since any subgroup of index 2 is normal, A_n has no subgroup of order $\dfrac{n!}{2}$ when $n \geq 5$.

It can be proved that A_5 is the smallest non-abelian simple group. Thus the only simple groups of order less than 60 are cyclic groups of prime order: $C_2, C_3, \ldots, C_{53}, C_{59}$.

Proposition 2.29 *Suppose G is simple and has a proper subgroup H of index m. Then G is isomorphic to a subgroup of S_m and so $|G|$ divides $m!$*

Proof

By Proposition 2.5 there is a group homomorphism $\phi : G \to S_m$ with $\operatorname{Ker}\phi \subset H$. If G is simple, it has no non-trivial normal subgroups so $\operatorname{Ker}\phi = \{e\}$ and thus $G \cong \phi(G)$, which is a subgroup of S_m. The order of G must then divide $|S_m| = m!$ ☐

By contraposition, if G is simple and $|G|$ does *not* divide $m!$ then G cannot have a subgroup of index m. We can use this to find the possible orders of subgroups of simple groups.

2.6.2 Examples

(i) Suppose G is a *simple* group of order 1092 and has a subgroup of index m. Then G is isomorphic to a subgroup of S_m, so $1092 \mid m!$ Now $1092 = 2^2 \times 3 \times 7 \times 13$ so the smallest possible value of m is 13. Hence the largest possible order of *any* subgroup of G is 84.

(ii) Suppose G is a simple group of order $60 = 2^2 \times 3 \times 5$. We know that A_5 is such a group, but have not yet shown that there are no others.

$w_5 \equiv 1 \pmod 5$ and $w_5 \mid 12$, so $w_5 = 1$ or 6. Now $w_5 \neq 1$, as this would make the 5-Sylow normal in G, so $w_5 = 6$. Thus G has a subgroup of index 6 (namely the normaliser of any 5-Sylow) so G is isomorphic to a subgroup H, say, of S_6.

If H is not contained in A_6 then $H \cap A_6 \lhd H$ by Proposition 2.13. But H is simple, since G is, so $H \subset A_6$. We have $[A_6 : H] = 360/60 = 6$.

H acts on the set A_6/H of six left cosets by $\sigma(\tau H) = \sigma\tau H$. For this action, each element of H fixes the coset H and permutes the other five.

Thus H is isomorphic to a 60-element subgroup of S_5. By Proposition 2.26 this subgroup is contained in A_5 so, up to isomorphism, A_5 is the only simple group of order 60.

The following examples illustrate how the Sylow theorems can be used to show that there are no simple groups of certain orders.

2.6.3 Examples

(i) Suppose $|G| = 245 = 5 \times 7^2$. The number of 7-Sylows is congruent to 1 modulo 7 and divides 5, so it can only be 1. Thus the 7-Sylow is a normal subgroup of G, so G is not simple. See Exercises 2.6 Question 5 for a generalisation of this.

(ii) Suppose $|G| = 48 = 2^4 \times 3$. Then G has at least one 2-Sylow H of order 16.

H has index 3 in G so by Proposition 2.5, if G is simple then G is isomorphic to a subgroup of S_3 so $|G|$ divides $3! = 6$. But $48 > 6$. Hence G cannot be simple.

(iii) Suppose $|G| = 65,625 = 3 \times 5^5 \times 7$.

A 5-Sylow, of order 5^5, is a subgroup of index 21 in G. By Proposition 2.5, if G is simple then it is isomorphic to a subgroup of S_{21} and so so $|G|$ divides 21!

But the highest power of 5 that divides 21! is 5^4 (since 5 divides 5, 10, 15 and 20 but no other integer less than or equal to 21). Hence G cannot be simple.

(iv) Suppose $|G| = 12 = 2^2 \times 3$.

A 2-Sylow has order 4. $w_2 \equiv 1 \pmod 2$ and $w_2 \mid 3$. Hence $w_2 = 1$ or 3.

A 3-Sylow has order 3. $w_3 \equiv 1 \pmod 3$ and $w_3 \mid 4$. Hence $w_3 = 1$ or 4.

If $w_3 = 4$, the four 3-Sylows intersect only in the identity (as they have prime order) so together they contain 9 elements. There is then room for only one 2-Sylow, which must be normal in G.

Thus either $w_2 = 1$ or $w_3 = 1$, so no group of order 12 is simple.

(v) Suppose $|G| = 56 = 2^3 \times 7$. Then $w_7 = 1$ or 8.

Suppose $w_7 = 8$. The eight 7-element subgroups intersect only in $\{e\}$, so together they contain 49 elements. This leaves room for only one 2-Sylow, of order 8, so $w_2 = 1$. Thus either $w_2 = 1$ or $w_7 = 1$, so G has a normal subgroup and hence is not simple.

(vi) Let $|G| = pq$ where p and q are prime and $q < p$. The number w_p of p-Sylows in G is congruent to 1 (mod p) and divides q. Thus $w_p = 1$, so there is only one p-Sylow. This must be normal in G, so G is not simple.

(vii) Suppose $|G| = 1125 = 3^2 \times 5^3$. $w_5 \equiv 1 \pmod 5$ and $w_5 \mid 9$, so $w_5 = 1$. Hence the 5-Sylow H (of order 125) is normal, so G is not simple. Now $|G/H| = 9$, so G/H is isomorphic to either C_9 or $C_3 \times C_3$. If $G/H \cong C_9$ then by Proposition 1.8, G has a subgroup J isomorphic to C_9. In that case the 3-Sylows of G are cyclic.

Exercises 2.6

1. Find the largest possible order of a proper subgroup of a simple group of order (a) 60, (b) 360, (c) 660, (d) 2520, (e) 5616.

2. Suppose a group G has a conjugacy class with exactly two elements. Show that G is not a simple group.

3. Identify simple groups of order 20,160 and 20,161.

4. Determine whether A_6 has a subgroup of order (a) 60, (b) 180.

5. A group G has order $p^k m$ where p is prime, $k \in \mathbb{N}$, $1 \leq m < p$ and k, m are not both 1. Show that G cannot be simple. List all the orders between 1 and 60 to which this applies.

6. G is a simple group of order $p^k m$ where p is prime, $m > 1$ and p does not divide m. Show that $|G|$ divides $m!$ and hence that $p^k \mid (m-1)!$ List some more numbers less than 60 which cannot be orders of simple groups.

7. Suppose G is a group whose order is divisible by a prime p, and $w_p > 1$. Give a reason why G has a subgroup of index w_p. Deduce that if G is simple then $|G|$ divides $w_p!$ Hence show that there is no simple group of order 36.

8. A group G has order $p^2 q^2$ where p and q are distinct odd primes and $p > q$. Show that G has only one p-Sylow subgroup. Deduce that G is not simple.

9. G is a group of order pqr where p, q, r are distinct primes with $p > q > r$. Suppose w_p, w_q, w_r are all greater than 1. Show that $w_p = qr$, $w_q \geq p$ and $w_r \geq q$. By considering the total number of elements in the Sylow subgroups, obtain a contradiction and hence show that G is not simple.

10. Show that there are no simple groups of the following orders. Any results from the notes, examples and exercises may be used where appropriate.

    (a) 30, (b) 40, (c) 66, (d) 143, (e) 225, (f) 132, (g) 2430.

Chapter 3

3.1 Rings and fields

Recall that a **ring** $(R, +, \times)$ is a non-empty set R with binary operations $+$ and $\times$ such that $(R, +)$ is an abelian group, $(R, \times)$ is a semigroup, and multiplication is distributive over addition.

A **commutative ring** is a ring in which the semigroup $(R, \times)$ is abelian.

A **ring with unity** is a ring in which $(R, \times)$ is a monoid.

A **unit** of a ring with unity R is a unit of the multiplicative monoid $(R, \times)$. These units form the **group of units** $U(R)$ under multiplication.

We usually refer to 'the ring R', rather than $(R, +, \times)$. $a \times b$ is normally written as ab. The additive identity is denoted by 0_R and the multiplicative identity ('unity'), when it exists, by 1_R. If no confusion arises, we may simply call them 0 and 1. It is straightforward to show that they are unique, and that $a0_R = 0_R a = 0_R$ for all $a \in R$.

Note that in some books, existence of a unity 1_R is part of the definition of a ring.

A **division ring** is a non-trivial ring R with unity in which *every* non-zero element has a multiplicative inverse, so $(R^*, \times)$ is a group.

A **field** is a commutative division ring K, so $(K^*, \times)$ is an abelian group.

3.1.1 Examples

(i) $\mathbb{Z}$ is a commutative ring with unity. $\mathbb{N}$ is not a ring.

(ii) $\mathbb{Q}, \mathbb{R}$ and $\mathbb{C}$ are infinite fields.

(iii) $(\mathbb{Z}_n, +_n, \times_n)$ is a commutative ring with unity. Its units are the elements of the group U_n.

(iv) If p is prime then $(\mathbb{Z}_p, +_p, \times_p)$ is a field, denoted by $\mathbb{F}_p$.

(v) $M_n(R)$ is the set of all $n \times n$ square matrices with entries from a ring R. Under matrix addition and multiplication, $M_n(R)$ is a non-commutative ring. Its units are all the invertible $n \times n$ matrices over R.

(vi) Let $R_1, \ldots, R_n$ be rings. The (external) **direct product** $R_1 \times \cdots \times R_n$ is the set $\{(a_1, \ldots, a_n) : a_i \in R_i\}$ with operations defined component-wise, i.e. $(a_1, \ldots, a_n) + (b_1, \ldots, b_n) = (a_1 + b_1, \ldots, a_n + b_n)$ and $(a_1, \ldots, a_n)(b_1, \ldots, b_n) = (a_1 b_1, \ldots, a_n b_n)$. This is easily shown to be a ring.

A **subring** of a ring R is a non-empty subset $S \subset R$ which is itself a ring under the same operations (restricted to S). If $S \neq R$ then S is a **proper subring** of R. If $S = \{0_R\}$ then S is the **trivial subring** of R.

If S is contained in the ring R then the associative and distributive properties of R must hold in S. Thus to prove that S is a subring of R, we need only show that S is an additive subgroup of R and is closed under the ring multiplication.

This gives the **subring test**: Let S be a non-empty subset of a ring R such that for all $a, b \in S$, (i) $a - b \in S$ and (ii) $ab \in S$. Then S is a subring of R.

3.1.2 Examples

(i) For any $n \in \mathbb{N}$, let $n\mathbb{Z} = \{nk \ : \ k \in \mathbb{Z}\}$, e.g. $2\mathbb{Z}$ is the set of all even integers. The subring test shows that $n\mathbb{Z}$ is a subring of $\mathbb{Z}$, so it is a commutative ring in its own right. It does not have a unity unless $n = 1$, in which case it equals $\mathbb{Z}$, so is not a proper subring.

(ii) The **centre** of a ring R is the set $Z(R) = \{a \in R \ : \ ar = ra \text{ for all } r \in R\}$. $Z(R)$ is easily shown to be a commutative subring of R.
$Z(R) = R$ if and only if R is a commutative ring.

Let R_1 and R_2 be rings.

A **ring homomorphism** is a mapping $\phi : R_1 \to R_2$ such that for all $a, b \in R_1$,

$\phi(a + b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$.

A **ring isomorphism** is a bijective ring homomorphism.

The **kernel** of ϕ is $\operatorname{Ker}\phi = \{a \in R_1 \ : \ \phi(a) = 0_{R_2}\}$.

The **image** of ϕ is $\phi(R_1) = \{\phi(a) \ : \ a \in R_1\}$.

It is routine to show that if $\phi : R_1 \to R_2$ is a ring homomorphism then $\phi(0_{R_1}) = 0_{R_2}$; $\operatorname{Ker}\phi$ is a subring of R_1; $\phi(R)$ is a subring of R_2; ϕ is injective if and only if $\operatorname{Ker}\phi = \{0_{R_1}\}$.

Note that a ring homomorphism ϕ is a group homomorphism from $(R_1, +)$ to $(R_2, +)$. However, it is possible for two sets to be isomorphic as groups but not as rings. For example, the map $\phi : \mathbb{Z} \to n\mathbb{Z}$ given by $\phi : k \mapsto nk$ is a group isomorphism between $(\mathbb{Z}, +)$ and $(n\mathbb{Z}, +)$, but for $n > 1$ it is *not* a ring isomorphism between $(\mathbb{Z}, +, \times)$ and $(n\mathbb{Z}, +, \times)$ since $\phi(k_1k_2) = nk_1k_2$ but $\phi(k_1)\phi(k_2) = (nk_1)(nk_2)$.

As for groups, $\cong$ is used to mean 'is isomorphic to'. The type of isomorphism will normally be clear from the context; if it is not, we can write '$R_1 \cong R_2$ (as rings)'.

Wedderburn's theorem states that every finite division ring is a field, i.e. for finite sets, commutativity of multiplication is a consequence of the other field axioms. Hence there are no non-commutative finite division rings. However, there are infinite division rings which are not fields, as we now show.

Let $\mathbb{H}$ be a real vector space with basis $\{1, i, j, k\}$, so $\mathbb{H} = \{a + bi + cj + dk \ : \ a, b, c, d \in \mathbb{R}\}$. If $u = a + bi + cj + dk$, $v = p + qi + rj + sk \in \mathbb{H}$ then $u + v = (a + p) + (b + q)i + (c + r)j + (d + s)k$. Clearly $(\mathbb{H}, +)$ is an abelian group.

Multiplication is now defined on $\mathbb{H}$ as follows: $i^2 = j^2 = k^2 = -1$, $ij = -ji = k$, $jk = -kj = i$, $ki = -ik = j$, and i, j, k commute with all real numbers. This is extended to the whole of $\mathbb{H}$ so as to be distributive over addition, i.e. we 'multiply out' expressions in the usual way. Clearly 1 is the multiplicative identity in $\mathbb{H}$.

3.1.3 Example

Let $u = 2-3i+4j+k$, $v = -1+i+2j-3k$. Then $u+v = 1-2i+6j-2k$, $uv = -4-9i-8j-17k$ and $vu = -4+19i+8j+3k$.

$\{1, i, j, k\}$ is not a group under the multiplication in $\mathbb{H}$. To make it closed we must include $-1, -i, -j, -k$. The set $\{\pm 1, \pm i, \pm j, \pm k\}$ is then a group of order 8 with identity 1 and presentation $\langle i, j \ : \ i^4 = 1, i^2 = j^2, iji = j\rangle$, so it is isomorphic to the quaternion group Q_4. As the group operation is associative, the multiplication in $\mathbb{H}$ is associative on the basis elements and hence on all of $\mathbb{H}$. Thus $(\mathbb{H}, +, \times)$ is a non-commutative ring with unity.

Definition 3.1 *The ring of real* **quaternions** *is the set* $\mathbb{H} := \{a + bi + cj + dk \ : \ a, b, c, d \in \mathbb{R}\}$, *with the operations defined above.*

The elements of the form a, where $a \in \mathbb{R}$, form a subring of $\mathbb{H}$ isomorphic to $\mathbb{R}$. Each of $\{a + bi \ : \ a, b \in \mathbb{R}\}$, $\{a + cj \ : \ a, c \in \mathbb{R}\}$, $\{a + dk \ : \ a, d \in \mathbb{R}\}$ is a subring of $\mathbb{H}$, isomorphic to the field of complex numbers $\mathbb{C}$.

It can be shown that $\mathbb{H}$ is isomorphic to $\left\{\begin{pmatrix} a+b\mathrm{i} & -c+d\mathrm{i} \\ c+d\mathrm{i} & a-b\mathrm{i}\end{pmatrix} \ : \ a, b, c, d \in \mathbb{R}\right\}$, a subring of $M_2(\mathbb{C})$. (Here the non-italic 'i' denotes the 'square root of -1' in $\mathbb{C}$.)

Definition 3.2 *Let* $u = a + bi + cj + dk \in \mathbb{H}$.

The **conjugate** *of u is* $\overline{u} := a - bi - cj - dk$. *The* **norm** *of u is* $||u|| := \sqrt{u\overline{u}}$.

Note: some books define the norm of u to be $u\overline{u}$ rather than $\sqrt{u\overline{u}}$.

We find that $u\overline{u} = \overline{u}u = a^2 + b^2 + c^2 + d^2$, so $||u|| = \sqrt{a^2 + b^2 + c^2 + d^2}$, which is real and is strictly positive provided $u \neq 0$.

For any $u \neq 0$, let $u' = \dfrac{1}{||u||^2}\overline{u}$. Then $uu' = u'u = \dfrac{u\overline{u}}{||u||^2} = 1$, so u' is a multiplicative inverse of u. Hence we have:

Proposition 3.1 *The real quaternions $\mathbb{H}$ form a non-commutative division ring in which each non-zero element u has multiplicative inverse* $\dfrac{1}{||u||^2}\overline{u}$.

If $u, v \in \mathbb{H}$ and $k, \ell \in \mathbb{R}$ then clearly $\overline{ku + \ell v} = k\overline{u} + \ell\overline{v}$. Using this it can be shown (see exercises) that $\overline{uv} = \overline{v}\,\overline{u}$ for all $u, v \in H$ and hence that $||uv|| = ||u|| \,.\, ||v||$. Thus any product of two sums of four squares is itself a sum of four squares. (More generally, it can be shown that every positive integer is a sum of four squares.)

Proposition 3.2 (Euler's four-square identity) *If $a, b, c, d, e, f, g, h \in \mathbb{R}$ then $(a^2 + b^2 + c^2 + d^2)(e^2 + f^2 + g^2 + h^2) = w^2 + x^2 + y^2 + z^2$ where w, x, y, z are the coefficients of $1, i, j, k$ respectively in the expansion of $(a + bi + cj + dk)(e + fi + gj + hk)$.*

3.1.4 Example

Find $w, x, y, z \in \mathbb{N}$ such that $(1^2+2^2+3^2+4^2)(5^2+6^2+7^2+8^2) = w^2+x^2+y^2+z^2$.

Let $u = 1+2i+3j+4k,\ v = 5+6i+7j+8k$.

Then $uv = -60+12i+30j+24k$, so $\{w, x, y, z\} = \{12, 24, 30, 60\}$.

Note that the answer is not unique. We could also take, for instance,
$u = 1+2i+3j+4k, v = 5+6i+7j-8k$.

Then $uv = 4-36i+62j+8k$, so $\{w, x, y, z\} = \{4, 8, 36, 62\}$.

Exercises 3.1

1. Let R be a ring. Use the subring test to show that the centre $Z(R) = \{a \in R \ :\ ar = ra \text{ for all } r \in R\}$ is a subring of R.

2. Let S be the set of all diagonal 2×2 real matrices. Show that S is a subring of $M_2(\mathbb{R})$ and that the map $\phi : \mathbb{R} \times \mathbb{R} \to S$ given by $\phi : (a, b) \mapsto \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ is a ring isomorphism.

3. Let $\mathrm{O} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, $\mathrm{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\mathrm{J} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\mathrm{K} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\mathrm{L} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $\mathrm{M} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, where 0 and 1 are the elements of the field $\mathbb{F}_2$.

   Draw up the Cayley tables for (i) {O, I, J, K} and (ii) {O, I, L, M} under matrix addition and multiplication, all entries being evaluated modulo 2.

   In each case state whether the set forms a field.

4. Let $u = 2-3i+4j+k,\ v = -1+i+2j-3k \in \mathbb{H}$. Find u^{-1} and v^{-1}.

5. Show that $\{a+bi \ :\ a, b \in \mathbb{R}\}$ is a subring of $\mathbb{H}$. Is it commutative?

6. Show that the map $\psi : \mathbb{H} \to M_2(\mathbb{C})$ defined by $\psi : a+bi+cj+dk \mapsto \begin{pmatrix} a+b\mathrm{i} & -c-d\mathrm{i} \\ c-d\mathrm{i} & a-b\mathrm{i} \end{pmatrix}$ is an injective ring homomorphism.

7. Let $u = a+bi+cj+dk \in \mathbb{H}$. Express u as $v+wj$ where each of v and w has the form $p+qi$. Show that $\overline{u} = \overline{v} - j\overline{w}$.

8. Let $u = a+bi+cj+dk \in \mathbb{H}$. Show that $u\overline{u} = a^2+b^2+c^2+d^2$.

   Show that for all $v \in \mathbb{H}$, $\overline{iv} = \overline{v}\,\overline{i}$, $\overline{jv} = \overline{v}\,\overline{j}$ and $\overline{kv} = \overline{v}\,\overline{k}$. Hence show that $\overline{uv} = \overline{v}\,\overline{u}$ and deduce that $||uv|| = ||u||\,.\,||v||$.

9. Find $w, x, y, z \in \mathbb{N}$ such that $(2^2+4^2+6^2+8^2)(3^2+5^2+7^2+9^2) = w^2+x^2+y^2+z^2$.

3.2 Ideals and ring decompositions

Definition 3.3 *Let R be a ring (not necessarily commutative).*

A **left ideal** *of R is a subring I of R such that $ra \in I$ for all $a \in I$ and $r \in R$.*

A **right ideal** *of R is a subring I of R such that $ar \in I$ for all $a \in I$ and $r \in R$.*

An **ideal** *(or* **two-sided ideal***) of R is a subring I of R which is both a left and right ideal of R.*

A **simple ring** *is a non-trivial ring which has no non-trivial proper two-sided ideals.*

When R is commutative, a left or right ideal is necessarily a two-sided ideal.

From the subring test, a non-empty subset I of a ring R is a left/right/two-sided ideal if and only if $a - b \in I$ for all $a, b \in I$ and the conditions in the above definition hold.

3.2.1 Examples

(i) The $n \times n$ matrices whose entries are all even integers form an ideal of $M_n(\mathbb{Z})$.

(ii) The kernel of a ring homomorphism $\phi : R_1 \to R_2$ is an ideal of R_1.

(iii) The set of all real matrices of the form $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$ is a left ideal but not a right ideal of $M_2(\mathbb{R})$. That is, it is a subring closed under left multiplication, but not under right multiplication, by arbitrary elements of $M_2(\mathbb{R})$.

(iv) Let x be a fixed element of a ring R.

Let $xR = \{xy \ : \ y \in R\}$. This is clearly a non-empty subset of R.

Let $a, b \in xR$, so $a = xy$ and $b = xz$ for some $y, z \in R$.

Then $a - b = xy - xz = x(y - z)$. As R is a ring, $y - z \in R$, so $a - b \in xR$.

Also, for any $r \in R$, $ar = (xy)r = x(yr)$. As R is a ring, $yr \in R$, so $ar \in xR$.

Thus xR is a right ideal of R. If R is commutative, xR is an ideal of R.

Proposition 3.3 *If D is a division ring then $M_n(D)$ is a simple ring for each $n \in \mathbb{N}$.*

Proof

Suppose $M_n(D)$ has a non-trivial two-sided ideal J. Let A be a non-zero matrix in J, so it has some entry $a_{rs} \neq 0$. As D is a division ring, a_{rs} has a multiplicative inverse.

Let E_{ij} be the $n \times n$ matrix with (i, j) entry 1 and all other entries zero.

Then $a_{rs}^{-1}\mathrm{E}_{ir}\mathrm{A}\mathrm{E}_{si} = \mathrm{E}_{ii}$. As J is an ideal, it follows that $\mathrm{E}_{ii} \in J$ for $i = 1, \ldots, n$.

Thus the identity matrix $\mathrm{I}_n = \mathrm{E}_{11} + \mathrm{E}_{22} + \cdots + \mathrm{E}_{nn}$ is in J, and then for every matrix $\mathrm{X} \in M_n(D)$ we have $\mathrm{X}\mathrm{I}_n \in J$, so $\mathrm{X} \in J$. Thus $M_n(D) \subset J \subset M_n(D)$.

Hence $J = M_n(D)$ so $M_n(D)$ has no non-trivial proper ideals, i.e. it is simple. $\square$

Definition 3.4

An **idempotent** *in a ring R is an element $e \in R$ such that $e^2 = e$.*

The **trivial idempotents** *are* 0 *and (if it exists in R)* 1.

An idempotent e of R is **central** *if it is in $Z(R)$, i.e. $er = re$ for all $r \in R$.*

Idempotents e and f in a ring R are **orthogonal** *if $ef = fe = 0$.*

An idempotent which is not *the sum of two non-zero orthogonal idempotents is called* **primitive**.

In ring theory, the letter e is usually used for an idempotent. This should not be confused with the use of e for the identity element of a group (although that certainly *is* idempotent).

Proposition 3.4 *Let R be a ring with unity and let e be an idempotent in R. Then $1 - e$ is also idempotent, and orthogonal to e. If e is central, so is $1 - e$.*

Proof

As e is idempotent, $e^2 = e$.

$(1-e)^2 = 1 - e - e + e^2 = 1 - e - e + e = 1 - e$, so $1 - e$ is idempotent.

$e(1-e) = e - e^2 = e - e = 0$, so e and $1 - e$ are orthogonal.

Suppose $e \in Z(R)$. For any $r \in R, r(1-e) = r - re = r - er = (1-e)r$ so $1 - e \in Z(R)$.

$\square$

An ideal of a ring R may have a unity which is not that of R itself, if indeed R has one. For example, $\{0, 2, 4, 6, 8\}$ is an ideal of $\mathbb{Z}_{10}$ in which 6, not 1, is the unity. Notice that $6^2 = 6$ in $\mathbb{Z}_{10}$, so 6 is idempotent, and $\{0, 2, 4, 6, 8\} = 6\mathbb{Z}_{10}$. The following result shows why this must be so:

Proposition 3.5 *Let I be an ideal of a ring R, such that I is itself a ring with unity e. Then e is a central idempotent of R, and $I = eR$.*

Proof As $e = 1_I$ we have $e^2 = e$ so e is idempotent. Clearly $e \in R$.

For all $r \in R$, $er \in I$ so $er = ere$ and $re = ere$.

Thus $er = re$ for all $r \in R$ so $e \in Z(R)$, i.e. e is central.

For any $b \in I$ we have $b = eb \in eR$, so $I \subset eR$. Also $eR \subset I$, so $I = eR$. $\square$

For any ideals I and J of a ring R, the intersection $I \cap J$ and the sum $I + J = \{a + b \ : \ a \in I, b \in J\}$ are also ideals of R (see Exercises).

Proposition 3.6 *Let I and J be ideals of a ring R with unity, such that $R = I + J$ and $I \cap J = \{0\}$. Then $R \cong I \times J$ (as rings), and $1_R = e + f$ where $e = 1_I$, $f = 1_J$ are central idempotents of R such that $I = eR$, $J = fR$.*

Proof

By Proposition 1.5, R and $I \times J$ are isomorphic as additive groups, the map $\phi : I \times J \to R$ by $\phi : (a, b) \mapsto a + b$ being a group isomorphism and hence bijective.

Also $\phi(a, b)\phi(c, d) = (a + b)(c + d) = ac + bd$ (since $ad, bc \in I \cap J = \{0\}$)
$= \phi(ac, bd) = \phi((a, b)(c, d))$. Thus ϕ is a ring isomorphism, i.e. $R \cong I \times J$ (as rings).

As $R = I + J$ we have $1 = e + f$ for some $e \in I, f \in J$ so for any $a \in I$, $\quad a - ea \in I$ and $a - ea = (1 - e)a = fa \in J$, as both I and J are ideals. As $I \cap J = \{0\}$ it follows that $a - ea = 0$, so $a = ea$. Similarly $a = ae$.

Hence e is a unity in I, so by Proposition 3.5 e is a central idempotent of R and $I = eR$. Similarly, f is a unity in J, $f \in Z(R)$ and $J = fR$. □

Definition 3.5 *Let I and J be ideals of a ring R such that $R = I + J$ and $I \cap J = \{0_R\}$. Then R is the (internal)* **direct sum** *of I and J, denoted by $I \oplus J$*

By Proposition 3.6, $I \oplus J \cong I \times J$ (as rings).

When $R = I \oplus J$, every element is uniquely expressible as the sum of an element of I and an element of J. For all $i \in I$ and $j \in J$ we have $ij \in I \cap J$, so $ij = 0$.

As for groups and vector spaces, the definition can be extended to give the direct sum $I_1 \oplus \cdots \oplus I_k$, where $I_1, \ldots, I_k$ are ideals, $R = I_1 + \cdots + I_k$ and $(I_1 + \cdots + I_{j-1}) \cap I_j = \{0\}$ for $j = 2, \ldots, k$.

Proposition 3.7 *Let e be a central idempotent in a ring with unity R.*

Then $R = eR \oplus (1 - e)R$.

Proof By Proposition 3.4, $1 - e$ is also a central idempotent of R.

Let $I = eR$, $J = (1 - e)R$. Then I and J are easily shown to be ideals of R.

Clearly $I + J \subset R$. Any element $r \in R$ can be written as $er + (1 - e)r \in I + J$, so $R \subset I + J$. Thus $R = I + J$.

Suppose $r \in I \cap J$. Then $r = ea = (1 - e)b$ for some $a, b \in R$, so $ea = b - eb$. Thus $e^2a = eb - e^2b$, so $ea = eb - eb$, so $r = 0$.

Hence $I \cap J = \{0\}$ so by Proposition 3.6, $R = eR \oplus (1 - e)R$. □

3.2.2 Example

In $\mathbb{Z}_{20}$, $5^2 = 5$ so 5 is idempotent. $1 - 5 = -4 = 16$ and $16^2 = (-4)^2 = 16$ so 16 is also idempotent.

$5 \times 16 = 0$ in $\mathbb{Z}_{20}$, i.e. 5 and 16 are orthogonal idempotents. They are central, since $\mathbb{Z}_{20}$ is a commutative ring.

$5\mathbb{Z}_{20} = \{0, 5, 10, 15\}$ is an ideal of $\mathbb{Z}_{20}$, in which 5 is the unity.

$16\mathbb{Z}_{20} = \{0, 4, 8, 12, 16\}$ is an ideal of $\mathbb{Z}_{20}$, in which 16 is the unity.

We have $\mathbb{Z}_{20} = 5\mathbb{Z}_{20} \oplus 16\mathbb{Z}_{20}$. That is, every element of $\mathbb{Z}_{20}$ can be uniquely expressed as $a + b$ where $a \in 5\mathbb{Z}_{20}, b \in 16\mathbb{Z}_{20}$ and $ab = 0$ in $\mathbb{Z}_{20}$. For instance, $7 = 15 +_{20} 12$ and $15 \times_{20} 12 = 0$. Furthermore, $\mathbb{Z}_{20}$ is isomorphic to the direct product ring $5\mathbb{Z}_{20} \times 16\mathbb{Z}_{20}$.

Exercises 3.2

1. Show that $\left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} \ : \ a, b \in \mathbb{R} \right\}$ is a right but not left ideal of $M_2(\mathbb{R})$.

2. x is a fixed element of a (not necessarily commutative) ring R. Show that $Rx = \{rx \ : \ r \in R\}$ is a left ideal of R.

3. Let R be a simple ring with unity. Let z be a non-zero element of the centre $Z(R)$. Show that z has a multiplicative inverse in R.

4. Let I and J be ideals of a ring R. Show that $I \cap J$ and $I + J$ are ideals of R.

5. Suppose A is an idempotent square matrix, other than the zero or identity matrix. Show that both A and A $-$ I are singular.

6. e is an idempotent in a ring R with unity. Show that (a) $e + (1 - e)re$ is idempotent for all $r \in R$, (b) $1 - 2e$ is a unit of R.

7. e is an idempotent in a ring R. $eRe = \{ere \ : \ r \in R\}$. Show that eRe is a subring of R, that e is the unity of eRe and that $eRe = \{a \in R \ : \ ea = ae = a\}$. Find eRe when $R = M_2(\mathbb{R})$ and $e = \mathrm{E}_{11}$.

8. Show that 9 is idempotent in the ring $\mathbb{Z}_{18}$. Hence express $\mathbb{Z}_{18}$ as the direct sum of two proper ideals. Express 3 as a sum of elements from the two ideals, checking that the product of the two elements is zero.

3.3 Algebras

Definition 3.6 *Let K be a field and let A be a vector space over K with a* **bilinear** *multiplication, i.e. $a(kb + \ell c) = kab + \ell ac$ and $(ka + \ell b)c = kac + \ell bc$ for all $a, b, c \in A$ and $k, \ell \in K$. Then A is called an* **algebra** *over K, or a* **K-algebra**.

The **dimension** *of A, $\dim_K(A)$, is its dimension as a vector space over K.*

A is a **commutative algebra** *if $ab = ba$ for all $a, b \in A$ and an* **associative algebra** *if $a(bc) = (ab)c$ for all $a, b, c \in A$.*

A is an **Algebra with unity** *if there is a multiplicative identity $\mathbf{1}_{\mathrm{A}}$ (or $\mathbf{1}$) such that $a\mathbf{1}_{\mathrm{A}} = \mathbf{1}_{\mathrm{A}}a = a$ for all $a \in A$ and $\mathbf{1}_{\mathrm{A}}$ is referred to as* **unity**.

A **division algebra** *is an associative algebra in which there is a multiplicative identity, and every non-zero element has a multiplicative inverse.*

An **algebra homomorphism** *is a map ϕ such that $\phi(ab) = \phi(a)\phi(b)$ and $\phi(ka+\ell b) = k\phi(a)+\ell\phi(b)$ for all $a, b \in A$ and $k, \ell \in K$. An* **algebra isomorphism** *is a bijective algebra homomorphism.*

A **subalgebra** *of an algebra A is a non-empty subset S of A which is an algebra under the same operations (restricted to S). An* **ideal** *of A is a subalgebra I of A such that $ai \in I$ and $ia \in I$ for all $a \in A, i \in I$.*

A **simple algebra** *is a non-trivial algebra which has no non-trivial proper ideals.*

Note that an associative algebra is both a vector space and a ring, and then an algebra homomorphism is both a linear map and a ring homomorphism.

All the following examples are associative algebras. We shall not consider non-associative algebras in this module.

3.3.1 Examples

(i) $\mathbb{R}$ and $\mathbb{C}$ are division algebras over $\mathbb{R}$, of dimensions 1 and 2 respectively.

(ii) For any field K, $M_n(K)$ is an n^2-dimensional algebra over K.

(iii) Let V be a vector space over K. The linear maps of V, i.e. the functions $T : V \to V$ such that $T(ku + \ell v) = kTu + \ell Tv$ for all $u, v \in V$ and $k, \ell \in K$, form an algebra $\mathcal{L}(V)$. The 'multiplication' in this algebra is composition of maps, i.e. $ST : v \mapsto S(Tv)$. If V has finite dimension n then $\mathcal{L}(V) \cong M_n(K)$.

(iv) Proposition 3.3 applied to algebras shows that if D is a division algebra then $M_n(D)$ is a simple algebra.

(v) The ring of quaternions $\mathbb{H}$ is a 4-dimensional non-commutative division algebra over $\mathbb{R}$. This can be generalised: for any field, let $\mathbb{H}(K)$ be the vector space over K with basis $\{1, i, j, k\}$ and multiplication defined as in $\mathbb{H}$. Then $\mathbb{H}(K)$ is a K-algebra which is 4-dimensional over K.

(vi) Given K-algebras A and B, the direct product $A \times B = \{(a, b) \ : \ a \in A, b \in B\}$ is an algebra with component-wise addition and multiplication, and scalar multiplication given by $k(a, b) = (ka, kb)$.

(vii) Let A be an algebra over a field K, with unity 1_A. Then $k1_A \in A$ for all $k \in K$ so, identifying $k1_A$ with k, we have $K \subset A$. Further, $a(k.1_A) = k(a.1_A)$ so $ak = ka$, i.e. $K \subset Z(A)$.

Let A be a finite-dimensional associative algebra with unity, over a field K. For each element $x \in A$, define $\lambda_x : A \to A$ by $\lambda_x : a \mapsto xa$, so λ_x is left multiplication by x.

It is straightforward to show that λ_x is a linear map of A and that for all $x, y \in A$ and $k, \ell \in K$ we have $\lambda_{kx+\ell y} = k\lambda_x + \ell\lambda_y$ and $\lambda_{xy} = \lambda_x\lambda_y$.

Let C_x be the matrix of λ_x relative to an ordered basis β for A over K. Define the map $\theta : A \to M_n(K)$ by $\theta(x) = \mathrm{C}_x$.

Then $\theta(kx + \ell y) = k\mathrm{C}_x + \ell\mathrm{C}_y = k\theta(x) + \ell\theta(y)$ and $\theta(xy) = \mathrm{C}_{xy} = \mathrm{C}_x\mathrm{C}_y = \theta(x)\theta(y)$.

Thus θ is an algebra homomorphism. If $\theta(x) = \mathrm{O}$ then $\mathrm{C}_x = \mathrm{O}$, so λ_x is the zero map, i.e. $xa = 0$ for all $a \in A$. Taking $a = 1_A$ gives $x = 0$. Hence θ is injective. Its image $\theta(A)$ is a subalgebra of $M_n(K)$, so we have:

Proposition 3.8 *Let A be an n-dimensional associative algebra with unity over a field K. Then A is isomorphic to a subalgebra of $M_n(K)$.*

Definition 3.7 *The (left)* **regular representation** *of the algebra A is the injective homomorphism $\theta : A \to M_n(K)$ which maps each $x \in A$ to the matrix representing the map $a \mapsto xa$ relative to a given ordered basis.*

The regular representation is not unique, as the matrices will depend on the choice of ordered basis for A. Given such a basis $(v_1, \dots, v_n)$, the matrix representing an element x is found as follows. For $i = 1, \dots, n$, express xv_i as a linear combination of $v_1, \dots, v_n$ and write the coefficients in column i of the matrix.

3.3.2 Example

The division algebra $\mathbb{H}$ has typical element $u = a + bi + cj + dk$.

$u1 = a + bi + cj + dk$, $ui = -b + ai + dj - ck$, $uj = -c - di + aj + bk$, $uk = -d + ci - bj + ak$ so relative to the ordered basis $(1, i, j, k)$ for $\mathbb{H}$, the elements $u1, ui, uj, uk$ have coordinates (a, b, c, d), $(-b, a, d, -c)$, $(-c, -d, a, b)$, $(-d, c, -b, a)$ respectively. Thus the matrix which corresponds to u is $\begin{pmatrix} a & -b & -c & -d \\ b & a & -d & c \\ c & d & a & -b \\ d & -c & b & a \end{pmatrix}$.

The set of all real matrices of this form is a subalgebra of $M_4(\mathbb{R})$, isomorphic to $\mathbb{H}$. This gives the regular representation of $\mathbb{H}$.

On page 52 we mentioned a different matrix representation of $\mathbb{H}$, using 2×2 complex matrices.

Now let G be a finite group with elements $x_1, \dots, x_n$, the operation being written multiplicatively, and let K be a field.

Define KG to be the set of all formal sums $\sum_{i=1}^{n} a_i x_i$ where $a_1, \dots, a_n \in K$. (A 'formal sum' is simply an expression formed in this way, as we do with polynomials and abstract vector spaces. It need not be derived from any property of the group G.)

Addition in KG is defined term-by-term: $\sum_{i=1}^{n} a_i x_i + \sum_{i=1}^{n} b_i x_i = \sum_{i=1}^{n} (a_i + b_i) x_i$. This operation is clearly commutative and associative. There is an additive identity $0 = \sum_{i=1}^{n} 0 x_i$. The additive inverse of $\sum_{i=1}^{n} a_i x_i$ is $\sum_{i=1}^{n} (-a_i) x_i$. Hence $(KG, +)$ is an abelian group, and is closed under multiplication by scalars in K.

Define multiplication in KG in the 'obvious' ways: multiply distributively using the rule $(a_i x_i)(b_j x_j) = (a_i b_j) x_k$, where $x_k = x_i x_j$ in G. Then KG is closed under this multiplication, which is associative due to the associativity in G.

$G \subset KG$, since we can identify x_j with the sum $\sum_{i=1}^{n} a_i x_i$ where $a_j = 1$ and $a_i = 0$ for $i \neq j$. The identity of G is a multiplicative identity in KG. To avoid confusion with an idempotent e, denote the identity of G by 1_G, or just 1. We conclude that KG is an algebra over K, spanned by the set of elements of G. It is a commutative algebra if and only if G is abelian.

$1_K x = x$ for all $x \in G$, so 1_K also acts as a multiplicative identity in KG. To avoid confusion with an idempotent e, we shall now write the identity of G as 1.

Identifying each $k \in K$ with $k.1$, we also have $K \subset KG$.

Definition 3.8 *Let G be a finite group and let K be a field.*

The set KG with the operations defined above is the **group algebra** *of G over K.*

3.3.3 Examples

(i) Let $G = \{1, g\}$ be a two-element group, so $g^2 = 1$.

We know that $\mathbb{F}_2 = \{0, 1\}$ is a field.

The group algebra $\mathbb{F}_2 G$ is the 2-dimensional vector space over $\mathbb{F}_2$ with basis $\{1, g\}$ so its elements are $0.1 + 0.g, 0.1 + 1.g, 1.1 + 0.g, 1.1 + 1.g$, i.e. $0, 1, g, 1 + g$.

$1 + 1 = 0$ in $\mathbb{F}_2$, so also in $\mathbb{F}_2 G$. Then $g + g = (1 + 1)g = 0g = 0$. Also $(1 + g)(1 + g) = 1 + g + g + 1 = 0$.

The Cayley tables for the algebra $\mathbb{F}_2 G$ are:

$+$	0	1	g	$1+g$
0	0	1	g	$1+g$
1	g	0	$1+g$	g
g	g	$1+g$	0	1
$1+g$	$1+g$	g	1	0

$\times$	0	1	g	$1+g$
0	0	0	0	0
1	0	1	g	$1+g$
g	0	g	1	$1+g$
$1+g$	0	$1+g$	$1+g$	0

We see that $(\mathbb{F}_2 G, +) \cong V$.

$(\mathbb{F}_2 G^*, \times)$ is not a group as it is not closed $(0 \notin \mathbb{F}_2 G^*)$.

(ii) Let K be any field and let $G = \{1, g, g^2\}$ be a group of order 3. The group algebra KG consists of all sums of the form $a + bg + cg^2$ where $a, b, c \in K$. If K is a finite field with q elements then KG has q^3 elements. If K is an infinite field then KG is infinite.

As $g^3 = 1$ and $g^4 = g$, the product of two elements in KG is given by $(a + bg + cg^2)(\ell + mg + ng^2) = (a\ell + bn + cm) + (am + b\ell + cn)g + (an + bm + c\ell)g^2$.

Let $x = a + bg + cg^2$ be any element of KG. Then $x1 = a + bg + cg^2$, $xg = c + ag + bg^2$, $xg^2 = b + cg + ag^2$ so by Proposition 3.8, KG is isomorphic to the algebra over K of all 3×3 matrices of the form $\begin{pmatrix} a & c & b \\ b & a & c \\ c & b & a \end{pmatrix}$. That is, we can represent $1, g, g^2$ by the matrices $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ respectively.

If A is an associative algebra then, ignoring the scalar multiplication, $(A, +, \times)$ is a ring. Thus a group algebra can be decomposed as in Proposition 3.7, provided the necessary idempotents exist. We now apply this to finite group algebras over the rational, real or complex numbers.

Note that if $n \in \mathbb{N}$ then $\frac{1}{n}$ is meaningful in any field K such that $\chi(K)$ does not divide n; it means the multiplicative inverse in K of $n.1_K$.

Proposition 3.9 *Let G be a finite group and K a field whose characteristic does not divide $|G|$. Let $e = \frac{1}{|G|}\sum_{x\in G} x$. Then e is a central idempotent in the group algebra KG and so $KG = eKG \oplus (1-e)KG$. Furthermore,* $\dim_K(eKG) = 1$.

Proof

Let $G = \{x_1, \ldots, x_n\}$ and let $e = \frac{1}{n}(x_1 + \cdots + x_n)$.

For each x_i, the elements $x_ix_1, \ldots, x_ix_n$ are $x_1, \ldots, x_n$ in some order, so $x_ie = e$. Similarly $ex_i = e$, so $e \in Z(KG)$, i.e. e is central.

$e^2 = \frac{1}{n}(x_1e + \cdots + x_ne) = \frac{1}{n}(e + \cdots + e) = e$, so e is an idempotent in KG.

Thus by Proposition 3.7, $KG = eKG \oplus (1-e)KG$.

$eKG = \{e(c_1x_1 + \cdots + c_nx_n) \ : \ c_1, \ldots, c_n \in K\} = \{(c_1 + \cdots + c_n)e \ : \ c_1, \ldots, c_n \in K\} = \text{span}_K\{e\}$, which has dimension 1 over K. $\square$

Note that $eKG \oplus (1-e)KG$ is a direct sum of subspaces of the vector space KG, as well as a direct sum of subrings of the ring KG.

3.3.4 Example

Let A be the group algebra $\mathbb{C}C_3 = \{a + bg + cg^2 \ : \ a, b, c \in \mathbb{C}\}$ as in Examples 3.3.3 (ii).

$e = \frac{1}{3}(1 + g + g^2)$ is an idempotent. This can be verified as follows:

$e^2 = \frac{1}{9}(1 + g + g^2 + g + g^2 + g^3 + g^2 + g^3 + g^4) = \frac{1}{9}(3 + 3g + 3g^2) = e$.

In the regular representation, e is represented by $\frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$.

Then $1 - e = \frac{1}{3}(2 - g - g^2)$ is an idempotent which is orthogonal to e. It is represented by $\frac{1}{3}\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$.

We have $A = eA \oplus (1-e)A$.

As eA is one-dimensional over $\mathbb{C}$, it is isomorphic to $\mathbb{C}$.

A has dimension 3, so $(1 - e)A$ has dimension $3 - 1 = 2$ over $\mathbb{C}$.

Exercises 3.3

1. Find the matrix which represents the element x of the algebra A over the field $\mathbb{R}$ in the left regular representation relative to the given ordered basis β, in each of the cases:

   (a) $A = \mathbb{C}$, $x = a + b\mathrm{i}$, $\beta = (1, \mathrm{i})$ (where $\mathrm{i}^2 = -1$).

   (b) $A = \{a + bv \;:\; a, b \in \mathbb{R}\}$, $x = a + bv$, $\beta = (1, v)$, where $v^2 = 0$.

   (c) $A = M_2(\mathbb{R})$, $x = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\beta = (\mathrm{E}_{11}, \mathrm{E}_{12}, \mathrm{E}_{21}, \mathrm{E}_{22})$.

   (d) $A = \mathbb{H}$, $x = a + bi + cj + dk$, $\beta = (k, j, i, 1)$.

2. For each of the following group algebras, state its order and its dimension over the field:

   (a) $\mathbb{F}_3 C_2$, (b) $\mathbb{F}_2 S_3$, (c) $\mathbb{F}_5 A_4$, (d) $\mathbb{Q} D_4$.

3. Make Cayley tables for addition and multiplication in the algebra $\mathbb{F}_2 C_3$. Identify the group $(\mathbb{F}_2 C_3, +)$. Is $(\mathbb{F}_2 {C_3}^*, \times)$ a group?

4. Let A be the group algebra $\mathbb{R} C_5$, where C_5 is a cyclic group with generator g.

   Write down an idempotent e in A. Hence express A as a direct sum of two proper ideals and state their dimensions over $\mathbb{R}$.

   Give the matrix of e in the regular representation.

5. In Example 3.3.4, let $f = \dfrac{1}{3}(1 + \omega g + \omega^2 g^2)$, where $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$.

   Show that f is an idempotent in A, f is orthogonal to e and $f \in (1 - e)A$.

6. Let g be a generator of the cyclic multiplicative group C_n, where $n \geq 2$. Show that $g - 1$ is a zero-divisor in the group algebra KC_n (where K is a field). Deduce that KC_n is not a division algebra.

7. Show that the idempotent e in the proof of Proposition 3.9 is primitive.

8. Let A be the associative algebra with basis $\{1, x, x^2\}$ over the finite field $\mathbb{F}_5$, where x satisfies the equation $x^3 = x$.

   (a) State the number of elements in A.

   (b) Find the matrix which represents a general element $a + bx + cx^2$ (where $a, b, c \in \mathbb{F}_5$) in the regular representation of A relative to $(1, x, x^2)$.

   (c) Find how many elements of A have multiplicative inverses in A. Also find the number of self-inverse elements and identify the group which they form.